A LOCAL ANALOGUE OF THE GHOST CONJECTURE OF BERGDALL–POLLACK

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Abstract. We formulate a local analogue of the ghost conjecture of Bergdall and Pollack, which essentially relies purely on the representation theory of $\mathrm{GL}_2(\mathbb{Q}_p)$. We further study the combinatorial properties of the ghost series as well as its Newton polygon, in particular, giving a characterization of the vertices of the Newton polygon and proving an integrality result of the slopes. In a forthcoming sequel, we will prove this local ghost conjecture under some mild hypothesis and give arithmetic applications.

Contents

1. Introduction

Let p be a prime number and let N be a positive integer relatively prime to p. The central object of this paper is *U_p-slopes*, that is, the p-adic valuations of the eigenvalues of the U_p-operator acting on the space of (overconvergent) modular forms of level $\Gamma_1(Np)$, or on more general space of (overconvergent) automorphic forms essentially of $\mathrm{GL}_2(\mathbb{Q}_p)$-type. In this paper, the p-adic valuation is normalized so that $v_p(p) = 1$.

Question 1.1. What the slopes of modular forms are?

This seemingly naive question has surprisingly definite answers in some cases. Inspired by the numerical pioneering works of Gouvêa and Mazur [GM92, Go01], as well as many foundational works of Buzzard–Calegari and many others (to name a few: [BC04], [Bu05], [Cl05], and [Lo07]), a significant breakthrough on this topic was made in the recent series of works by Bergdall and Pollack [BP19a, BP19b], in which they proposed a new conjecture on slopes: (under some regular conditions), the slopes of modular forms should be given

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by the slopes of the Newton polygon of a combinatorially defined power series, called *ghost series*. On the one hand, this conjecture unifies all previous works, and on the other hand its combinatorial structure provides theoretical insight into these numerical data of slopes. The purpose of this manuscript is to reproduce this idea in a purely local setup and investigate several important properties of the corresponding ghost series.

In what follows, we shall first introduce our local version of ghost conjecture and then discuss its relation to conjectures that are historically important.

To motivate our construction, we first fix an absolutely irreducible residual Galois representation $\bar{r} : \mathrm{Gal}_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F})$ with $\mathbb{F}$ a finite extension of $\mathbb{F}_p$. For an open compact subgroup $K \subseteq \mathrm{GL}_2(\mathbb{A}_f)$, write $Y(K)$ for the corresponding open modular curve over $\mathbb{Q}$. For $\mathcal{O}$ a complete discrete valuation ring with residue field $\mathbb{F}$ and uniformizer ϖ, let

$$\mathrm{H}_1^{\text{ét}}\big(Y(K)_{\overline{\mathbb{Q}}}, \mathcal{O}\big)_{\mathfrak{m}_{\bar{r}}}^{\mathrm{cplx}=1}$$

denote the subspace of the first étale homology of the modular curve $Y(K)$, localized at the maximal Hecke ideal $\mathfrak{m}_{\bar{r}}$ corresponding to $\bar{r}$, on which a fixed complex conjugation acts by 1. Fix a neat tame level $K^p \subseteq \mathrm{GL}_2(\mathbb{A}_f^p)$. The *$\bar{r}$-localized completed homology* with tame level K^p is defined to be

$$\widetilde{\mathrm{H}}(K^p)_{\mathfrak{m}_{\bar{r}}} := \varprojlim_m \mathrm{H}_1^{\text{ét}}\Big(Y\big(K^p(1+p^m\mathrm{M}_2(\mathbb{Z}_p))\big)_{\overline{\mathbb{Q}}}, \mathcal{O}\Big)_{\mathfrak{m}_{\bar{r}}}^{\mathrm{cplx}=1}. \tag{1.1.1}$$

This completed homology is a finitely generated projective (right) $\mathcal{O}[\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$-module whose $\mathrm{GL}_2(\mathbb{Z}_p)$-action extends to a $\mathrm{GL}_2(\mathbb{Q}_p)$-action. As discovered by Emerton [Em06], one can recover the information of classical modular forms from $\widetilde{\mathrm{H}}(K^p)_{\mathfrak{m}_{\bar{r}}}$ as follows: for each weight $k \geq 2$, let $\mathrm{Sym}^{k-2}\,\mathcal{O}^{\oplus 2}$ denote the $(k-2)$th symmetric power of the standard representation of $\mathrm{GL}_2(\mathbb{Z}_p)$ (the right action of $\mathrm{GL}_2(\mathbb{Z}_p)$ is the transpose of the standard left action), then

$$\mathrm{Hom}_{\mathcal{O}[\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]}\big(\widetilde{\mathrm{H}}(K^p)_{\mathfrak{m}_{\bar{r}}},\ \mathrm{Sym}^{k-2}\,\mathcal{O}^{\oplus 2}\big) \cong \mathrm{H}^1_{\text{ét}}\big(Y(K^p\,\mathrm{GL}_2(\mathbb{Z}_p))_{\overline{\mathbb{Q}}},\ \mathrm{Sym}^{k-2}(R^1\pi_*\underline{\mathcal{O}})\big)_{\mathfrak{m}_{\bar{r}}}^{\mathrm{cplx}=1}, \tag{1.1.2}$$

where $R^1\pi_*\underline{\mathcal{O}}$ is the usual rank two étale local system on the open modular curve. It is well-known that the right hand side of (1.1.2) is isomorphic to the $\mathfrak{m}_{\bar{r}}$-localized space of modular forms as a Hecke module (e.g. [Em06, Proposition 4.4.2]). Moreover, a similar isomorphism holds for $\mathrm{GL}_2(\mathbb{Z}_p)$ replaced by the Iwahori subgroup $\mathrm{Iw}_p = \big(\begin{smallmatrix} \mathbb{Z}_p^\times & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{smallmatrix}\big)$, and the U_p-action on the space of classical forms can be realized easily on the left hand side of (1.1.2). So in a sense, the completed homology $\widetilde{\mathrm{H}}(K^p)_{\mathfrak{m}_{\bar{r}}}$ captures all information we need to study p-adic properties of classical modular forms. This motivates the following.

Definition 1.2. Consider a residual representation $\bar{\rho} : \mathrm{I}_{\mathbb{Q}_p} \to \mathrm{GL}_2(\mathbb{F})$ of the *inertia subgroup at p* that is *reducible*, nonsplit, and generic. More precisely,

$$\bar{\rho} \simeq \begin{pmatrix} \omega_1^{a+b+1} & * \neq 0 \\ 0 & \omega_1^b \end{pmatrix} \qquad \text{for } 1 \leq a \leq p-4 \text{ and } 0 \leq b \leq p-2, \tag{1.2.1}$$

where $\omega_1 : \mathrm{I}_{\mathbb{Q}_p} \to \mathrm{Gal}(\mathbb{Q}_p(\mu_p)/\mathbb{Q}_p) \cong \mathbb{F}_p^\times$ is the first fundamental character.

A *primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar{\rho}$* is a finitely generated projective (right) $\mathcal{O}[\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$-module $\widetilde{\mathrm{H}}$ with a (right) $\mathrm{GL}_2(\mathbb{Q}_p)/\big(\begin{smallmatrix} p & \\ & p \end{smallmatrix}\big)^{\mathbb{Z}}$-action extending the $\mathrm{GL}_2(\mathbb{Z}_p)$-action from the module structure, such that

- $\overline{\mathrm{H}} := \widetilde{\mathrm{H}}/(\varpi, \mathrm{I}_{1+p\mathrm{M}_2(\mathbb{Z}_p)})$ is isomorphic to the projective envelope of $\mathrm{Sym}^a\,\mathbb{F}^{\oplus 2}\otimes \det^b$ as an $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$-module, where $\mathrm{I}_{1+p\mathrm{M}_2(\mathbb{Z}_p)}$ is the augmentation ideal of the Iwasawa algebra $\mathcal{O}[\![1+p\mathrm{M}_2(\mathbb{Z}_p)]\!]$.

By Serre weight conjecture, in the setup of modular forms, if $\bar{r}|_{\mathrm{I}_{\mathbb{Q}_p}} \cong \bar{\rho}$, $\widetilde{\mathrm{H}}(K^p)_{\mathfrak{m}_{\bar{r}}}$ is a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar{\rho}$, if the following mod-p-multiplicity-one condition is met:

$$\mathrm{rank}_{\mathcal{O}}\Big(\mathrm{H}_1^{\text{ét}}\big(Y(K^p\mathrm{Iw}_{p,1})_{\overline{\mathbb{Q}}}, \mathcal{O}\big)^{\mathrm{cplx}=1}_{\mathfrak{m}_{\bar{r}}}\Big) = 2; \tag{1.2.2}$$

here $\mathrm{Iw}_{p,1} = \begin{pmatrix} 1+p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & 1+p\mathbb{Z}_p \end{pmatrix}$ and the rank 2 module is the direct sum of two rank-one eigenspaces for the diamond operators in $(\mathbb{F}_p^\times)^2$ with eigenvalues $\omega^a\times 1$ and $1\times\omega^a$, respectively, where $\omega : \mathbb{F}_p^\times \to \mathcal{O}^\times$ is the Teichmüller lift.

One of the key features of our new setup is that: it is purely local, and does not depend on the global automorphic input.

Remark 1.3. There are several origins of primitive $\mathcal{O}[\![K_p]\!]$-projective augmented modules.

(1) For any $U(2)$-Shimura varieties or quaternionic Shimura varieties (where p splits completely in the corresponding fields), taking its completed homology (localized at an appropriate $\bar{r}$) similarly to (1.1.1), we obtain a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module provided a mod-p-multiplicity-one assumption similar to (1.2.2) hold.

(2) In a global setup as in (1) with the mod-p-multiplicity-one assumption, assuming some further Taylor–Wiles conditions, one may construct a patched completed homology as in [CEGGPS, § 2], which is a projective module over $\mathcal{O}[\![z_1,\dots,z_g]\!][\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$ and whose $\mathrm{GL}_2(\mathbb{Z}_p)$-action extends to an $\mathcal{O}[\![z_1,\dots,z_g]\!]$-linear action of $\mathrm{GL}_2(\mathbb{Q}_p)/p^{\mathbb{Z}}$ (assuming a certain central character condition). Evaluating the patching variables $z_1,\dots,z_g$ arbitrarily will give rise to a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar{\rho}$, which does not have any automorphic origin.

(3) Unfortunately, we do not know, for a fixed $\bar{\rho}$ in (1.2.1), whether one can find a global $\bar{r}$ such that the restriction of $\bar{r}$ to some p-adic place isomorphic to $\bar{\rho}$ and that $\bar{r}$ satisfies the mod-p-multiplicity-one assumption similar to (1.2.2). Fortunately, the Paškūnas functor provides yet another example of primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module.

   Let $\tilde{\rho} = \begin{pmatrix} \omega_1^{a+b+1}\mathrm{ur}(\alpha_1) & *\neq 0 \\ 0 & \omega_1^b\mathrm{ur}(\alpha_2) \end{pmatrix} : \mathrm{Gal}_{\mathbb{Q}_p} \to \mathrm{GL}_2(\mathbb{F})$ be a residual Galois representation whose restriction to $\mathrm{I}_{\mathbb{Q}_p}$ is $\bar{\rho}$. Here ω_1 can be extended to a character of $\mathrm{Gal}_{\mathbb{Q}_p}$ in a natural way, and $\mathrm{ur}(\alpha_i)$ for $i=1,2$ is the unramified representation of $\mathrm{Gal}_{\mathbb{Q}_p}$ sending the geometric Frobenius to $\alpha_i \in \mathbb{F}^\times$. Let $\pi(\tilde{\rho})$ be the smooth admissible representation associated to $\pi(\tilde{\rho})$ a la p-adic local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_p)$. Paškūnas [Pa13] constructed an injective envelope $\mathbf{J}_{\tilde{\rho}}$ of $\pi(\tilde{\rho})$ in the category of locally admissible torsion $\mathcal{O}$-representations of $\mathrm{GL}_2(\mathbb{Q}_p)/p^{\mathbb{Z}}$. Its Pontryagin dual $\mathbf{P}_{\tilde{\rho}}$ can be viewed as a projective module over $\mathcal{O}[\![x]\!][\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$, where x is a certain element in the universal deformation ring of $\tilde{\rho}$ (see [Pa15, Theorem 5.2]). Then any evaluation of x gives a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar{\rho}$.

This example is extremely important as it allows us to relate the local ghost conjecture with questions on trianguline deformation space of Breuil–Hellmann-Schraen [BHS17], using the known Breuil–Mézard conjecture for $\mathrm{GL}_2(\mathbb{Q}_p)$.

One can immediately reduce the discussion to the case when $b = 0$, which we assume from now on.

Definition 1.4. Given a "relevant" character $\varepsilon : (\mathbb{F}_p^\times)^2 \to \mathcal{O}^\times$ (which is used to parametrize weight disks; see Notation 2.24 (1)), define the space of "abstract p-adic forms"

$$\mathrm{S}^{(\varepsilon)}_{p\text{-adic}} := \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}(\widetilde{\mathrm{H}},\ \mathcal{C}^0(\mathbb{Z}_p, \mathcal{O}[\![w]\!])^{(\varepsilon)}) \quad \text{for } \varepsilon \text{ a character of } \Delta^2,$$

where $\mathcal{C}^0(\mathbb{Z}_p; \mathcal{O}[\![w]\!])^{(\varepsilon)}$ is the space of continuous functions on $\mathbb{Z}_p$, and the space is equipped with an action by Iw_p twisted by ε in some way (see § 2.4). This space carries an $\mathcal{O}[\![w]\!]$-linear action of the U_p-operator, whose characteristic power series is denoted by $C^{(\varepsilon)}_{\widetilde{\mathrm{H}}}(w,t) \in \mathcal{O}[\![w,t]\!]$ (see § 2.10 for the precise definitions).

We can define *abstract classical forms* and *abstract overconvergent forms* similarly; see § 2.13 and 2.17.

The first goal of this article is to formulate the following.

Conjecture 1.5 (Local ghost Conjecture 2.29)**.** *Let $\widetilde{\mathrm{H}}$ be a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar\rho$ and let $\varepsilon : (\mathbb{F}_p^\times)^2 \to \mathcal{O}^\times$ be a relevant character. Then there exists a combinatorially defined* ghost series

$$G^{(\varepsilon)}_{\bar\rho}(w,t) := \sum_{n\geq 0} g^{(\varepsilon)}_n(w)t^n \in \mathbb{Z}_p[w][\![t]\!] \quad \textit{with} \quad g^{(\varepsilon)}_n(w) = \prod_{\substack{k\geq 2\\ k\equiv k_\varepsilon \bmod p-1}} (w-w_k)^{m^{(\varepsilon)}_n(k)} \in \mathbb{Z}_p[w],$$

where $w_k := \exp((k-2)p) - 1$, $k_\varepsilon \in \{2, \dots, p\}$ is an integer defined in Notation 2.24 and the exponents $m^{(\varepsilon)}_n(k)$ are defined combinatorially in Definition 2.25, which only depends on $\bar\rho$, ε, (and k and n), such that for any $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$, the Newton polygon of $C^{(\varepsilon)}_{\widetilde{\mathrm{H}}}(w_\star, -)$ is the same as $G^{(\varepsilon)}_{\bar\rho}(w_\star, -)$.

We advertise that we shall prove this conjecture under some mild hypotheses in a sequel to this paper.

Remark 1.6. (1) The definition of the exponents $m^{(\varepsilon)}_n(k)$ follows exactly the recipe laid out by [BP19a], by harvesting input from the newform/old form theory. We refer to their paper as well as Remark 2.27 for a more intuitive explanation.

(2) When $\widetilde{\mathrm{H}}$ comes from the case of modular forms as discussed above (and assuming certain mod-p-multiplicity-one condition), the local ghost conjecture is precisely the $\bar r$-ghost conjecture raised by Bergdall–Pollack [BP19a, BP19b].

(3) The condition that $\bar\rho$ is generic and reducible is essential to Conjecture 1.5. If $\bar r|_{\mathrm{Gal}_{\mathbb{Q}_p}}$ is irreducible, the situation is significantly different. We hope to address that in a future work. The genericity is also important as indicated in [BP21$^+$]. These conditions on $\bar\rho$ first appeared in the literature in the form of so-called *Buzzard-regular condition* [Bu05]; we refer to [BP21$^+$, § 2] for a detailed discussion of Buzzard-regular condition in terms of $\bar r|_{\mathrm{Gal}_{\mathbb{Q}_p}}$. See Remark 2.32 for more discussion.

(4) The primitive hypothesis is not a serious constraint: for each $\bar{\rho}$, one can always globalize it to a case in (1) above by a construction in [GK14, Appendix A]; using the known Breuil–Mézard conjecture, one can "remember the slope information" on the Galois deformation side, and then bootstrap from it in a general situation. The same argument may be able to treat the case with reducible generic split case. We shall give more details on this argument in a subsequent paper.

1.7. **Main results of this paper.** The focus of this paper is to investigate the following basic properties of the ghost series.

(1) We give an explicit and asymptotic computation of $m_n^{(\varepsilon)}(k)$. (Propositions 4.1 and 4.7), making the ghost series explicit.
(2) The local ghost conjecture would require that the slopes of the Newton polygon of $G_{\bar{\rho}}^{(\varepsilon)}(w_k, -)$ for $k \in \mathbb{Z}$ to include slopes of abstract classical forms. We check its compatibility with theta maps, the Atkin–Lehner involutions, and the p-stabilizations (Proposition 4.18 (1)–(3)).
(3) The *ghost duality*, which is the most important property of ghost series, near a weight point where the space of abstract p-new forms is nonzero. We refer to Proposition 4.18 (4) for the statement.
(4) For $k \in \mathbb{Z}_{\geq 2}$, the "old form slopes" of $\mathrm{NP}(G(w_k, -))$ are less than or equal to $\lfloor \frac{k-1}{p+1} \rfloor$, as predicted by a conjecture of Gouvêa [Go01].
(5) For $n \in \mathbb{Z}_{\geq 0}$ and $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$, the point $(n, v_p(g_n(w_\star)))$ is a vertex on the Newton Polygon $\mathrm{NP}(G(w_\star, -))$ if and only if $(n, w_\star)$ is not near-Steinberg, a condition that roughly says that $w_\star$ is p-adically close to a weight point w_k at which the nth slope is the slope of an abstract p-new form. (Theorem 5.19).
(6) For $k \in \mathbb{Z}$, the slopes of $\mathrm{NP}(G(w_k, -))$ are all integers if a is even and belong to $\frac{1}{2}\mathbb{Z}$ if a is odd (Corollary 5.24).

The proofs of (1)–(4) are relatively straightforward, as soon as we setup the theory for abstract classical and overconvergent forms properly. The proofs of (5) and (6) are rather combinatorially involved; they are the main content of this paper.

1.8. **Relation with historical conjectures and results on slopes of modular forms.** The general study of slopes of (overconvergent) modular forms began with extensive computer enumeration by Gouvêa and Mazur [GM92, Go01] in 1990's. Let N be a positive integer that is relatively prime to p. The space $S_k(\Gamma_0(Np))$ of modular forms of weight $k \in \mathbb{N}_{\geq 2}$ contains two copies of $S_k(\Gamma_0(N))$ as the space of p-old forms $S_k(\Gamma_0(Np))^{p\text{-old}}$, and the Hecke complement is the space of p-new forms $S_k(\Gamma_0(Np))^{\text{new}}$. The eigenvalues of U_p-action on $S_k(\Gamma_0(Np))^{\text{new}}$ are $\pm p^{(k-2)/2}$, so U_p has slope $(k-2)/2$ on this space. On the other hand, Atkin–Lehner theory implies that the U_p-slopes on $S_k(\Gamma_0(Np))^{p\text{-old}}$ can be paired so that the sum of each pair is $k-1$. Based on significant computation data, Gouvêa raised the following conjecture in [Go01].

Conjecture 1.9 (Gouvêa)**.** *Let μ_k denote the density measure on $[0,1]$ given by the set of U_p-slopes on $S_k(\Gamma_0(Np))$, which are normalized by being divided by $k-1$ and are counted with multiplicities. As $k \to \infty$, μ_k converges to the sum of the delta measure of mass $\frac{p-1}{p+1}$ at $\frac{1}{2}$, and the uniform measure on $[0, \frac{1}{p+1}] \cup [\frac{p}{p+1}, 1]$ with total mass $\frac{2}{p+1}$.*

The delta measure at $\frac{1}{2}$ clearly comes from the p-new forms, the p-old form part (supposedly responsible for the uniform measure) does not seem to have any theoretical explanation. In sharp contrast, numerically, Gouvêa observes that the lesser of the p-old form slopes (in the pairs mentioned above) are almost always less than or equal to $\lfloor\frac{k-1}{p+1}\rfloor$; this is known as Gouvêa's $\lfloor\frac{k-1}{p+1}\rfloor$-conjecture.

From modern point of view, it is best to study the two conjectures above after first localizing at a residual Galois representation $\bar{r}$. Then as proved in [BP19b], the local ghost Conjecture 1.5 implies the $\bar{r}$-part of Conjecture 1.9 (at least under the mod-p-multiplicity-one assumption). On the other hand, § 1.7(6) implies the $\bar{r}$-part of Gouvêa's $\lfloor\frac{k-1}{p+1}\rfloor$-conjecture (at least under the mod-p-multiplicity-one assumption).

Remark 1.10.
(1) There has been a couple of interesting results on Gouvêa's $\lfloor\frac{k-1}{p+1}\rfloor$-conjecture, obtained by Berger–Li–Zhu [BLZ04] and Bergdall–Levin [BL19+], respectively. They used (φ,Γ)-modules and Kisin modules techniques to show that if the slopes of the crystalline Frobenius is greater than or equal to $\lfloor\frac{k-1}{p-1}\rfloor$ and $\lfloor\frac{k-1}{p}\rfloor$, respectively, then the reduction of such representation is irreducible.
(2) We also mention a conjecture by Gouvêa–Mazur [GM92, Conjecture 1] on radii of constancy of slopes on spectral curves. We do not know whether our local ghost conjecture implies the $\bar{r}$-part of Gouvêa–Mazur's conjecture.

1.11. **Implication on Kisin's crystalline Galois deformation space.** As indicated by Remark 1.3(2) that applying local ghost Conjecture 1.5 to the patched completed homology, we may deduce results on Kisin's crystalline deformation space of $\tilde{\rho}$. One interesting conjecture in this direction was originated in emails among Breuil, Buzzard, and Emerton in 2005 ([BeGe16]).

Conjecture 1.12 (Breuil–Buzzard–Emerton)**.** *Assume that $\tilde{\rho}:\mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)\to\mathrm{GL}_2(\mathbb{F})$ is reducible, nonsplit, and generic, then the p-adic valuations of the eigenvalues of the crystalline Frobenius action on the Kisin's framed crystalline deformation space of $\tilde{\rho}$ are integers (when k is even) or half-integers (when k is odd).*

Our property § 1.7 is related to this conjecture, and suggests that local ghost conjecture should imply the conjecture above.

Remark 1.13. It seems to be very difficult to directly compute the reduction of a crystalline representations when the (difference of) Hodge–Tate weights become larger than $2p$ (see for example, [Br03]); retroactively speaking, this is expected as the ghost series is itself rather combinatorially involved.

When the slopes are less than equal to 3, there have been several works on the computation of reduction of crystalline representations using mod p local Langlands correspondence, see [Br03, BuGe13, BhGh15, GG15, BGR18].

Our approach may be viewed as coming from the automorphic side of the p-adic local Langlands correspondence, which harnesses the advantage of working with a family of abstract forms parametrized by the weight space. Working purely with the setup of $\mathcal{O}[\![K_p]\!]$-projective augmented modules allows us to relate the slope question to Kisin's crystalline deformation space via the known Breuil–Mézard conjecture for $\mathrm{GL}_2(\mathbb{Q}_p)$.

1.14. **Relation to a refined spectral halo conjecture.** Note that if one evaluates the ghost series at $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$ with $v_p(w_\star) \in (0,1)$, we quickly deduce that

$$v_p(g_n^{(\varepsilon)}(w_\star)) = v_p(w_\star) \cdot \deg g_n^{(\varepsilon)}.$$

In Proposition 4.11 below, we compute explicitly $\deg g_{n+1}^{(\varepsilon)} - \deg g_n^{(\varepsilon)}$ and show that, under our assumption $1 \leq a \leq p-4$, the degree differences is strictly increasing in n. Thus, the local ghost conjecture can imply that the spectral curve associated to $\widetilde{\mathrm{H}}$, when restricted to the boundary of the weight disc: $\mathcal{W}^{(0,1)} := \big\{ w_\star \in \mathfrak{m}_{\mathbb{C}_p} \mid v_p(w_\star) \in (0,1) \big\}$ is an infinite disjoint union of copies of $\mathcal{W}^{(0,1)}$ and the slopes on each such copy is proportional to $v_p(w_\star)$. We also compute the predicted slope ratios in Proposition 4.11. This may be viewed as a refined version of the spectral halo conjecture proved in [LWX14] (but only for the $\bar{r}$-part of the spectral curve) for which $\bar{r}$ satisfies the reducible, nonsplit, and generic conditions.

Roadmap of the paper. Section 2 is devoted to formulating the local ghost conjecture (Conjecture 2.29). In the next section, we make standard constructions of abstract classical forms and abstract p-adic forms from an $\mathcal{O}[\![K_p]\!]$-projective augmented module. In Section 4, we prove basic properties of the ghost series, including § 1.7(1)–(3), most notably the *ghost duality*. Section 5 is the most technical part of this paper, in which we prove many important results regarding vertices of Newton polygon of ghost series, namely § 1.7(4)–(6). In the appendix, we recollect some basic facts on modular representations of $\mathrm{GL}_2(\mathbb{F}_p)$ that we use throughout the paper.

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Notation.

(1) For a field k, we write $\overline{k}$ for its algebraic closure.
(2) Throughout the paper, we fix a prime number $p \geq 5$ unless otherwise specified. Although some statements in this paper may also hold for smaller primes, we insist this assumption to avoid unnecessarily complicated argument. Let $\mathrm{I}_{\mathbb{Q}_p} \subset \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p)$ denote the inertia subgroup, and $\omega_1 : \mathrm{I}_{\mathbb{Q}_p} \twoheadrightarrow \mathrm{Gal}(\mathbb{Q}_p(\mu_p)/\mathbb{Q}_p) \cong \mathbb{F}_p^\times$ the *the 1st fundamental character*.
(3) Let $\Delta \cong (\mathbb{Z}/p\mathbb{Z})^\times$ be the torsion subgroup of $\mathbb{Z}_p^\times$, and let $\omega : \Delta \to \mathbb{Z}_p^\times$ be the Teichmüller character. For an element $\alpha \in \mathbb{Z}_p^\times$, we often use $\bar{\alpha} \in \Delta$ to denote

its reduction modulo p. In particular, the Greek letters $\alpha, \beta, \gamma, \delta$ are reserved for elements in $\mathbb{Z}_p$, and $\bar{\alpha}, \bar{\beta}, \bar{\gamma}, \bar{\delta}$ denote elements in $\mathbb{F}_p$.

(4) Let E be a finite extension of $\mathbb{Q}_p(\sqrt{p})$, which serves as the coefficient field so that E contains $p^{\frac{k-2}{2}}$ for every integer $k \geq 2$. Let $\mathcal{O}$, $\mathbb{F}$, and ϖ denote its ring of integers, residue field, and a uniformizer, respectively.

(5) The p-adic valuation $v_p(-)$ is normalized so that $v_p(p) = 1$.

(6) We use $\lceil x \rceil$ to denote the ceiling function and $\lfloor x \rfloor$ to denote the floor function.

(7) For M a topological $\mathcal{O}$-module, we write $\mathcal{C}^0(\mathbb{Z}_p; M)$ for the space of continuous functions on $\mathbb{Z}_p$ with values in M. It is endowed with the compact-open topology.

    All maps and characters are continuous with respect to the relevant topology. All hom spaces in this paper refer to the spaces of continuous homomorphisms.

(8) We shall encounter both p-adic logs $\log(x) = (x-1) - \frac{(x-1)^2}{2} + \cdots$ for x a p-adic or formal element, and the natural logs $\ln(-)$ in the real analysis. (The usual p-based logarithm will always be denoted by $\ln(-)/\ln(p)$.)

(9) For each $m \in \mathbb{Z}$, we write $\{m\}$ for the unique integer satisfying the conditions

$$0 \leq \{m\} \leq p-2 \quad \text{and} \quad m \equiv \{m\} \bmod p-1.$$

    For $m \in \mathbb{Z}_{\geq 0}$, let $\mathrm{Dig}(m)$ denotes the sum of all digits in the p-based expression of m.

(10) For a square (possibly infinite) matrix M with coefficients in a commutative ring R, its *characteristic power series* (if it exists) is denoted by $\mathrm{Char}(M; t) := \det(I - Mt) \in R[\![t]\!]$, where I is the identity matrix. When an R-linear action U on an R-module N is given by such a matrix M, we write $\mathrm{Char}(U, N; t)$ for $\mathrm{Char}(M; t)$.

(11) For a power series $F(t) = \sum_{n \geq 0} c_n t^n \in \mathbb{C}_p[\![t]\!]$ with $c_0 = 1$, we use $\mathrm{NP}(F)$ to denote its *Newton polygon*, i.e. the convex hull of points $(n, v_p(c_n))$ for all n; the slopes the segments of $\mathrm{NP}(F)$ are often referred to as *slopes* of $F(t)$.

2. Local ghost conjecture: formulation

In this section, we give a detailed formulation of Conjecture 1.5, which is a local analogue of the ghost conjecture of Bergdall and Pollack [BP19a, BP19b]. The novelty here is to work with a representation of $\mathrm{GL}_2(\mathbb{Q}_p)$, without referencing to a global automorphic setup. Such idea was inspired by a discussion with Matthew Emerton; we thank him heartily.

To be more precise, we start with a projective module $\widetilde{\mathrm{H}}$ over the Iwasawa algebra of $\mathrm{GL}_2(\mathbb{Z}_p)$ whose group action extends to an action of $\mathrm{GL}_2(\mathbb{Q}_p)$; we often encounter such $\widetilde{\mathrm{H}}$ as the completed homology in the automorphic setup. The information of $\bar{\rho}|_{G_{\mathbb{Q}_p}}$ is "manually" inserted by requiring $\widetilde{\mathrm{H}}$ to be the projective envelope as a $\mathcal{O}[\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$-module of the corresponding Serre weight. In this purely local setup, we can still properly define "abstract classical forms", "abstract overconvergent forms", and "abstract p-adic forms" associated to $\widetilde{\mathrm{H}}$. This allows us to define the corresponding characteristic power series of the U_p-operator and the ghost series. We state a more concrete statement of the local ghost conjecture (Conjecture 2.29) at the end of the section.

2.1. **Recollection of Iwasawa algebras.** Let G be topological group that contains a finite-index *analytic* pro-p-Sylow subgroup H (in the sense of Lazard [La65, § III.3]). We write

$$\mathcal{O}[\![G]\!] := \varprojlim_{N \trianglelefteq G} \mathcal{O}[G/N]$$

for the corresponding (possibly non-commutative) Iwasawa algebra. It is left and right noetherian by [La65, Proposition V.2.2.4]. We write I_G for its augmentation ideal. For an element $g \in G$, we write $[g]$ for the group element in $\mathcal{O}[\![G]\!]$.

In this paper, we shall consider right $\mathcal{O}[\![G]\!]$-modules as opposed to left ones because the completed homology groups are naturally right modules over the Iwasawa algebra of $\mathrm{GL}_2(\mathbb{Z}_p)$.

Lemma 2.2. *Keep the notation from above. The following are equivalent for a finitely generated* right *$\mathcal{O}[\![G]\!]$-module M:*

1. *M is a finite projective right $\mathcal{O}[\![G]\!]$-module,*
2. *M is a finite projective right $\mathcal{O}[\![H]\!]$-module,*
3. *M is a finite free right $\mathcal{O}[\![H]\!]$-module, and*
4. $\mathrm{Ext}^1_{\mathcal{O}[\![H]\!]}(M, \mathbb{F}) = 0$.

Proof. Indeed, (1) obviously implies (2). Conversely, for any (right) $\mathcal{O}[\![G]\!]$-module N, the composition

$$\mathrm{Ext}^i_{\mathcal{O}[\![G]\!]}(M, N) \xrightarrow{\text{Restriction}} \mathrm{Ext}^i_{\mathcal{O}[\![H]\!]}(M, N) \xrightarrow{\text{Trace}} \mathrm{Ext}^i_{\mathcal{O}[\![G]\!]}(M, N)$$

is the multiplication by the index $[G : H]$ ([Serre97, Chapter I, § 2.4]), which is relatively prime to p. So the vanishing of $\mathrm{Ext}^{>0}_{\mathcal{O}[\![H]\!]}(M, N)$ implies that of $\mathrm{Ext}^{>0}_{\mathcal{O}[\![G]\!]}(M, N)$.

(3) obviously implies (2). The converse implication holds because $\mathcal{O}[\![H]\!]$ is a noetherian local ring (whose maximal ideal is generated by ϖ and the augmentation ideal).

(3) obviously implies (4). Conversely, since $\mathrm{Ext}^i_{\mathcal{O}[\![H]\!]}(M, -)$ commutes with taking inverse limit on finite objects, it reduces to showing that $\mathrm{Ext}^1_{\mathcal{O}[\![H]\!]}(M, N) = 0$ for any finite length $\mathcal{O}[\![H]\!]$-modules N. But all such N are successive extensions of $\mathbb{F}$ as $\mathcal{O}[\![H]\!]$ is a local ring. So (4) implies (3). $\square$

Definition 2.3. We will mostly work with the following subgroups of $\mathrm{GL}_2(\mathbb{Q}_p)$:

$$\mathrm{K}_p := \mathrm{GL}_2(\mathbb{Z}_p) \supset \mathrm{Iw}_p := \begin{pmatrix} \mathbb{Z}_p^\times & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{pmatrix} \supset \mathrm{Iw}_{p,1} := \begin{pmatrix} 1 + p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & 1 + p\mathbb{Z}_p \end{pmatrix}.$$

By an *$\mathcal{O}[\![K_p]\!]$-projective augmented module*, we mean a finite projective *right* $\mathcal{O}[\![\mathrm{K}_p]\!]$-module $\widetilde{\mathrm{H}}$ together with an action of $\mathrm{GL}_2(\mathbb{Q}_p)$ extending the K_p-action. Such a module is endowed with the canonical topology as a finitely generated $\mathcal{O}[\![\mathrm{K}_p]\!]$-module defined in [Em10, Definition 2.1.4]. (The notation is an adaptation of the notion of pro-augmented representations in [Em10, Definition 2.1.5].) We fix one for this and the next four sections.

By [Br-notes, Corollary 3.4], extending the K_p-action on $\widetilde{\mathrm{H}}$ to a $\mathrm{GL}_2(\mathbb{Q}_p)$-action is equivalent to giving a (right) action of $\Pi = \left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)$ on $\widetilde{\mathrm{H}}$ which satisfies the relation

$$m\Pi\left(\begin{smallmatrix} \alpha & \beta \\ p\gamma & \delta \end{smallmatrix}\right) = m\left(\begin{smallmatrix} \delta & \gamma \\ p\beta & \alpha \end{smallmatrix}\right)\Pi, \quad \text{for all } m \in \widetilde{\mathrm{H}} \text{ and } \left(\begin{smallmatrix} \alpha & \beta \\ p\gamma & \delta \end{smallmatrix}\right) \in \mathrm{Iw}_p.$$

2.4. **Abstract p-adic forms.** We follow the setup of [LWX14, § 2.3] with small variation. In particular, we consider *right* inductions from the lower triangular matrices (as opposed to the upper triangular matrices) but with the natural action (as opposed to the twisted action used in [LWX14, § 2.3]). Luckily, we end up with the same formula.

Set $\Delta := \mathbb{F}_p^\times$ and let $\omega : \Delta \to \mathbb{Z}_p^\times$ be the Teichmüller character. Write $\log(-)$ and $\exp(-)$ for the formal p-adic logarithmic function and formal p-adic exponential function, respectively. For an element $\alpha \in \mathbb{Z}_p$, write $\bar{\alpha} \in \mathbb{F}_p$ for its reduction modulo p.

Fix a character $\varepsilon : \Delta^2 \to \mathcal{O}^\times$ and put

$$\mathcal{O}[\![w]\!]^{(\varepsilon)} := \mathcal{O}[\![\Delta \times \mathbb{Z}_p^\times]\!] \otimes_{\mathcal{O}[\Delta^2],\varepsilon} \mathcal{O}.$$

It is isomorphic to the ring of formal power series ring $\mathcal{O}[\![w]\!]$ with w corresponding to $[(1, \exp(p))] - 1$. (In [LWX14], we used the letter T for w. In this paper, we follow the convention of [BP19a, BP19b].)

Consider the universal character

$$\begin{aligned} \chi_{\mathrm{univ}}^{(\varepsilon)} : B^{\mathrm{op}}(\mathbb{Z}_p) = \begin{pmatrix} \mathbb{Z}_p^\times & 0 \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{pmatrix} &\longrightarrow \big(\mathcal{O}[\![w]\!]^{(\varepsilon)}\big)^\times \\ \begin{pmatrix} \alpha & 0 \\ \gamma & \delta \end{pmatrix} &\longmapsto [(\bar{\alpha}, \delta)] \otimes 1 = \varepsilon(\bar{\alpha}, \bar{\delta}) \cdot (1+w)^{\log(\delta/\omega(\bar{\delta}))/p}, \end{aligned}$$

where the superscript in $\mathcal{O}[\![\omega]\!]^{(\varepsilon)}$ indicates the twisted action. The induced representation (for the *right* action convention)

$$\begin{aligned} \mathrm{Ind}_{B^{\mathrm{op}}(\mathbb{Z}_p)}^{\mathrm{Iw}_p}(\chi_{\mathrm{univ}}^{(\varepsilon)}) := \big\{ &\text{continuous functions } f : \mathrm{Iw}_p \to \mathcal{O}[\![w]\!]^{(\varepsilon)}; \\ &f(gb) = \chi_{\mathrm{univ}}^{(\varepsilon)}(b) \cdot f(g) \text{ for } b \in B^{\mathrm{op}}(\mathbb{Z}_p) \text{ and } g \in \mathrm{Iw}_p \big\} \end{aligned}$$

carries a *right* action of Iw_p given by $f|_{\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)}(g) := f\big(\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) g\big)$ for $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathrm{Iw}_p$. The Iwasawa decomposition allows us to make the following identification:

$$\begin{aligned} \mathrm{Ind}_{B^{\mathrm{op}}(\mathbb{Z}_p)}^{\mathrm{Iw}_p}(\chi_{\mathrm{univ}}^{(\varepsilon)}) &\xrightarrow{\cong} \mathcal{C}^0(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}) \\ f &\longmapsto h(z) := f\big(\left(\begin{smallmatrix} 1 & z \\ 0 & 1 \end{smallmatrix}\right)\big). \end{aligned}$$

The natural Iw_p-action on the left hand side then translates to the following action on the right hand side

$$\begin{aligned} (2.4.1)\quad h|_{\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)}(z) &= \big[(\bar{\alpha}, \gamma z + \delta)\big] \cdot h\Big(\frac{\alpha z + \beta}{\gamma z + \delta}\Big) \\ &= \varepsilon(\bar{\alpha}, \bar{\delta}) \cdot (1+w)^{\log\left((\gamma z+\delta)/\omega(\bar{\delta})\right)/p} \cdot h\Big(\frac{\alpha z + \beta}{\gamma z + \delta}\Big) \quad \text{for } \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathrm{Iw}_p. \end{aligned}$$

This action extends to the action of a bigger monoid (where the notation comes from [Bu04, § 4])

$$\mathbf{M}_1 = \big\{ \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathrm{M}_2(\mathbb{Z}_p);\ p|\gamma,\ p \nmid \delta,\ \alpha\delta - \beta\gamma \neq 0 \big\}$$

through the following formula analogous to (2.4.1): for $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathbf{M}_1$ with determinant $\alpha\delta - \beta\gamma = p^r d$ ($d \in \mathbb{Z}_p^\times$),

$$(2.4.2)\qquad h|_{\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)}(z) = \varepsilon(\bar{d}/\bar{\delta}, \bar{\delta}) \cdot (1+w)^{\log\left((\gamma z+\delta)/\omega(\bar{\delta})\right)/p} \cdot h\Big(\frac{\alpha z + \beta}{\gamma z + \delta}\Big).$$

For the fixed $\mathcal{O}[\![K_p]\!]$-projective augmented module $\widetilde{\mathrm{H}}$ and a character ε as above, we define the space of *abstract p-adic forms* to be

$$\mathrm{S}^{(\varepsilon)}_{p\text{-adic}} = \mathrm{S}^{(\varepsilon)}_{\widetilde{\mathrm{H}},p\text{-adic}} := \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\big(\widetilde{\mathrm{H}},\, \mathrm{Ind}^{\mathrm{Iw}_p}_{B^{\mathrm{op}}(\mathbb{Z}_p)}(\chi^{(\varepsilon)}_{\mathrm{univ}})\big) \cong \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\big(\widetilde{\mathrm{H}},\, \mathcal{C}^0(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)})\big).$$

Remark 2.5. (1) Recall that the hom spaces in this paper refer to the space of continuous homomorphisms. We recall the topologies of the source and target in the definition of $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ for the audience's convenience. The topology on $\widetilde{\mathrm{H}}$ is the canonical topology as a finitely generated $\mathcal{O}[\![K_p]\!]$-module and the topology on $\mathcal{C}^0(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)})$ is the open-compact topology. In particular, if we have a subset S such that the $\mathcal{O}[K_p]$-submodule of $\widetilde{\mathrm{H}}$ generated by S is dense in $\widetilde{\mathrm{H}}$, then any element in $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ is uniquely determined by its values on S.

(2) When $\widetilde{\mathrm{H}} = \varprojlim_n \mathcal{O}\big[D^\times\backslash D(\mathbb{A}_f)^\times/K^p(1+p^n\mathrm{M}_2(\mathbb{Z}_p))\big]$ for a definite quaternion algebra D over $\mathbb{Q}$ split at p and an open compact neat subgroup $K^p \subset (D\otimes\mathbb{A}_f^{(p)})^\times$ (we refer to [LWX14, §2.4] for the precise definition), $\widetilde{\mathrm{H}}$ is an $\mathcal{O}[\![K_p]\!]$-projective augmented module and (the direct sum over characters ε of) $\mathrm{S}^{(\varepsilon)}_{\widetilde{\mathrm{H}},p\text{-adic}}$ is the same as the space of p-adic automorphic forms S^D_{int} of [LWX14, § 2.7], after properly adjusted for normalizations.

The space $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ is a countable infinite completed direct sum of copies of $\mathcal{O}[\![w]\!]$, carrying an $\mathcal{O}[\![w]\!]$-linear U_p-action: fixing a decomposition of the double coset $\mathrm{Iw}_p\big(\begin{smallmatrix} p^{-1} & 0 \\ 0 & 1\end{smallmatrix}\big)\mathrm{Iw}_p = \coprod_{j=0}^{p-1} v_j\mathrm{Iw}_p$ (e.g. $v_j = \big(\begin{smallmatrix} p^{-1} & 0 \\ j & 1\end{smallmatrix}\big)$ with $v_j^{-1} = \big(\begin{smallmatrix} p & 0 \\ -jp & 1\end{smallmatrix}\big) \in \mathbf{M}_1$), the U_p-operator sends $\varphi \in \mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ to

$$U_p(\varphi)(x) = \sum_{j=0}^{p-1} \varphi(xv_j)|_{v_j^{-1}} \quad \text{for all } x \in \widetilde{\mathrm{H}}. \tag{2.5.1}$$

The U_p-operator does not depend on the choice of coset representatives.

Remark 2.6. The next logical step is show that the characteristic power series of the U_p-action on $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ has coefficients in $\mathcal{O}[\![w]\!]$. Unfortunately, such well-believed result is not documented in the literature under our abstract setup. The situation is further complicated by two subtleties: the U_p-action is *not* compact on $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ (see [LWX14, Example 5.2 and Remark 5.3]), and certain non-neat phenomenon appears because we only assumed $\widetilde{\mathrm{H}}$ to be finite projective (as opposed to finite free) over $\mathcal{O}[\![\mathrm{K}_p]\!]$. The argument we provide below in § 2.8–2.10 is tailored to the need of defining the characteristic power series. We shall revisit finer estimates of the U_p-actions analogous to [LWX14] in the next paper.

Convention 2.7. In what follows, we will frequently study the action of the U_p-operator on various $\mathcal{O}$-modules or vector spaces. Following the convention in this area, we say *finite U_p-slopes* to mean the p-adic valuations of the nonzero eigenvalues of the U_p-action on the corresponding space, usually counted with multiplicity.

2.8. **Explicit description of** $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$**.** Unlike in [LWX14], our $\widetilde{\mathrm{H}}$ may not be free over $\mathcal{O}[\![\mathrm{Iw}_p]\!]$. We now discuss this subtle point.

We may view $\bar{\mathrm{T}} = \big(\begin{smallmatrix} \Delta & 0 \\ 0 & \Delta\end{smallmatrix}\big) \cong \Delta^2$ as a subgroup of Iw_p via Teichmüller lifts. The $\mathcal{O}[\![K_p]\!]$-projective augmented module $\widetilde{\mathrm{H}}$ is a finite projective right $\mathcal{O}[\![\mathrm{K}_p]\!]$-module, and hence a finite projective (right) $\mathcal{O}[\![\mathrm{Iw}_p]\!]$-module. So there exists a (non-canonical) isomorphism of right $\mathcal{O}[\![\mathrm{Iw}_p]\!]$-modules:

$$\widetilde{\mathrm{H}} \simeq \bigoplus_{i=1}^{r} e_i \mathcal{O} \otimes_{\chi_i, \mathcal{O}[\bar{\mathrm{T}}]} \mathcal{O}[\![\mathrm{Iw}_p]\!], \tag{2.8.1}$$

where $\chi_i : \bar{\mathrm{T}} \to \mathcal{O}^\times$ is a character, and r is equal to the rank of $\widetilde{\mathrm{H}}$ as a free $\mathcal{O}[\![\mathrm{Iw}_{p,1}]\!]$-module. Using the evaluations at the elements e_i, we have identifications

$$\mathrm{S}^{(\varepsilon)}_{p\text{-adic}} \simeq \bigoplus_{i=1}^{r} e_i^* \cdot \mathcal{C}^0\big(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}\big)^{\bar{\mathrm{T}}=\chi_i}. \tag{2.8.2}$$

Here for a function $h(z) \in \mathcal{C}^0\big(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}\big)^{\bar{\mathrm{T}}=\chi_i}$, the notation $e_i^* \cdot h(z)$ denotes the (unique) element in $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ that sends e_i to $h(z)$, and e_j to 0 for all $j \neq i$.

In explicit terms, $\mathcal{C}^0\big(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}\big)^{\bar{\mathrm{T}}=\chi_i}$ consists of continuous functions $h(z)$ on $\mathbb{Z}_p$ with values in $\mathcal{O}[\![w]\!]^{(\varepsilon)}$ such that

$$\text{for any } \bar{\alpha}, \bar{\delta} \in \Delta, \quad \text{we have} \quad \varepsilon(\bar{\alpha}, \bar{\delta}) \cdot h(\bar{\alpha}z/\bar{\delta}) = \chi_i(\bar{\alpha}, \bar{\delta}) \cdot h(z).$$

So $\mathcal{C}^0\big(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}\big)^{\bar{\mathrm{T}}=\chi_i}$ is non-zero precisely when $\varepsilon\chi_i^{-1}(\bar{x}, \bar{x}) = 1$ for all $\bar{x} \in \Delta$, and in this case, it is the subspace of continuous functions on $\mathbb{Z}_p$ such that

$$h(\bar{\alpha}z) = \chi_i\varepsilon^{-1}(\bar{\alpha}, 1)h(z), \quad \text{for all } \bar{\alpha} \in \Delta. \tag{2.8.3}$$

This can be viewed as an eigenspace for the Δ-action on $\mathcal{C}^0\big(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}\big)$.

The reason that we see this eigenspace (as opposed to the whole space) is analogous to the situation of non-neat level structure.

2.9. **Explicit description of the U_p-action.** Using the explicit description (2.8.2) of $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$, we rewrite the action of U_p on $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ as follows. For each e_i and v_j, we write

$$e_i v_j = \sum_{k=1}^{r} e_k c_{ijk} \quad \text{for} \quad c_{ijk} \in \mathcal{O} \otimes_{\chi_k, \mathcal{O}[\bar{\mathrm{T}}]} \mathcal{O}[\![\mathrm{Iw}_p]\!].$$

Then via the isomorphism (2.8.2), the U_p-action on $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ is the same as the $\mathcal{O}[\![w]\!]$-linear endomorphism

$$\bigoplus_{i=1}^{r} e_i^* \cdot \mathcal{C}^0\big(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}\big)^{\bar{\mathrm{T}}=\chi_i} \longrightarrow \bigoplus_{i=1}^{r} e_i^* \cdot \mathcal{C}^0\big(\mathbb{Z}_p; \mathcal{O}[\![w]\!]^{(\varepsilon)}\big)^{\bar{\mathrm{T}}=\chi_i}$$

represented by an $r \times r$ matrix, whose (i,k)-entry is given by

$$h(z) \longmapsto \sum_{j=0}^{p-1} h|_{c_{ijk}v_j^{-1}}(z). \tag{2.9.1}$$

Each expression in (2.9.1) is a uniform limit of finite sums of actions given by elements from $\left(\begin{smallmatrix} p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{smallmatrix}\right)^{\det \in p\mathbb{Z}_p^\times}$, and hence converges to a well-defined $\mathcal{O}[\![w]\!]$-linear map.

2.10. **Characteristic power series of U_p.** To define the characteristic power series of the U_p-operator, we need to circumvent the issue that the U_p-action on $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ is not compact (see [LWX14, § 5] for further discussion). This is well know to experts; we include it here for completeness.

In this subsection, we write $\Lambda = \mathcal{O}[\![w]\!]$ to simplify notation. We put

$$\Lambda^{\geq 1/p} := \mathcal{O}[\![w]\!]\langle p/w\rangle, \quad \Lambda^{=1/p} := \mathcal{O}\langle w/p, p/w\rangle, \quad \text{and} \quad \Lambda^{\leq 1/p} := \mathcal{O}\langle w/p\rangle.$$

Then $\Lambda^{\geq 1/p}$ and $\Lambda^{\leq 1/p}$ are noetherian Banach–Tate rings (in the sense of [AIP, Annexe B]) with pseudo-uniformizers w and p, respectively. In particular, we note that the intersection of $\Lambda^{\geq 1/p}$ and $\Lambda^{\leq 1/p}$ in $\Lambda^{=1/p}$ is exactly Λ. We shall define an $\Lambda^?$-linear subspace $\mathrm{S}^{(\varepsilon),?}_{p\text{-adic}}$ of $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}} \otimes_\Lambda \Lambda^?$ where ? stands for $\geq 1/p$ or $\leq 1/p$, such that we have a natural equality

$$\mathrm{S}^{(\varepsilon),\geq 1/p}_{p\text{-adic}} \otimes_{\Lambda^{\geq 1/p}} \Lambda^{=1/p} = \mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}} \otimes_{\Lambda^{\leq 1/p}} \Lambda^{=1/p} \tag{2.10.1}$$

inside $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}} \otimes_\Lambda \Lambda^{=1/p}$.

Set $\Lambda^{(\varepsilon),?} := \mathcal{O}[\![w]\!]^{(\varepsilon)} \otimes_\Lambda \Lambda^?$. Recall that $\mathcal{C}^0(\mathbb{Z}_p;\mathcal{O})$ has a natural orthonormal Mahler basis $1, z, \binom{z}{2}, \dots$. We consider two subspaces:

$$\mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\geq 1/p}\big)^{\mathrm{mod}} := \widehat{\bigoplus}_{n\geq 0} w^{\lfloor n/p\rfloor}\binom{z}{n}\cdot \Lambda^{(\varepsilon),\geq 1/p} \subseteq \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\geq 1/p}\big), \quad \text{and} \tag{2.10.2}$$

$$\mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod}} := \widehat{\bigoplus}_{n\geq 0} p^{\lfloor n/p\rfloor}\binom{z}{n}\cdot \Lambda^{(\varepsilon),\leq 1/p} \subseteq \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big). \tag{2.10.3}$$

(Our choice of basis elements differs from [LWX14, § 5.4], where a basis $w^n\binom{z}{n}$ is used. This to circumvent a technical issue as indicated in the errata of the paper in § B.2.) Following the argument in [LWX14, § 5.4], we now show that both subspaces (2.10.2) and (2.10.3) are stable under the $\mathbf{M}_1$-action, and the action of $\left(\begin{smallmatrix} p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times\end{smallmatrix}\right)^{\det\neq 0}$ is compact on these two subspaces.

Indeed, by [LWX14, Proposition 3.14 (2)], if $P = (P_{m,n})_{m,n\geq 0}$ denotes the infinite matrix for the (right) action of $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta\end{smallmatrix}\right) \in \mathbf{M}_1$ on $\mathcal{C}^0(\mathbb{Z}_p;\Lambda^{(\varepsilon),\geq 1/p})$ (resp. $\mathcal{C}^0(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p})$) with respect to the basis $w^{\lfloor n/p\rfloor}\binom{z}{n}$ (resp. $p^{\lfloor n/p\rfloor}\binom{z}{n}$), i.e. we have $w^{\lfloor n/p\rfloor}\binom{z}{n}|_{\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta\end{smallmatrix}\right)} = \sum_{m\geq 0} P_{m,n}\cdot w^{\lfloor m/p\rfloor}\binom{z}{m}$ (resp. $p^{\lfloor n/p\rfloor}\binom{z}{n}|_{\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta\end{smallmatrix}\right)} = \sum_{m\geq 0} P_{m,n}\cdot p^{\lfloor m/p\rfloor}\binom{z}{m}$) then

$$P_{m,n} \in w^{\lfloor n/p\rfloor - \lfloor m/p\rfloor}(p,w)^{\max\{m-n,0\}} \subseteq w^{\lfloor n/p\rfloor - \lfloor m/p\rfloor + \max\{0,m-n\}}\Lambda^{\geq 1/p}$$

$$(\text{resp. } P_{m,n} \in p^{\lfloor n/p\rfloor - \lfloor m/p\rfloor}(p,w)^{\max\{m-n,0\}} \subseteq p^{\lfloor n/p\rfloor - \lfloor m/p\rfloor + \max\{0,m-n\}}\Lambda^{\leq 1/p} \quad).$$

For a fixed n, we let $m\to\infty$ and then the exponents on w (resp. p) in the above estimations tend to ∞; so the subspace $\mathcal{C}^0(\mathbb{Z}_p;\Lambda^{(\varepsilon),?})^{\mathrm{mod}}$ is stable under the Iw_p-action. Now, if $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta\end{smallmatrix}\right) \in \left(\begin{smallmatrix} p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times\end{smallmatrix}\right)^{\det\neq 0}$, [LWX14, Proposition 3.14 (1)] gives a stronger estimate:

$$P_{m,n} \in w^{\lfloor n/p\rfloor - \lfloor m/p\rfloor}(p,w)^{\max\{m-\lfloor n/p\rfloor,0\}} \subseteq w^{m-\lfloor m/p\rfloor}\Lambda^{\geq 1/p}$$

$$(\text{resp. } P_{m,n} \in p^{\lfloor n/p\rfloor - \lfloor m/p\rfloor}(p,w)^{\max\{m-\lfloor n/p\rfloor,0\}} \subseteq p^{m-\lfloor m/p\rfloor}\Lambda^{\leq 1/p} \quad). \tag{2.10.4}$$

This shows the compactness of the $\left(\begin{smallmatrix} p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times\end{smallmatrix}\right)^{\det\neq 0}$-action on (2.10.2) and (2.10.3).

We further remark that, since the order of Δ is prime-to-p, an eigensubspace for the Δ-action on $\mathcal{C}^0(\mathbb{Z}_p;\Lambda^?)^{\mathrm{mod}}$ is a direct summand integrally. So the the action of a uniform limit of finite $\mathbb{Z}$-linear combinations of $\left(\begin{smallmatrix} p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{smallmatrix}\right)^{\det\neq 0}$-actions on the Δ-eigenspaces of (2.10.2) and (2.10.3) is compact. (The action of $\left(\begin{smallmatrix} p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{smallmatrix}\right)^{\det\neq 0}$ may not preserve the Δ-eigenspaces; here we meant the action after projecting back to the Δ-eigenspaces.)

Now, we define

$$\mathrm{S}^{(\varepsilon),?}_{p\text{-adic}} := \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\left(\widetilde{\mathrm{H}}, \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),?}\big)^{\mathrm{mod}}\right) \cong \bigoplus_{i=1}^r e_i^*\cdot \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),?}\big)^{\mathrm{mod},\bar{\mathrm{T}}=\chi_i}.$$

Then $\mathrm{S}^{(\varepsilon),\geq 1/p}_{p\text{-adic}}$ and $\mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}}$ clearly satisfy the condition (2.10.1), because in $\Lambda^{=1/p}$, p and w are differed by a unit. Moreover, the actions of U_p on $\mathrm{S}^{(\varepsilon),\geq 1/p}_{p\text{-adic}}$ and $\mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}}$ are represented by $r\times r$-matrices whose entries are uniform limit of finite sums of actions given by elements from $\left(\begin{smallmatrix} p\mathbb{Z}_p & \mathbb{Z}_p \\ p\mathbb{Z}_p & \mathbb{Z}_p^\times \end{smallmatrix}\right)^{\det\neq 0}$. So they are compact.

In conclusion, the U_p-actions on $\mathrm{S}^{(\varepsilon),\geq 1/p}_{p\text{-adic}}$ and $\mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}}$ are compact and are compatible when base changed to $\Lambda^{=1/p}$. Since both spaces are completed (infinite) direct sum of copies of $\Lambda^{\geq 1/p}$ and $\Lambda^{\leq 1/p}$, respectively, the characteristic power series of U_p (by choosing any orthonormal basis of $\mathrm{S}^{(\varepsilon),?}_{p\text{-adic}}$ over $\Lambda^?$) is well-defined and is independent of the choice of the basis. They are

$$\mathrm{Char}\left(U_p,\,\mathrm{S}^{(\varepsilon),\geq 1/p}_{p\text{-adic}};t\right)\in\Lambda^{\geq 1/p}[\![t]\!] \quad\text{and}\quad \mathrm{Char}\left(U_p,\,\mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}};t\right)\in\Lambda^{\leq 1/p}[\![t]\!]. \tag{2.10.5}$$

They are Fredholm series in the sense of [AIP, Définition B.1], and agree in $\Lambda^{=1/p}[\![t]\!]$ because of the isomorphism (2.10.1). So both power series in (2.10.5) are equal and lie in $\Lambda[\![t]\!]$. We denote it by

$$C^{(\varepsilon)}(w,t) = C^{(\varepsilon)}_{\widetilde{\mathrm{H}}}(w,t) = \sum_{n\geq 0} c_n^{(\varepsilon)}(w)t^n \in \Lambda[\![t]\!] = \mathcal{O}[\![w,t]\!]. \tag{2.10.6}$$

It is called the *characteristic power series of the U_p-action on the abstract p-adic forms* $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$. This is the central object we study in this paper.

Notation 2.11. Let $\mathcal{W}^{(\varepsilon)}$ denote the rigid analytic open unit disk associated to $\mathcal{O}[\![w]\!]^{(\varepsilon)} := \mathcal{O}[\![\Delta\times\mathbb{Z}_p^\times]\!]\otimes_{\mathcal{O}[\Delta^2],\varepsilon}\mathcal{O}$. It is the *weight disk of character ε*. The set of points $\kappa\in\mathcal{W}^{(\varepsilon)}(\mathbb{C}_p)$ is the same as the set of continuous characters $\kappa:\Delta\times\mathbb{Z}_p^\times\to\mathcal{O}_{\mathbb{C}_p}^\times$ such that $\kappa|_{\Delta^2}=\varepsilon$.

The *spectral curve* $\mathrm{Spc}^{(\varepsilon)}_{\widetilde{\mathrm{H}}} = \mathrm{Spc}^{(\varepsilon)}$ is the scheme theoretical zero locus of the characteristic power series $C^{(\varepsilon)}(w,t)$ in $\mathcal{W}^{(\varepsilon)}\times\mathbb{G}_m^{\mathrm{rig}}$. The spectral curve, as a closed subspace of $\mathcal{W}^{(\varepsilon)}\times\mathbb{G}_m^{\mathrm{rig}}$, carries two natural maps, the *weight map* and the *slope map*

$$\mathrm{wt}:\mathrm{Spc}^{(\varepsilon)}\to\mathcal{W}^{(\varepsilon)} \quad\text{and}\quad a_p:\mathrm{Spc}^{(\varepsilon)}\to\mathbb{G}_m^{\mathrm{rig}},$$

sending a point $(w,t)\in\mathrm{Spc}^{(\varepsilon)}$ to $w\in\mathcal{W}^{(\varepsilon)}$ and $t^{-1}\in\mathbb{G}_m^{\mathrm{rig}}$, respectively.

Remark 2.12. Following Andreatta–Iovita–Pilloni's adic Fredholm theory [AIP], one can reinterpret our construction as follows. The U_p-action on $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}}$ is not compact, but thinking of it as a Banach sheaf over $\mathrm{Spa}(\Lambda,\Lambda)$, we constructed a Banach subsheaf by gluing $\mathrm{S}^{(\varepsilon),\geq 1/p}_{p\text{-adic}}$ over $\mathrm{Spa}(\Lambda^{\geq 1/p},\Lambda^{\geq 1/p})$ and $\mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}}$ over $\mathrm{Spa}(\Lambda^{\leq 1/p},\Lambda^{\leq 1/p})$, such that the U_p-action on this

Banach subsheaf is compact. Then the characteristic power series of the U_p-action on this Banach subsheaf is well-defined and has coefficients in Λ.

2.13. **Family of abstract overconvergent forms.** The construction of abstract p-adic forms using $\mathcal{C}^0(\mathbb{Z}_p;\Lambda^{(\varepsilon),?})^{\mathrm{mod}}$ above allows us to prove that the characteristic power series $C^{(\varepsilon)}(w,t)$ has coefficients in $\mathcal{O}[\![w]\!]$. To relate the (abstract) p-adic forms with the (abstract classical) forms, it is more natural to consider a construction using the space $\Lambda^{(\varepsilon),\leq 1/p}\langle z\rangle = \mathcal{O}\langle w/p, z\rangle \otimes_\Lambda \Lambda^{(\varepsilon)}$, that is, the space of analytic functions on the closed unit disk with values in $\mathcal{O}\langle w/p\rangle$.

We define the space of *family of abstract overconvergent forms* to be

$$\mathrm{S}^{\dagger,(\varepsilon)} = \mathrm{S}^{\dagger,(\varepsilon)}_{\widetilde{\mathrm{H}}} := \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\big(\widetilde{\mathrm{H}}, \Lambda^{(\varepsilon),\leq 1/p}\langle z\rangle\big),$$

where $\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in \mathbf{M}_1$ acts on the right hand side by the same formula as (2.4.2), i.e. if $\alpha\delta - \beta\gamma = p^r d$ with $d \in \mathbb{Z}_p^\times$, then

$$\begin{aligned}(2.13.1)\quad h\big|_{\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)}(z) &= \varepsilon(\bar d/\bar\delta,\bar\delta)\cdot(1+w)^{\log\left((\gamma z+\delta)/\omega(\bar\delta)\right)/p}\cdot h\Big(\frac{\alpha z+\beta}{\gamma z+\delta}\Big)\\ &= \varepsilon(\bar d/\bar\delta,\bar\delta)\cdot h\Big(\frac{\alpha z+\beta}{\gamma z+\delta}\Big)\cdot\sum_{n\geq 0}\binom{\log\left((\gamma z+\delta)/\omega(\bar\delta)\right)/p}{n}p^n\Big(\frac{w}{p}\Big)^n \;\in\; \mathcal{O}\langle w/p, z\rangle.\end{aligned}$$

We define the U_p-operator on $\mathrm{S}^{\dagger,(\varepsilon)}$ via the same formula as in (2.5.1).

Note that we have a natural inclusion

$$(2.13.2)\qquad \Lambda^{(\varepsilon),\leq 1/p}\langle z\rangle \subset \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod}},$$

because each power basis element z^i can be written as a $\mathbb{Z}_p$-linear combination of $p^{\lfloor n/p\rfloor}\binom{z}{n}$'s. In addition, this inclusion is equivariant for the $\mathbf{M}_1$-action, inducing a natural U_p-equivariant inclusion

$$\mathrm{S}^{\dagger,(\varepsilon)} \subseteq \mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}}.$$

Lemma 2.14. *The U_p-action on $\mathrm{S}^{\dagger,(\varepsilon)}$ is compact, and the characteristic power series of this U_p-action agrees with $C^{(\varepsilon)}(w,t)$ in $\Lambda^{\leq 1/p}[\![t]\!]$.*

Proof. Consider a different subspace:

$$\mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod},\flat} := \widehat{\bigoplus}_{n\geq 0} p^{\lfloor n/p\rfloor + n/2}\binom{z}{n}\cdot\Lambda^{(\varepsilon),\leq 1/p} \subset \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)$$

with basis $p^{\lfloor n/p\rfloor + n/2}\binom{z}{n}$ instead. It is easy to see the inclusions

$$\mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod},\flat} \subset \Lambda^{(\varepsilon),\leq 1/p}\langle z\rangle \subset \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod}},$$

which is compatible with the $\mathbf{M}_1$-actions. Moreover, if $\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in \left(\begin{smallmatrix}p\mathbb{Z}_p & \mathbb{Z}_p\\ p\mathbb{Z}_p & \mathbb{Z}_p^\times\end{smallmatrix}\right)^{\det\neq 0}$, then the estimate (2.10.4) implies that $|_{\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)}$ takes $\mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod}}$ into the subspace $\mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod},\flat}$ because $n - \lfloor n/p\rfloor \geq n/2$.

The sequence of inclusions above induces U_p-equivariant and compact inclusions

$$\mathrm{S}^{(\varepsilon),\leq 1/p,\flat}_{p\text{-adic}} := \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\big(\widetilde{\mathrm{H}}, \mathcal{C}^0\big(\mathbb{Z}_p;\Lambda^{(\varepsilon),\leq 1/p}\big)^{\mathrm{mod},\flat}\big) \subseteq \mathrm{S}^{\dagger,(\varepsilon)} \subseteq \mathrm{S}^{(\varepsilon),\leq 1/p}_{p\text{-adic}}.$$

Yet the U_p-operator on $\mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p}$ factors through the inclusion $\mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p,\flat} \subseteq \mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p}$; namely the following diagram commutes

$$\begin{tikzcd}
\mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p,\flat} \arrow[r, hook] \arrow[d, "U_p"] & \mathrm{S}^{\dagger,(\varepsilon)} \arrow[r, hook] & \mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p} \arrow[d, "U_p"] \arrow[dll, dashed, "\psi"'] \\
\mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p,\flat} \arrow[r, hook] & \mathrm{S}^{\dagger,(\varepsilon)} \arrow[r, hook] & \mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p}
\end{tikzcd}$$

so that the compositions $\mathrm{S}^{\dagger,(\varepsilon)} \to \mathrm{S}^{\dagger,(\varepsilon)}$ is the U_p-operator. Now, the lemma follows from the link property of Buzzard [Bu07, Lemma 2.7]. $\square$

Remark 2.15. There is no analogous construction of $\mathrm{S}^{\dagger,(\varepsilon)}$ over the entire boundary of the weight disk $\mathrm{Spa}(\Lambda^{(\varepsilon),\geq 1/p}, \Lambda^{(\varepsilon),\geq 1/p})$, because for the expression (2.4.2) to converge in power series, we need to require $v_p(w)$ to be bigger than at least a fixed positive number.

Notation 2.16. For the rest of this paper, *the weights of our classical forms will be k, and the corresponding point on the weight disk is $w_k := \exp(p(k-2)) - 1$.* (This may appear cumbersome in this and the next sections, yet this choice seems to be the best for the rest of the paper.)

For a character η of Δ, we write $\tilde{\eta}$ for the character $\eta \times \eta$ of Δ^2. For an integer $k \geq 2$, we write $\mathcal{O}[z]^{\leq k-2}$ for the space of polynomials with coefficients in $\mathcal{O}$ of degree $\leq k-2$.

2.17. **Abstract classical and overconvergent forms.** For an integer $k \geq 2$ and a character $\psi : \Delta^2 \to \mathcal{O}^\times$, the space of *abstract overconvergent forms of weight k and character ψ* is defined to be

$$\mathrm{S}_k^\dagger(\psi) = \mathrm{S}_{\widetilde{\mathrm{H}},k}^\dagger(\psi) := \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\big(\widetilde{\mathrm{H}},\, \mathcal{O}\langle z\rangle \otimes \psi\big),$$

and its subspace of *abstract classical forms of weight k and character ψ* is

$$\mathrm{S}_k^{\mathrm{Iw}}(\psi) = \mathrm{S}_{\widetilde{\mathrm{H}},k}^{\mathrm{Iw}}(\psi) := \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\big(\widetilde{\mathrm{H}},\, \mathcal{O}[z]^{\leq k-2} \otimes \psi\big).$$

Here a matrix $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathrm{Iw}_p$, or more generally $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathbf{M}_1$, with determinant $p^r d$ (for $r \in \mathbb{Z}$ and $d \in \mathbb{Z}_p^\times$) acts *from the right* on $\mathcal{O}\langle z\rangle$ by

$$f|_{\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)}(z) := (\gamma z + \delta)^{k-2} \cdot f\big(\frac{\alpha z + \beta}{\gamma z + \delta}\big), \tag{2.17.1}$$

stabilizing the subspace $\mathcal{O}[z]^{\leq k-2}$; and we view ψ as a character of $\mathbf{M}_1$ by setting

$$\psi\big(\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)\big) = \psi(\bar{d}/\bar{\delta}, \bar{\delta}). \tag{2.17.2}$$

The space $\mathcal{O}[z]^{\leq k-2} \otimes \psi$ is a finite free $\mathcal{O}$-module, and is endowed with the p-adic topology.

We define the U_p-operator on $\mathrm{S}_k^{\mathrm{Iw}}(\psi) \subset \mathrm{S}_k^\dagger(\psi)$ via the same formula as in (2.5.1). In particular, $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ is a subspace of $\mathrm{S}_k^\dagger(\psi)$ invariant under the U_p-operator.

Set $\varepsilon := \varepsilon(k,\psi) := \psi \cdot (1 \times \omega^{k-2})$ as a character of Δ^2. Then we have a natural isomorphism

$$\mathrm{S}_k^\dagger(\psi) \cong \mathrm{S}^{\dagger,(\varepsilon)} \otimes_{\Lambda^{\leq 1/p}, w \mapsto w_k} \mathcal{O}, \tag{2.17.3}$$

because the action (2.13.1) exactly specializes to the tensor product of (2.17.1) and (2.17.2) per the definition of ε. An immediate consequence of this is that the characteristic power series of the U_p-action on $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ (which is in fact a polynomial) divides the characteristic power series $C^{(\varepsilon)}(w_k,t)$. We defer further discussions of these inclusions and classicality to § 3.6 and Proposition 3.10.

For a character $\eta:\Delta\to\mathcal{O}^\times$, we define the space of *abstract classical forms of weight k with full level structure* to be

$$\mathrm{S}_k^{\mathrm{ur}}(\eta):=\mathrm{Hom}_{\mathcal{O}[\mathrm{K}_p]}\big(\widetilde{\mathrm{H}},\mathcal{O}[z]^{\leq k-2}\otimes\eta\circ\det\big),$$

where a matrix $\left(\begin{smallmatrix}\alpha&\beta\\ \gamma&\delta\end{smallmatrix}\right)\in\mathrm{M}_2(\mathbb{Z}_p)^{\det\neq0}$ with determinant p^rd (for $r\in\mathbb{Z}$ and $d\in\mathbb{Z}_p^\times$) acts from the right on $\mathcal{O}[z]^{\leq k-2}$ by

$$f|_{\left(\begin{smallmatrix}\alpha&\beta\\ \gamma&\delta\end{smallmatrix}\right)}(z):=\eta(\bar d)\cdot(\gamma z+\delta)^{k-2}\cdot f\big(\frac{\alpha z+\beta}{\gamma z+\delta}\big),\tag{2.17.4}$$

and we view $\eta\circ\det$ as a character of $\mathrm{M}_2(\mathbb{Z}_p)^{\det\neq0}$ by setting

$$\eta\circ\det\big(\left(\begin{smallmatrix}\alpha&\beta\\ \gamma&\delta\end{smallmatrix}\right)\big)=\eta(\bar d).\tag{2.17.5}$$

As before, we endow the space $\mathcal{O}[z]^{\leq k-2}\otimes\eta$ with the p-adic topology.

This space carries an action of T_p-operator: taking a coset decomposition $\mathrm{K}_p\left(\begin{smallmatrix}p&0\\ 0&1\end{smallmatrix}\right)\mathrm{K}_p=\coprod_{j=0}^p\mathrm{K}_pu_j$ (e.g. $u_j=\left(\begin{smallmatrix}1&j\\ 0&p\end{smallmatrix}\right)$ for $j=0,\dots,p-1$ and $u_p=\left(\begin{smallmatrix}p&0\\ 0&1\end{smallmatrix}\right)$), then

$$T_p(\varphi)(m)=\sum_{j=0}^p\varphi(mu_j^{-1})|_{u_j}\quad\text{for all }m\in\widetilde{\mathrm{H}}.$$

The T_p-operator does not depend on the choice of the coset decomposition.

Notation 2.18. Since $\widetilde{\mathrm{H}}$ is a projective $\mathcal{O}[\![\mathrm{K}_p]\!]$-module, these spaces $\mathrm{S}_k^{\mathrm{ur}}(\eta)$ and $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ are finite free $\mathcal{O}$-modules (but possibly zero). We record their ranks:

$$d_k^{\mathrm{ur}}(\eta):=\mathrm{rank}_{\mathcal{O}}\,\mathrm{S}_k^{\mathrm{ur}}(\eta)\quad\text{and}\quad d_k^{\mathrm{Iw}}(\psi):=\mathrm{rank}_{\mathcal{O}}\,\mathrm{S}_k^{\mathrm{Iw}}(\psi).$$

The character η induces a character $\tilde\eta$ of Δ^2 per Notation 2.16; it is straightforward to see that $\mathrm{S}_k^{\mathrm{ur}}(\eta)\subseteq\mathrm{S}_k^{\mathrm{Iw}}(\tilde\eta)$. (We defer to the forthcoming paper for a detailed study of the corresponding p-stabilization process, which is a key input to proving local ghost conjecture.) For now, we simply put

$$d_k^{\mathrm{new}}(\eta):=d_k^{\mathrm{Iw}}(\tilde\eta)-2d_k^{\mathrm{ur}}(\eta).$$

We will call this the *rank of abstract p-new forms.*

Remark 2.19. If D is a definite quaternion algebra D over $\mathbb{Q}$ split at p, and if $\widetilde{\mathrm{H}}=\varprojlim_n\mathcal{O}\big[D^\times\backslash D(\mathbb{A}_f)^\times/K^p(1+p^n\mathrm{M}_2(\mathbb{Z}_p))\big]$, then $\mathrm{S}_k^{\mathrm{ur}}(\mathrm{triv})$ is the same as the space of automorphic forms $S_k^D(K^p\mathrm{K}_p)$ on D with level $K^p\mathrm{K}_p$ and weight k defined in [Bu04, § 4]. Similarly, $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ is the same as $S_k^D(K^p\mathrm{Iw}_p;\psi)$, the space of automorphic forms of level $K^p\mathrm{Iw}_p$ and nebentypus character ψ (cf. [Bu04, *loc.cit.*]). The similar statement holds for the space of abstract overconvergent forms.

One can also define abstract classical forms of higher level and more general nebentypus characters. We will not make essential use of them in this paper; so we leave that to interested readers.

Remark 2.20. One can consider a homogenized version of the space of abstract classical forms; explicitly, we have natural isomorphisms:

$$\mathrm{S}_k^{\mathrm{Iw}}(\psi) \cong \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\big(\widetilde{\mathrm{H}}, \mathrm{Sym}^{k-2}\mathcal{O}^{\oplus 2}\otimes\psi\big) \text{ and } \mathrm{S}_k^{\mathrm{ur}}(\eta) \cong \mathrm{Hom}_{\mathcal{O}[\mathrm{K}_p]}(\widetilde{\mathrm{H}}, \mathrm{Sym}^{k-2}\mathcal{O}^{\oplus 2}\otimes\eta\circ\det).$$

Here we view ψ as a character of Iw_p (resp. $\eta\circ\det$ as a character of $\mathrm{GL}_2(\mathbb{Z}_p)$) via (2.17.2) (resp. (2.17.5)). The space $\mathrm{Sym}^{k-2}\mathcal{O}^{\oplus 2}$ is provided with the natural action of K_p from the right, namely,

$$\tilde{h}\big|_{\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)}(X,Y) := \tilde{h}(\alpha X+\beta Y, \gamma X+\delta Y). \tag{2.20.1}$$

(This is the same as the transpose of the left action considered in [Br-notes] and [Pa04].)

The first isomorphism above follows from the identification

$$\begin{aligned} \mathrm{Sym}^{k-2}\mathcal{O}^{\oplus 2}\otimes\psi &\longleftrightarrow \mathcal{O}[z]^{\leq k-2}\otimes\psi \\ \sum_{i=0}^{k-2} a_i X^i Y^{k-2-i} &\longleftrightarrow \sum_{i=0}^{k-2} a_i z^i. \end{aligned} \tag{2.20.2}$$

equivariant for the Iw_p-action (2.20.1) above on the left hand side and the action (2.17.1) on the right hand side. The second isomorphism follows from similar identifications.

Notation 2.21. For a pair of non-negative integers (a,b), we use $\sigma_{a,b}$ to denote the *right* $\mathbb{F}$-representation $\mathrm{Sym}^a\mathbb{F}^{\oplus 2}\otimes\det^b$ of $\mathrm{GL}_2(\mathbb{F}_p)$; (in particular, b is taken modulo $p-1$.) As argued in Remark 2.20, we may identify the representation $\sigma_{a,b}$ with the $\mathbb{F}$-vector space $\mathbb{F}[z]^{\deg\leq a}$ of polynomials of degree less than or equal to a, where the $\mathrm{GL}_2(\mathbb{F}_p)$-action is given by

$$h\big|_{\left(\begin{smallmatrix}\bar\alpha & \bar\beta\\ \bar\gamma & \bar\delta\end{smallmatrix}\right)}(z) = (\bar\alpha\bar\delta-\bar\beta\bar\gamma)^b\cdot(\bar\gamma z+\bar\delta)^a\cdot h\Big(\frac{\bar\alpha z+\bar\beta}{\bar\gamma z+\bar\delta}\Big), \quad \text{for } \left(\begin{smallmatrix}\bar\alpha & \bar\beta\\ \bar\gamma & \bar\delta\end{smallmatrix}\right)\in\mathrm{GL}_2(\mathbb{F}_p).$$

When $a\in\{0,\dots,p-1\}$ and $b\in\{0,\dots,p-2\}$, $\sigma_{a,b}$ is irreducible. These representations exhaust all irreducible (right) $\mathbb{F}$-representations of $\mathrm{GL}_2(\mathbb{F}_p)$ ([BL94, Proposition 1]). We call them the *Serre weights*. We use $\mathrm{Proj}_{a,b}$ to denote the projective envelope of $\sigma_{a,b}$ as a (right) $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$-module. We refer to Lemma A.2 in the appendix for an explicit description of $\mathrm{Proj}_{a,b}$.

Now, we switch to stating the local version of Bergdall and Pollack's ghost conjecture. This is largely inspired by their work [BP19a, BP19b], with one difference: we confine ourselves to the reducible, non-split, and generic type below.

Definition 2.22. For the rest of this paper, fix a *reducible, nonsplit, and generic* residual representation $\bar\rho:\mathrm{I}_{\mathbb{Q}_p}\to\mathrm{GL}_2(\mathbb{F})$ of the inertia subgroup:

$$\bar\rho\simeq\begin{pmatrix}\omega_1^{a+b+1} & *\neq 0\\ 0 & \omega_1^b\end{pmatrix} \qquad \text{for } 1\leq a\leq p-4 \text{ and } 0\leq b\leq p-2, \tag{2.22.1}$$

where ω_1 is the first fundamental character (recall $p\geq 5$). We say an $\mathcal{O}[\![K_p]\!]$-projective augmented module $\widetilde{\mathrm{H}}$ is of *type* $\bar\rho$ with *multiplicity* $m(\widetilde{\mathrm{H}})\in\mathbb{N}$ if

(1) (Serre weight) $\overline{\mathrm{H}} := \widetilde{\mathrm{H}}/(\varpi, \mathrm{I}_{1+p\mathrm{M}_2(\mathbb{Z}_p)})$ is isomorphic to a direct sum of $m(\widetilde{\mathrm{H}})$ copies of $\mathrm{Proj}_{a,b}$ as a right $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$-module, and

(2) (Central character I) the action of $\left(\begin{smallmatrix} p & 0 \\ 0 & p \end{smallmatrix}\right)$ on $\widetilde{\mathrm{H}}$ is the multiplication by an invertible element $\xi \in \mathcal{O}^\times$.
(3) (Central character II) there exists an isomorphism $\widetilde{\mathrm{H}} \cong \widetilde{\mathrm{H}}_0 \widehat{\otimes}_{\mathcal{O}} \mathcal{O}[\![(1+p\mathbb{Z}_p)^\times]\!]$ of $\mathcal{O}[\mathrm{GL}_2(\mathbb{Q}_p)]$-modules, where $\widetilde{\mathrm{H}}_0$ carries an action of $\mathrm{GL}_2(\mathbb{Q}_p)$ which is trivial on elements of the form $\left(\begin{smallmatrix} \alpha & 0 \\ 0 & \alpha \end{smallmatrix}\right)$ for $\alpha \in (1+p\mathbb{Z}_p)^\times$, and the latter factor $\mathcal{O}[\![(1+p\mathbb{Z}_p)^\times]\!]$ carries the natural action of $\mathrm{GL}_2(\mathbb{Q}_p)$ through the map $\mathrm{GL}_2(\mathbb{Q}_p) \xrightarrow{\det} \mathbb{Q}_p^\times \xrightarrow{p^r\delta \mapsto \delta/\omega(\bar{d})} (1+p\mathbb{Z}_p)^\times$.

We say $\widetilde{\mathrm{H}}$ is *primitive* if $m(\widetilde{\mathrm{H}}) = 1$.

Remark 2.23. (1) If D is a definite quaternion algebra over $\mathbb{Q}$ that is split at p and if $\bar{r} : \mathrm{Gal}_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{F})$ is an absolutely irreducible global residual Galois representation such that $\bar{r}|_{\mathrm{I}_{\mathbb{Q}_p}} \cong \bar{\rho}$, then the $\bar{r}$-localized completed homology

$$\widetilde{\mathrm{H}} := \varprojlim_n \mathcal{O}\big[D^\times \backslash D(\mathbb{A}_f)^\times / K^p(1+p^n\mathrm{M}_2(\mathbb{Z}_p))\big]_{\mathfrak{m}_{\bar{r}}}$$

is an $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar{\rho}$; the primitive condition is equivalent to the mod-p multiplicity-one hypothesis: $\mathrm{S}_2^D(K^p\mathrm{Iw}_p; \omega^{a+b} \times \omega^b)$ has rank 1 over $\mathcal{O}$. Indeed, condition (1) of Definition 2.22 is determined by the Serre weight conjecture. The other two conditions are natural properties of central characters. One have similar construction for modular curves, as indicated in the introduction.

(2) If $\tilde{\mathrm{H}}$ is primitive, we will see in Proposition A.7 that the space $\mathrm{S}_2^{\mathrm{Iw}}(\omega^b \times \omega^{a+b})$ has rank 1 over $\mathcal{O}$ and the U_p-action on it is given by an element in $\mathcal{O}^\times$. In particular, this space is ordinary. Since we will not use this result in this paper, we leave it to the Appendix (Proposition A.7). For a general $\tilde{\mathrm{H}}$, we still expect the space $\mathrm{S}_2^{\mathrm{Iw}}(\omega^b \times \omega^{a+b})$ is ordinary although we cannot give a proof in this paper.

One can see why this should be true as follows: in the automorphic situation above, the mod-p-multiplicity-one condition automatically puts us in the situation with one Serre weight, which (in many cases) means $\bar{r}|_{\mathrm{I}_{\mathbb{Q}_p}} \cong \bar{\rho}$ for some $\bar{\rho}$ above; in this case, the ordinary condition is automatic. Thus, it is conceivable that such ordinarity result can be deduced directly on purely on $\mathcal{O}[\![K_p]\!]$-projective augmented modules.

We shall primarily focus on the case of a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented modules, but we shall indicate from time to time how our theory adapts to the non-primitive case.

Notation 2.24. For the rest of this paper, we fix a *primitive* $\mathcal{O}[\![K_p]\!]$-projective augmented module $\widetilde{\mathrm{H}}$ of type $\bar{\rho}$, unless otherwise specified. In particular, $\overline{\mathrm{H}} = \widetilde{\mathrm{H}}/(\varpi, \mathrm{I}_{1+p\mathrm{M}_2(\mathbb{Z}_p)}) \cong \mathrm{Proj}_{a,b}$.

(1) A character ε of Δ^2 is called *relevant* to $\bar{\rho}$ if $\varepsilon(\bar{\alpha}, \bar{\alpha}) = \bar{\alpha}^{a+2b}$ for all $\bar{\alpha} \in \Delta$. By comparing the central action of $\left(\begin{smallmatrix} \omega(\bar{\alpha}) & 0 \\ 0 & \omega(\bar{\alpha}) \end{smallmatrix}\right)$ for $\bar{\alpha} \in \Delta$ on $\widetilde{\mathrm{H}}$ and on $\mathrm{Ind}_{B^{\mathrm{op}}(\mathbb{Z}_p)}^{\mathrm{Iw}_p}(\chi_{\mathrm{univ}}^{(\varepsilon)})$, one sees that $\mathrm{S}_{p\text{-adic}}^{(\varepsilon)}$ as defined in (2.4) is nonzero if and only if ε is relevant to $\bar{\rho}$.

(2) We write each relevant character ε of Δ^2 in the form

$$\varepsilon = \varepsilon_1 \times \varepsilon_2 = \omega^{-s_\varepsilon + b} \times \omega^{a+s_\varepsilon+b}$$

for some $s_\varepsilon \in \{0, \dots, p-2\}$. A very important invariant we shall use later is

$$k_\varepsilon := 2 + \{a + 2s_\varepsilon\} \ \in \ \{2, 3, \dots, p\}. \tag{2.24.1}$$

here $\{a+2s_\varepsilon\}$ is the unique integer in $\{0,\dots,p-2\}$ satisfying the condition

$$a+2s_\varepsilon \equiv \{a+2s_\varepsilon\} \mod p-1.$$

If we write $\tilde{\varepsilon}_1$ for the character $\varepsilon_1 \times \varepsilon_1$ of Δ^2 (per Notation 2.16), then

$$\varepsilon = \tilde{\varepsilon}_1 \cdot (1 \times \omega^{k_\varepsilon - 2}). \tag{2.24.2}$$

(3) By the discussion in § 2.17 and (2.24.2), for integer $k \geq 2$ such that $k \equiv k_\varepsilon \bmod (p-1)$, the space of abstract classical forms $\mathrm{S}_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1)$ and hence $\mathrm{S}_k^{\mathrm{ur}}(\varepsilon_1)$ are subspaces of $\mathrm{S}^{(\varepsilon)}_{p\text{-adic}} \otimes_{\mathcal{O}[\![w]\!], w \mapsto w_k} \mathcal{O}$, where we recall that $w_k = \exp((k-2)p) - 1$.

For the rest of this section, we fix a character ε of Δ^2 relevant to $\bar{\rho}$.

Definition 2.25. Following [BP19a, BP19b], we define the *ghost series* for $\widetilde{\mathrm{H}}$ over $\mathcal{W}^{(\varepsilon)}$ to be the formal power series

$$G^{(\varepsilon)}(w,t) = G^{(\varepsilon)}_{\widetilde{\mathrm{H}}}(w,t) = 1 + \sum_{n=1}^{\infty} g_n^{(\varepsilon)}(w) t^n \in \mathcal{O}[w][\![t]\!],$$

where each coefficient $g_n^{(\varepsilon)}(w)$ is a product

$$g_n^{(\varepsilon)}(w) = \prod_{\substack{k \geq 2 \\ k \equiv k_\varepsilon \bmod p-1}} (w - w_k)^{m_n^{(\varepsilon)}(k)} \in \mathbb{Z}_p[w] \tag{2.25.1}$$

with exponents $m_n^{(\varepsilon)}(k)$ given by the following recipe

$$m_n^{(\varepsilon)}(k) = \begin{cases} \min\big\{n - d_k^{\mathrm{ur}}(\varepsilon_1), d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1) - n\big\} & \text{if } d_k^{\mathrm{ur}}(\varepsilon_1) < n < d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1) \\ 0 & \text{otherwise.} \end{cases}$$

In particular, for a fixed k, the sequence $(m_n^{(\varepsilon)}(k))_{n \geq 1}$ is given by the following palindromic pattern

$$\underbrace{0,\dots,0}_{d_k^{\mathrm{ur}}(\varepsilon_1)}, 1, 2, 3, \dots, \tfrac{1}{2}d_k^{\mathrm{new}}(\varepsilon_1) - 1, \tfrac{1}{2}d_k^{\mathrm{new}}(\varepsilon_1), \tfrac{1}{2}d_k^{\mathrm{new}}(\varepsilon_1) - 1, \dots, 3, 2, 1, 0, 0, \dots, \tag{2.25.2}$$

where the maximum $\frac{1}{2}d_k^{\mathrm{new}}(\varepsilon_1)$ appears at the $\frac{1}{2}d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1)$th place. (We shall see later in Corollary 4.4 that $d_k^{\mathrm{new}}(\varepsilon_1) = d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - 2d_k^{\mathrm{ur}}(\varepsilon_1)$ is always an even integer.) As we shall late prove in Corollary 4.8 that, for a fixed n, $m_n^{(\varepsilon)}(k)$ is only nonzero for finitely many k. So the product (2.25.1) defining $g_n^{(\varepsilon)}(w)$ is a finite product.

We remark that, the ghost pattern (2.25.2) implies that, for any integer $k \geq 2$ such that $k \equiv k_\varepsilon \bmod (p-1)$, we have, for $n \geq 1$,

$$m_{n+1}^{(\varepsilon)}(k) - m_n^{(\varepsilon)}(k) = \begin{cases} 1 & \text{if } d_k^{\mathrm{ur}}(\varepsilon_1) \leq n < \frac{1}{2}d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) \\ -1 & \text{if } \frac{1}{2}d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) \leq n < d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1) \\ 0 & \text{otherwise,} \end{cases} \tag{2.25.3}$$

and for $n \geq 2$,

$$m_{n+1}^{(\varepsilon)}(k) - 2m_n^{(\varepsilon)}(k) + m_{n-1}^{(\varepsilon)}(k) = \begin{cases} -2 & \text{if } n = \frac{1}{2}d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) \\ 1 & \text{if } n = d_k^{\mathrm{ur}}(\varepsilon_1) \text{ or } d_k^{\mathrm{ur}}(\varepsilon_1) + d_k^{\mathrm{new}}(\varepsilon_1) \\ 0 & \text{otherwise.} \end{cases} \tag{2.25.4}$$

Example 2.26. We give an example of the ghost series and refer to § 4 for the computation of the dimensions. Suppose that $p = 7$, $a = 2$, $b = 0$, and that $\widetilde{\mathrm{H}}$ is primitive of type $\bar{\rho} = \left(\begin{smallmatrix} \omega_1^3 & *\neq 0 \\ 0 & 1 \end{smallmatrix}\right)$; so $\overline{\mathrm{H}} \cong \mathrm{Proj}_{2,0}$. We list below the dimensions $d_k^{\mathrm{Iw}}(\psi)$ for small k's.

ε	k	2	3	4	5	6	7	8	9	10	11	12	13	14
$1 \times \omega^2$	$d_k^{\mathrm{Iw}}(1 \times \omega^{4-k}) = \lfloor \frac{k+2}{6} \rfloor + \lfloor \frac{k+4}{6} \rfloor$	1	1	2^*	2	2	2	3	3	4^*	4	4	4	5
$\omega^5 \times \omega^3$	$d_k^{\mathrm{Iw}}(\omega^5 \times \omega^{5-k}) = \lfloor \frac{k+1}{6} \rfloor + \lfloor \frac{k+3}{6} \rfloor$	0	1	1	2	2^*	2	2	3	3	4	4^*	4	4
$\omega^4 \times \omega^4$	$d_k^{\mathrm{Iw}}(\omega^4 \times \omega^{-k}) = \lfloor \frac{k}{6} \rfloor + \lfloor \frac{k+2}{6} \rfloor$	0^*	0	1	1	2	2	2^*	2	3	3	4	4	4^*
$\omega^3 \times \omega^5$	$d_k^{\mathrm{Iw}}(\omega^3 \times \omega^{1-k}) = \lfloor \frac{k-1}{6} \rfloor + \lfloor \frac{k+1}{6} \rfloor$	0	0	0^*	1	1	2	2	2	2^*	3	3	4	4
$\omega^2 \times 1$	$d_k^{\mathrm{Iw}}(\omega^2 \times \omega^{2-k}) = \lfloor \frac{k+4}{6} \rfloor + \lfloor \frac{k}{6} \rfloor$	1	1	1	1	2^*	2	3	3	3	3	4^*	4	5
$\omega \times \omega$	$d_k^{\mathrm{Iw}}(\omega \times \omega^{3-k}) = \lfloor \frac{k+3}{6} \rfloor + \lfloor \frac{k-1}{6} \rfloor$	0^*	1	1	1	1	2	2^*	3	3	3	3	4	4^*

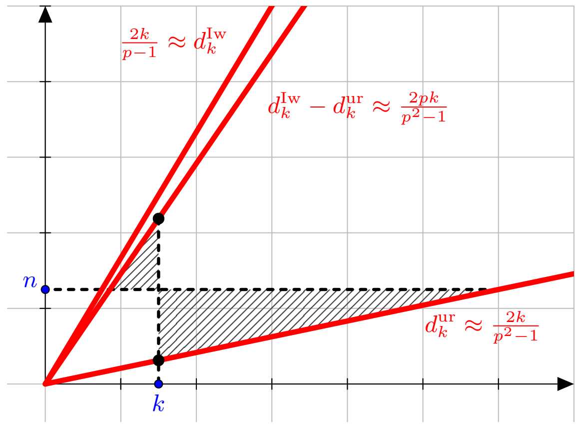
The superscript $*$ indicates where the character ψ is equal to $\tilde{\varepsilon}_1$, in which case $d_k^{\mathrm{ur}}(\varepsilon_1)$ makes sense. In the table below, we list the information on dimensions of abstract classical forms with level K_p and Iw_p.

ε	Triples $\big(k,\ d_k^{\mathrm{ur}}(\varepsilon_1),\ d_k^{\mathrm{new}}(\varepsilon_1)\big)$ on the corresponding weight disk						
$1 \times \omega^2$	$(4,1,0)$	$(10,1,2)$	$(16,1,4)$	$(22,1,6)$	$(28,2,6)$	$(34,2,8)$	$(40,2,10)$
$\omega^5 \times \omega^3$	$(6,0,2)$	$(12,1,2)$	$(18,1,4)$	$(24,1,6)$	$(30,1,8)$	$(36,2,8)$	$(42,2,10)$
$\omega^4 \times \omega^4$	$(2,0,0)$	$(8,0,2)$	$(14,0,4)$	$(20,1,4)$	$(26,1,6)$	$(32,1,8)$	$(38,1,10)$
$\omega^3 \times \omega^5$	$(4,0,0)$	$(10,0,2)$	$(16,0,4)$	$(22,0,6)$	$(28,1,6)$	$(34,1,8)$	$(40,1,10)$
$\omega^2 \times 1$	$(6,0,2)$	$(12,1,2)$	$(18,1,4)$	$(24,1,6)$	$(30,1,8)$	$(36,2,8)$	$(42,2,10)$
$\omega \times \omega$	$(2,0,0)$	$(8,0,2)$	$(14,0,4)$	$(20,1,4)$	$(26,1,6)$	$(32,1,8)$	$(38,1,10)$

Unwinding the definition of ghost series in Definition 2.25, we give the first four terms of the ghost series on the $\varepsilon = (1 \times \omega^2)$-weight disk (corresponding to the first rows in the above two tables).

$$
\begin{aligned}
g_1^{(\varepsilon)}(w) &= 1, \\
g_2^{(\varepsilon)}(w) &= (w - w_{10})(w - w_{16})(w - w_{22}), \\
g_3^{(\varepsilon)}(w) &= (w - w_{16})^2 (w - w_{22})^2 (w - w_{28})(w - w_{34})(w - w_{40})(w - w_{46}), \\
g_4^{(\varepsilon)}(w) &= (w - w_{16})(w - w_{22})^3 (w - w_{28})^2 \cdots (w - w_{46})^2 (w - w_{52}) \cdots (w - w_{70}).
\end{aligned}
$$

The following graph gives the asymptotic behavior of the dimensions $d_k^{\mathrm{ur}} = \frac{2k}{p^2-1} + O(1)$ and $d_k^{\mathrm{Iw}} = \frac{2k}{p-1} + O(1)$ when $p = 7$. (The vertical axis was stretched approximately 5 times.) The degree of $g_n(w)$ is proportional to the area of the shaded region, which corresponds to the contribution of the ghost zeros.



Graph 1: Functions d_k^{ur}, d_k^{Iw}, and $d_k^{\mathrm{Iw}} - d_k^{\mathrm{ur}}$, and the contribution of ghost zeros.

The following table gives the increments in degrees $\deg g_n^{(\varepsilon)}$ over different weight disks.

ε	$\deg g_n^{(\varepsilon)} - \deg g_{n-1}^{(\varepsilon)}$ for $n \geq 1$														
$1 \times \omega^2$	0	3	5	8	10	13	15	18	21	23	26	28	31	33	36	...
$\omega^5 \times \omega^3$	1	3	6	9	11	14	16	19	21	24	27	29	32	34	37	...
$\omega^4 \times \omega^4$	2	4	7	9	12	15	17	20	22	25	27	30	33	35	38	...
$\omega^3 \times \omega^5$	3	5	8	10	13	15	18	21	23	26	28	31	33	36	39	...
$\omega^2 \times 1$	1	3	6	9	11	14	16	19	21	24	27	29	32	34	37	...
$\omega \times \omega$	2	4	7	9	12	15	17	20	22	25	27	30	33	35	38	...

These are expected to be the halo w-adic slopes "near the boundary" of the weight disks; see Corollary 4.14 and the surrounding remark for more discussions.

Remark 2.27. In the definition of ghost series, the palindromic pattern in the middle of (2.25.2) is what Bergdall and Pollack referred to as the "ghost pattern". They discovered this through a numerical computation. We refer to [BP19a, §1] for a discussion on the heuristic behind the construction of ghost series.

Notation 2.28. For a power series $F(t) = \sum_{n\geq 0} f_n t^n \in \mathbb{C}_p[\![t]\!]$ (with $f_0 = 1$), we use $\mathrm{NP}(F)$ to denote its *Newton polygon*, that is the convex hull of points $(n, v_p(f_n))$ for all n. The slopes of its segments are exactly the minuses of the p-adic valuations of zeros of $F(t)$, counted with multiplicities.

The following is the local analogue of the ghost conjecture of Bergdall and Pollack.

Conjecture 2.29 (Local ghost conjecture)**.** *Let* $\bar{\rho} = \begin{pmatrix} \omega_1^{a+b+1} & *\neq 0 \\ 0 & \omega_1^b \end{pmatrix} : \mathrm{I}_{\mathbb{Q}_p} \to \mathrm{GL}_2(\mathbb{F})$ *be a character with* $a \in \{1, \dots, p-4\}$ *and* $b \in \{0, \dots, p-2\}$*. Let* $\widetilde{\mathrm{H}}$ *be a primitive* $\mathcal{O}[\![K_p]\!]$*-projective augmented module of type* $\bar{\rho}$*, and let* ε *be a character of* Δ^2 *relevant to* $\bar{\rho}$*. We define the characteristic power series* $C^{(\varepsilon)}(w,t)$ *of* U_p*-action and the ghost series* $G^{(\varepsilon)}(w,t)$ *for* $\widetilde{\mathrm{H}}$ *as in this section. Then*

(1) *the ghost series* $G^{(\varepsilon)}(w,t)$ *depends only* $\bar{\rho}$*, and* ε*, and*
(2) *for every* $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$*, we have* $\mathrm{NP}(G^{(\varepsilon)}(w_\star, -)) = \mathrm{NP}(C^{(\varepsilon)}(w_\star, -))$*.*

Part (1) of this conjecture will be proved in Corollary 4.8; in fact the dimensions $d_k^{\mathrm{Iw}}(\varepsilon_1)$ and $d_k^{\mathrm{ur}}(\varepsilon_1)$ depend only on k, $\bar\rho$, and ε. Because of this, one may rename $G^{(\varepsilon)}_{\widetilde{\mathrm{H}}}(w,t)$ into $G^{(\varepsilon)}_{\bar\rho}(w,t)$, and call it *the ghost series attached to $\bar\rho$ and ε.*

We will address part (2) in a sequel to this paper.

Remark 2.30. (1) The ghost series $G^{(\varepsilon)}(w,t)$ may be viewed as a good enough approximation of the characteristic power series $C^{(\varepsilon)}(w,t)$. At least the p-adic valuations of their zeros are the same.

(2) The definition of the ghost series depends on the choice of the topological generator $\exp(p)$ of $(1+p\mathbb{Z}_p)^\times$. In other words, for any $\eta \in 1+p\mathbb{Z}_p^\times$, we may replace in (2.4.2) the exponent $\log\big((\gamma z+\delta)/\omega(\bar\delta)\big)/p$ by $\log\big((\gamma z+\delta)/\omega(\bar\delta)\big)/\log\eta$, and replace $w_k=\exp((k-2)p)-1$ by $\eta^{k-2}-1$, to get a essentially different ghost series. So ghost series is in this sense not canonical; yet the Newton polygon $\mathrm{NP}(G^{(\varepsilon)}(w_\star,t))$ (for all $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$) does not depend on the choice of the topological generator. So we may work with our "preferred" topological generator $\exp(p)$ of $(1+p\mathbb{Z}_p)^\times$.

(3) Recall that $\left(\begin{smallmatrix} p & 0\\ 0 & p\end{smallmatrix}\right)$ acts by multiplication with a unit $\xi\in\mathcal{O}^\times$ on $\widetilde{\mathrm{H}}$. Moreover, by Definition 2.22 (1), the diagonal matrices $\left(\begin{smallmatrix} \bar\alpha & 0\\ 0 & \bar\alpha\end{smallmatrix}\right)$ for $\bar\alpha\in\Delta$ acts on $\widetilde{\mathrm{H}}$ by multiplication with $\omega(\bar\alpha)^{a+2b}$. (By possibly enlarging $\mathcal{O}$), consider a character $\zeta:\mathbb{Q}_p^\times\to\mathcal{O}^\times$ such that $\zeta(p)^2=\xi^{-1}$ and $\zeta(\alpha)=\omega(\bar\alpha)^{-b}$ for all $\alpha\in\mathbb{Z}_p^\times$. If twist the $\mathrm{GL}_2(\mathbb{Q}_p)$-action on $\widetilde{\mathrm{H}}$ by the character $\zeta\circ\det$, then the resulting completed homology piece $\widetilde{\mathrm{H}}' := \widetilde{\mathrm{H}}\otimes\zeta\circ\det$ is of type $\left(\begin{smallmatrix} \omega_1^{a+1} & *\neq 0\\ 0 & 1\end{smallmatrix}\right)$, and the action of $\left(\begin{smallmatrix} p & 0\\ 0 & p\end{smallmatrix}\right)$ is trivial.

It follows easily from the construction of the characteristic power series of the U_p-action and the definition of the ghost series that

$$C^{(\varepsilon)}_{\widetilde{\mathrm{H}}}(w,t)=C^{(\varepsilon\cdot\tilde\omega^{-b})}_{\widetilde{\mathrm{H}}'}(w,\zeta(p)^{-1}t)\quad\text{and}\quad G^{(\varepsilon)}_{\widetilde{\mathrm{H}}}(w,t)=G^{(\varepsilon\cdot\tilde\omega^{-b})}_{\widetilde{\mathrm{H}}'}(w,t).$$

So Conjecture 2.29 can be reduced to the case when $b=0$ and $\xi(p)=1$.

Remark 2.31. Our definition of ghost series follows Bergdall and Pollack [BP19a, BP19b]. In fact, a variant of the ghost series already appeared in a much earlier paper by Buzzard and Calegari [BC04] (see also [Lo11]) but only in a special case: $p=2$ and level $N=1$. In our language, one can set $u=\log(1+w)+2p$ and then it is easy to check that $\mathcal{O}[\![w/p^{1/(p-1)}]\!]\cong\mathcal{O}[\![u/p^{1/(p-1)}]\!]$, where we assume that $\mathcal{O}$ contains a $(p-1)$th root of p. Then one can generalize Buzzard and Calegari's construction to define a variant of ghost series:

$$A^{(\varepsilon)}(u,t)=1+\sum_{n=1}^{\infty}a_n^{(\varepsilon)}(u)t^n\in\mathcal{O}[\![u/p^{1/(p-1)}]\!][\![t]\!]$$

with coefficients given by

$$a_n^{(\varepsilon)}(u):=\prod_{k\geq 2,k\equiv k_\varepsilon \bmod p-1}(u-pk)^{m_n^{(\varepsilon)}(k)}\in\mathbb{Z}_p[u].$$

(In Buzzard–Calegari case, the numbers $m_n^{(\varepsilon)}(k)$ are explicitly computed, as it is for one example $p=2$ and level $N=1$.) One can prove that for every $w_\star\in\mathcal{O}_{\mathbb{C}_p}$ with $v_p(w_\star)>\frac{1}{p-1}$, setting

$$u_\star=\log(1+w_\star)+2p$$

(which has the same p-adic valuation as $w_\star$), the two Newton polygons $\mathrm{NP}(G^{(\varepsilon)}(w_\star,-))$ and $\mathrm{NP}(A^{(\varepsilon)}(u_\star,-))$ coincide. Thus proving Conjecture 2.29 is equivalent to proving that, for every pair $w_\star$ and $u_\star$ above (with valuation $> \frac{1}{p-1}$),

$$\mathrm{NP}(C^{(\varepsilon)}(w_\star,-)) = \mathrm{NP}(A^{(\varepsilon)}(u_\star,-)).$$

Remark 2.32. It is natural to ask what one should expect when $\bar\rho$ is nongeneric, reducible split, or of the form $\bar\rho = \omega_2^a \oplus \omega_2^{pa}$, where ω_2 is second fundamental character. There are several places of the theory that needs modifications:

1. a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module $\widetilde{\mathrm{H}}$ should be instead isomorphic to the direct sum of the projective envelopes of Serre weights, viewed as an $\mathcal{O}[\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$-module (which would then change the dimension combinatorics);
2. some extra condition should be imposed to give initial "seeds" of the theory corresponding to the Hodge–Tate weight $(0,1)$ crystabelian deformation rings of $\bar\rho$ (in the Fontaine-Laffaille range, the crystalline deformations of a reducible $\bar\rho$ are ordinary, while for the irreducible ρ's, we need extra information to record the initial slopes); and
3. the combinatorics may be different and more complicated.

It seems that, with appropriate modifications, the ghost series framework might still partially work. We hope to address some of these cases in future works.

We do not know how to generalize the ghost conjecture framework beyond $\mathrm{GL}_2(\mathbb{Q}_p)$.

Remark 2.33. If $\widetilde{\mathrm{H}}_1$ and $\widetilde{\mathrm{H}}_2$ are two primitive $\mathcal{O}[\![K_p]\!]$-projective augmented modules of type $\bar\rho_1$ and $\bar\rho_2$, respectively, with $\det\bar\rho_1 = \det\bar\rho_2$, then one may use the recipe in Definition 2.25 to define a ghost series $G^{(\varepsilon)}_{\widetilde{\mathrm{H}}_1\oplus\widetilde{\mathrm{H}}_2}(w,t)$ for $\widetilde{\mathrm{H}}_1\oplus\widetilde{\mathrm{H}}_2$. While it is clear that $C^{(\varepsilon)}_{\widetilde{\mathrm{H}}_1\oplus\widetilde{\mathrm{H}}_2}(w,t) = C^{(\varepsilon)}_{\widetilde{\mathrm{H}}_1}(w,t)C^{(\varepsilon)}_{\widetilde{\mathrm{H}}_2}(w,t)$, it is not obvious whether $\mathrm{NP}\big(G^{(\varepsilon)}_{\widetilde{\mathrm{H}}_1\oplus\widetilde{\mathrm{H}}_2}(w_\star,-)\big) = \mathrm{NP}\big(G^{(\varepsilon)}_{\widetilde{\mathrm{H}}_1}(w_\star,t)G^{(\varepsilon)}_{\widetilde{\mathrm{H}}_2}(w_\star,-)\big)$ for all $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$.

In fact, as a recent preprint of Rufei Ren [Re22$^+$] suggests, this equality of Newton polygon of ghost series should hold if and only if $\bar\rho_1^{ss} = \bar\rho_2^{ss}$, where $\bar\rho_?^{ss}$ stands for the semisimplification of $\bar\rho_?$. When $\bar\rho_1^{ss} \neq \bar\rho_2^{ss}$, the two Newton polygons can disagree at some nth slope of a certain weight k, where both n and k are very large integers. In turn, this suggests the original ghost conjecture for the full space of modular forms may not be correct as stated in [BP19a, Conjecture 2.5], where the numerical computation is not able to reach very large n.

Now if $\widetilde{\mathrm{H}}'$ is an $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar\rho$ with multiplicity $m(\widetilde{\mathrm{H}}')$, then Rufei Ren's result [Re22$^+$] would imply that $\mathrm{NP}(G^{(\varepsilon)}_{\widetilde{\mathrm{H}}'}(w_\star,-))$ is the same as $\mathrm{NP}(G^{(\varepsilon)}_{\bar\rho}(w_\star,-))$ with x-axis stretched $m(\widetilde{\mathrm{H}}')$ times, for every $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$. It is natural to conjecture that $\mathrm{NP}(C^{(\varepsilon)}_{\widetilde{\mathrm{H}}'}(w_\star,-))$ agrees with this polygon.

3. Properties of abstract classical, overconvergent, and p-adic forms

In this section, we discuss finer properties of abstract classical, overconvergent, and p-adic forms:

1. a careful choice of the basis e_1 and e_2 of $\widetilde{\mathrm{H}}$ as a free $\mathcal{O}[\![\mathrm{Iw}_{p,1}]\!]$-module (Notation 3.4),
2. a description of the power basis of abstract classical, overconvergent forms (Proposition 3.7),

(3) classicality of abstract overconvergent forms, and the theta operator (Proposition 3.10), and
(4) the Atkin–Lehner involution for abstract classical forms (Proposition 3.12).

Property (1) is well known. Property (2) will be useful later for establishing the dimension formula for abstract classical forms with Iwahori level structure. Properties (3) and (4) are all well known in the context of modular forms, and more generally automorphic forms; we just reproduce them in our abstract setup.

In view of the discussion of Remark 2.30 (3), we assume the following.

Hypothesis 3.1. For the rest of this paper, we assume that $b=0$, $\xi=1$, and that $\widetilde{\mathrm{H}}$ is a *primitive* $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar\rho = \left(\begin{smallmatrix}\omega_1^{a+1} & *\neq 0\\ 0 & 1\end{smallmatrix}\right)$, on which $\left(\begin{smallmatrix}p & 0\\ 0 & p\end{smallmatrix}\right)$ acts trivially. In addition, we assume that $1 \le a \le p-4$.

We reserve the Greek letter ε to denote a character of Δ^2 relevant to $\bar\rho$.

3.2. **$\widetilde{\mathrm{H}}$ as an $\mathcal{O}[\![\mathrm{Iw}_p]\!]$-module.** Let $\widetilde{\mathrm{H}}$ be as above. As explained in § 2.8, we may write $\widetilde{\mathrm{H}}$ as an $\mathcal{O}[\![\mathrm{Iw}_p]\!]$-module

$$\widetilde{\mathrm{H}} \simeq e_1\mathcal{O}\otimes_{\chi_1,\mathcal{O}[\bar{\mathrm{T}}]}\mathcal{O}[\![\mathrm{Iw}_p]\!] \oplus e_2\mathcal{O}\otimes_{\chi_2,\mathcal{O}[\bar{\mathrm{T}}]}\mathcal{O}[\![\mathrm{Iw}_p]\!] \tag{3.2.1}$$

for two characters χ_1 and χ_2 of Δ^2, which we determine now.

Taking (3.2.1) modulo $(\varpi, \mathrm{I}_{1+p\mathrm{M}_2(\mathbb{Z}_p)})$ and using condition (1) of Definition 2.22, we obtain an isomorphism of (right) $\mathbb{F}[\bar{\mathrm{B}}]$-modules (for $\bar{\mathrm{B}} := \left(\begin{smallmatrix}\mathbb{F}_p^\times & \mathbb{F}_p\\ 0 & \mathbb{F}_p^\times\end{smallmatrix}\right)$)

$$\mathrm{Proj}_{a,0} \cong e_1\mathbb{F}\otimes_{\chi_1,\mathbb{F}[\bar{\mathrm{T}}]}\mathbb{F}[\bar{\mathrm{B}}] \oplus e_2\mathbb{F}\otimes_{\chi_2,\mathbb{F}[\bar{\mathrm{T}}]}\mathbb{F}[\bar{\mathrm{B}}].$$

Taking the $\bar{\mathrm{U}} := \left(\begin{smallmatrix}1 & \mathbb{F}_p\\ 0 & 1\end{smallmatrix}\right)$-invariants of both sides, the left hand side has 2-dimensional $\bar{\mathrm{U}}$-invariants with characters $1\times\omega^a$ and $\omega^a\times 1$ under the $\bar{\mathrm{T}}$-action by Lemma A.5; the $\bar{\mathrm{U}}$-invariants of the right hand side are spanned by $e_i\cdot\left(\left(\begin{smallmatrix}1 & 0\\ 0 & 1\end{smallmatrix}\right)+\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)+\cdots+\left(\begin{smallmatrix}1 & p-1\\ 0 & 1\end{smallmatrix}\right)\right)$ for $i=1,2$, which have characters χ_1 and χ_2, respectively. So it follows that (after possibly exchanging the χ_i's)

$$\chi_1 = 1\times\omega^a \quad \text{and} \quad \chi_2 = \omega^a\times 1.$$

Although the characters χ_1 and χ_2 are uniquely determined, the two generators e_1 and e_2 are far from unique. We further rigidify the generators e_1 and e_2 as follows.

Lemma 3.3. *In* (3.2.1)*, we may replace e_2 by $e_2' := e_1\left(\begin{smallmatrix}0 & 1\\ p & 0\end{smallmatrix}\right)$ (so that $e_1 = e_2'\left(\begin{smallmatrix}0 & 1\\ p & 0\end{smallmatrix}\right)$).*

Proof. Note that the action of $\bar{\mathrm{T}}$ on e_2 and e_2' are the same. Indeed, for the given basis e_1 and e_2, we can write

$$e_1\left(\begin{smallmatrix}0 & 1\\ p & 0\end{smallmatrix}\right) = e_1A + e_2B \quad \text{and} \quad e_2\left(\begin{smallmatrix}0 & 1\\ p & 0\end{smallmatrix}\right) = e_1C + e_2D, \tag{3.3.1}$$

with $A,B,C,D\in\mathcal{O}[\![\mathrm{Iw}_{p,1}]\!]$. Let $\bar A$, $\bar B$, $\bar C$, and $\bar D$ denote the reduction in $\mathbb{F}$ of A, B, C, and D modulo the maximal ideal $(\varpi, \mathrm{I}_{\mathrm{Iw}_{p,1}})$ of $\mathcal{O}[\![\mathrm{Iw}_{p,1}]\!]$. Consider the $\bar{\mathrm{T}}$-action on (3.3.1) modulo $(\varpi, \mathrm{I}_{\mathrm{Iw}_{p,1}})$ (noting that e_1 and e_2 have different characters $1\times\omega^a$ and $\omega^a\times 1$, respectively). We get $\bar A = \bar D = 0$ and $\bar B, \bar C \in \mathbb{F}^\times$. Thus it follows that e_1 and $e_1\left(\begin{smallmatrix}0 & 1\\ p & 0\end{smallmatrix}\right)$ spans $\widetilde{\mathrm{H}}$ over $\mathcal{O}[\![\mathrm{Iw}_{p,1}]\!]$. □

Notation 3.4. *For the rest of the paper, we fix an isomorphism*

$$\widetilde{\mathrm{H}} \simeq e_1\mathcal{O} \otimes_{1\times\omega^a,\mathcal{O}[\bar{\mathrm{T}}]} \mathcal{O}[\![\mathrm{Iw}_p]\!] \oplus e_2\mathcal{O} \otimes_{\omega^a\times 1,\mathcal{O}[\bar{\mathrm{T}}]} \mathcal{O}[\![\mathrm{Iw}_p]\!],$$

where the generators satisfy $e_1 = e_2\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)$ and $e_2 = e_1\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)$.

Remark 3.5. Choosing generators with the above property will simplify our proof in this paper; and will be essential for sequel.

3.6. **Power basis of abstract overconvergent and classical forms.** Fix a *relevant* character $\varepsilon = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon}$ of Δ^2 as in Notation 2.24 (in particular, $\varepsilon_1 = \omega^{-s_\varepsilon}$). We give a natural power basis of the space of the family of abstract overconvergent forms $\mathrm{S}^{\dagger,(\varepsilon)}$ defined in § 2.13; then by specializing the parameters, we obtain a power basis of the space of abstract overconvergent forms $\mathrm{S}_k^\dagger\big(\varepsilon\cdot(1\times\omega^{2-k})\big)$ of weight k and its subspace of abstract classical forms $\mathrm{S}_k^{\mathrm{Iw}}\big(\varepsilon\cdot(1\times\omega^{2-k})\big)$ as defined in § 2.17.

Following a discussion similar to § 2.8, we have

$$\begin{aligned}
&\mathrm{S}^{\dagger,(\varepsilon)} = \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\left(\widetilde{\mathrm{H}},\ \Lambda^{\leq 1/p}\langle z\rangle \otimes (\omega^{-s_\varepsilon}\times\omega^{a+s_\varepsilon})\right)\\
&\cong e_1^*\cdot\left(\Lambda^{\leq 1/p}\langle z\rangle \otimes (\omega^{-s_\varepsilon}\times\omega^{a+s_\varepsilon})\right)^{\bar{\mathrm{T}}=1\times\omega^a} \oplus e_2^*\cdot\left(\Lambda^{\leq 1/p}\langle z\rangle \otimes (\omega^{-s_\varepsilon}\times\omega^{a+s_\varepsilon})\right)^{\bar{\mathrm{T}}=\omega^a\times 1}.
\end{aligned}$$

The monomial $z^j \in \mathcal{O}\langle z\rangle$ is an eigenvector for the $\bar{\mathrm{T}}$-action with character $\omega^j\times\omega^{-j}$. So we deduce the following results on basis of $\mathrm{S}^{\dagger,(\varepsilon)}$.

Proposition 3.7. *For the relevant character $\varepsilon = \omega^{-s_\varepsilon}\times\omega^{a+s_\varepsilon}$ with $s_\varepsilon \in \{0,\dots,p-2\}$ and for k an integer, the following list is a basis of $\mathrm{S}^{\dagger,(\varepsilon)}$ as a Banach $\Lambda^{\leq 1/p}$-module and also a basis of $\mathrm{S}_k^\dagger\big(\varepsilon\cdot(1\times\omega^{2-k})\big)$ as a Banach $\mathcal{O}$-module:*

$$e_1^*z^{s_\varepsilon}, e_1^*z^{p-1+s_\varepsilon}, e_1^*z^{2(p-1)+s_\varepsilon},\dots;\ e_2^*z^{\{a+s_\epsilon\}}, e_2^*z^{p-1+\{a+s_\epsilon\}}, e_2^*z^{2(p-1)+\{a+s_\epsilon\}},\dots. \tag{3.7.1}$$

When $k\geq 2$, the subsequence consisting of terms whose power in z is less than or equal to $k-2$ forms a basis of $\mathrm{S}_k^{\mathrm{Iw}}\big(\varepsilon\cdot(1\times\omega^{2-k})\big)$ as an $\mathcal{O}$-module.

Proof. The list of basis (3.7.1) follows from the discussion above. The case for $\mathrm{S}_k^\dagger(\varepsilon\cdot(1\times\omega^{2-k}))$ follows from the family case and the isomorphism (2.17.3). ☐

This result will immediately give a dimension formula for $d_k^{\mathrm{Iw}}(\psi)$, which we defer to Proposition 4.1.

Notation 3.8.
(1) We call (3.7.1) the *power basis* of $\mathrm{S}^{\dagger,(\varepsilon)}$ and $\mathrm{S}_k^\dagger\big(\varepsilon\cdot(1\times\omega^{2-k})\big)$, denoted by $\mathbf{B}^{(\varepsilon)}$ (as it formally does not depend on k).
(2) The *degree* of each basis element $\mathbf{e} = e_i^*z^j \in \mathbf{B}^{(\varepsilon)}$ is the exponents on z, i.e., $\deg(e_i^*z^j) = j$. We order the elements in $\mathbf{B}^{(\varepsilon)}$ as $\mathbf{e}_1^{(\varepsilon)}, \mathbf{e}_2^{(\varepsilon)}, \dots$ with increasing degrees. In particular each $\mathbf{e}_\ell^{(\varepsilon)}$ is some $e_i^*z^j$ and i depends only on the parity of ℓ. Moreover, under our generic assumption $1\leq a\leq p-4$ (indeed $1\leq a\leq p-2$ is already enough), the degrees of elements of $\mathbf{B}^{(\varepsilon)}$ are pairwise distinct.
(3) Write $\mathbf{B}_k^{(\varepsilon)}$ for the subset of elements of $\mathbf{B}^{(\varepsilon)}$ with degree less than or equal to $k-2$. It is a basis of $\mathrm{S}_k^{\mathrm{Iw}}\big(\varepsilon\cdot(1\times\omega^{2-k})\big)$ by Proposition 3.7. When fixing the k, elements in $\mathbf{B}_k^{(\varepsilon)}$ are often called *classical* and elements in $\mathbf{B}^{(\varepsilon)}\backslash\mathbf{B}_k^{(\varepsilon)}$ are often called *non-classical.*

(4) We write $\mathrm{U}^{\dagger,(\varepsilon)} \in \mathrm{M}_\infty(\Lambda^{\leq 1/p})$ for the matrix of $\Lambda^{\leq 1/p}$-linear U_p-action on $\mathrm{S}^{\dagger,(\varepsilon)}$ with respect to the power basis $\mathbf{B}^{(\varepsilon)}$; its evaluation at $w = w_k$ is the matrix $\mathrm{U}_k^{\dagger,(\varepsilon)}$ of U_p-action on $\mathrm{S}_k^\dagger\big(\varepsilon \cdot (1 \times \omega^{2-k})\big)$ (with respect to $\mathbf{B}^{(\varepsilon)}$).

Since the U_p-operator on $\mathrm{S}_k^\dagger\big(\varepsilon \cdot (1 \times \omega^{2-k})\big)$ leaves the subspace $\mathrm{S}_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big)$ stable, the matrix $\mathrm{U}_k^{\dagger,(\varepsilon)}$ is a block-upper-triangular matrix, whose $d_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big) \times d_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big)$ upper-left block $\mathrm{U}_k^{\mathrm{Iw},(\varepsilon)}$ is the matrix for the U_p-action on $\mathrm{S}_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big)$ with respect to $\mathbf{B}_k^{(\varepsilon)}$.

Remark 3.9. As we proved in Lemma 2.14 that the U_p-action on $\mathrm{S}^{\dagger,(\varepsilon)}$ is compact, and the inclusion $\mathrm{S}_{p\text{-adic}}^{(\varepsilon),\leq 1/p} \subseteq \mathrm{S}^{\dagger,(\varepsilon)}$ induces an identification of characteristic power series:

$$\mathrm{Char}(\mathrm{U}^{\dagger,(\varepsilon)}, t) = C^{(\varepsilon)}(w,t) \quad \text{and} \quad \mathrm{Char}(\mathrm{U}_k^{\dagger,(\varepsilon)}, t) = C^{(\varepsilon)}(w_k,t). \tag{3.9.1}$$

The following is an analogue of the classicality result of overconvergent automorphic forms.

Proposition 3.10 (Theta map)**.** *Let $\varepsilon = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon}$ be a character of Δ^2 relevant to $\bar\rho$, and let $k \geq 2$ be an integer. Put $\psi = \varepsilon \cdot (1 \times \omega^{2-k})$, $\varepsilon' = \varepsilon \cdot (\omega^{k-1} \times \omega^{1-k})$, and $\psi' = \varepsilon' \cdot (1 \times \omega^k) = \psi \cdot \tilde\omega^{k-1}$.*

(1) *There is a short exact sequence*

$$0 \to \mathrm{S}_k^{\mathrm{Iw}}(\psi) \longrightarrow \mathrm{S}_k^\dagger(\psi) \xrightarrow{(\frac{d}{dz})^{k-1}\circ} \mathrm{S}_{2-k}^\dagger(\psi'), \tag{3.10.1}$$

which is equivariant for the usual U_p-action on the first two terms and the $p^{k-1}U_p$-action on the third term. Here the map $\big(\frac{d}{dz}\big)^{k-1}\circ$ sends an element $\varphi \in \mathrm{S}_k^\dagger(\psi)$ to the function that takes $m \in \widetilde{\mathrm{H}}$ to the $(k-1)$th derivative of the function $\varphi(m) \in \mathcal{O}\langle z\rangle$.

(2) *We have an equality of characteristic power series*

$$C^{(\varepsilon)}(w_k,t) = C^{(\varepsilon')}(w_{2-k}, p^{k-1}t) \cdot \mathrm{Char}\big(U_p;\, \mathrm{S}_k^{\mathrm{Iw}}(\psi)\big).$$

(3) *All finite U_p-slopes of $\mathrm{S}_k^\dagger(\psi)$ strictly less than $k-1$ are the same as the finite U_p-slopes of $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ (which are strictly less than $k-1$). The multiplicity of $k-1$ as U_p-slopes of $\mathrm{S}_k^\dagger(\psi)$ is the sum of the multiplicity of $k-1$ as U_p-slopes of $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ and the multiplicity of 0 as U_p-slopes of $\mathrm{S}_{2-k}^\dagger(\psi')$.*

Proof. The results of the proposition are well-known. But we do not know any proof in our abstract setup. We include one for completeness. *Only in this proof, for a an integer, we write $|^{(a)}_{\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)}$ for the action of $\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in \mathbf{M}_1$ on $h(z) \in \mathcal{O}\langle z\rangle$ given by*

$$h\big|^{(a)}_{\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)}(z) = (\gamma z+\delta)^a h\big(\frac{\alpha z+\beta}{\gamma z+\delta}\big).$$

For (1), it follows from the discussion in [Bu07, §7] that for $h(z) \in \mathcal{O}\langle z\rangle$, we have the following equality for $\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in \mathbf{M}_1$,

$$\begin{aligned}\frac{d^{k-1}}{dz^{k-1}}\Big(h\big|^{(k-2)}_{\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)}(z)\Big) &= \frac{d^{k-1}}{dz^{k-1}}\Big((\gamma z+\delta)^{k-2} h\big(\frac{\alpha z+\beta}{\gamma z+\delta}\big)\Big)\\ = (\alpha\delta-\beta\gamma)^{k-1}(\gamma z+\delta)^{-k}\big(\frac{d^{k-1}h}{dz^{k-1}}\big)\big(\frac{\alpha z+\beta}{\gamma z+\delta}\big) &= (\alpha\delta-\beta\gamma)^{k-1}\big(\frac{d^{k-1}h}{dz^{k-1}}\big)\big|^{(-k)}_{\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right)}.\end{aligned}$$

In other words, $\left(\frac{d}{dz}\right)^{k-1}$ induces a natural morphism (equivariant for the $\mathbf{M}_1$-action)

$$(3.10.2)\qquad \big(\frac{d}{dz}\big)^{k-1}: \Big(\mathcal{O}\langle z\rangle,\ \big|^{(k-2)}\Big) \longrightarrow \Big(\mathcal{O}\langle z\rangle,\ \big|^{(-k)}\otimes \det{}^{k-1}\Big).$$

Applying $\mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}(\widetilde{\mathrm{H}}, -\otimes\psi)$ to (3.10.2) gives a natural morphism

$$(3.10.3)\qquad \big(\frac{d}{dz}\big)^{k-1}\circ: \mathrm{S}_k^{\dagger,(\varepsilon)} \longrightarrow \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\Big(\widetilde{\mathrm{H}}, \big(\mathcal{O}\langle z\rangle,\ \big|^{(-k)}\otimes \det{}^{k-1}\otimes\psi\big)\Big).$$

Write the character det (on $\mathrm{GL}_2(\mathbb{Q}_p)$) as $\det_1\times\det_2$ so that if $\det\left(\begin{smallmatrix}\alpha&\beta\\ \gamma&\delta\end{smallmatrix}\right) = p^r d$ for $r\in\mathbb{Z}$ and $d\in\mathbb{Z}_p^\times$, then $\det_1\left(\begin{smallmatrix}\alpha&\beta\\ \gamma&\delta\end{smallmatrix}\right) = p^r\omega(\bar d)$ and $\det_2\left(\begin{smallmatrix}\alpha&\beta\\ \gamma&\delta\end{smallmatrix}\right) = d/\omega(\bar d)$. Then we may rewrite (3.10.3) as

$$(3.10.4)\qquad \big(\frac{d}{dz}\big)^{k-1}\circ: \mathrm{S}_k^{\dagger,(\varepsilon)} \longrightarrow \mathrm{Hom}_{\mathcal{O}[\mathrm{Iw}_p]}\Big(\widetilde{\mathrm{H}}\otimes\det{}_2^{1-k}, \big(\mathcal{O}\langle z\rangle,\ \big|^{(-k)}\otimes \det{}_1^{k-1}\otimes\psi\big)\Big).$$

By assumption (3) in Definition 2.22, the $\mathcal{O}[\![K_p]\!]$-projective augmented module $\widetilde{\mathrm{H}}$ is continuously and $\mathcal{O}[\mathrm{GL}_2(\mathbb{Q}_p)]$-equivariantly isomorphic to $\widetilde{\mathrm{H}}\otimes\det_2^{1-k}$. Using this isomorphism, we deduce that the right hand side of (3.10.4) is isomorphic to $\mathrm{S}_{2-k}^\dagger(\psi')$ with $\psi'=\psi\cdot\tilde\omega^{k-1}$, and the corresponding weight disk character

$$\varepsilon' = (1\times\omega^{-k})\cdot\psi' = (1\times\omega^{-k})\cdot\varepsilon\cdot(1\times\omega^{2-k})\cdot(\omega^{k-1}\times\omega^{k-1}) = \varepsilon\cdot(\omega^{k-1}\times\omega^{1-k}).$$

The U_p-action on the right hand side of (3.10.4) is p^{k-1} times the usual U_p-action on $\mathrm{S}_{2-k}^\dagger(\psi')$ (where the extra factor p^{k-1} comes from the image of p under $\det_1^{k-1}$). This proves (1) of the Proposition.

(2) Consider the standard power basis (3.7.1) of $\mathrm{S}_k^\dagger(\psi)$. The kernel of the map $(\frac{d}{dz})^{k-1}\circ$, namely $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$, is exactly spanned by those power basis elements $\mathbf{e}\in\mathbf{B}^{(\varepsilon)}$ such that $\deg(\mathbf{e})\leq k-2$, i.e $\mathbf{B}_k^{(\varepsilon)}$. Moreover, the image of a basis element $\mathbf{e}=e_i^*z^j\in\mathbf{B}^{(\varepsilon)}-\mathbf{B}_k^{(\varepsilon)}$ under $(\frac{d}{dz})^{k-1}\circ$ is exactly $j(j-1)\cdots(j-k+2)e_i^*z^{j-k+1}$, namely a multiple of the power basis element in $\mathbf{B}^{(\varepsilon')}$ for $\mathrm{S}_{2-k}^\dagger(\psi')$ (if nonzero). Therefore, it follows that

$$(3.10.5)\qquad \mathrm{U}_k^{\dagger,(\varepsilon)} = \begin{pmatrix}\mathrm{U}_k^{\mathrm{Iw},(\varepsilon)} & * \\ 0 & p^{k-1}D^{-1}\mathrm{U}_{2-k}^{\dagger,(\varepsilon')}D\end{pmatrix},$$

where D is the diagonal matrix whose diagonal entries are indexed by $\mathbf{e}=e_i^*z^j\in\mathbf{B}^{(\varepsilon)}$ with $i=1,2$ and $j\geq k-1$, and are given by $j(j-1)\cdots(j-k+2)$. From this and the equality (3.9.1), we see immediately that

$$\begin{aligned} C^{(\varepsilon)}(w_k,t) &\overset{(3.9.1)}{=} \mathrm{Char}(\mathrm{U}_k^{\dagger,(\varepsilon)},t) = \mathrm{Char}(\mathrm{U}_k^{\mathrm{Iw},(\varepsilon)},t)\cdot\mathrm{Char}(\mathrm{U}_{2-k}^{\dagger,(\varepsilon')},p^{k-1}t)\\ &\overset{(3.9.1)}{=} \mathrm{Char}\big(U_p;\ \mathrm{S}_k^{\mathrm{Iw}}(\psi)\big)\cdot C^{(\varepsilon')}(w_{2-k},p^{k-1}t).\end{aligned}$$

(3) By (2), the multiset of the finite U_p-slopes of $\mathrm{S}_k^\dagger(\psi)$ is the disjoint union of the multiset of U_p-slopes of $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ and the multiset of $k-1+\alpha$ for each U_p-slope α of $\mathrm{S}_{2-k}^\dagger(\psi')$, counted with multiplicity. (3) is a corollary of this and the fact that the U_p-slope on $\mathrm{S}_{2-k}^\dagger(\psi')$ are non-negative (as $C^{(\varepsilon')}(w_{2-k},t)$ has integral coefficients). □

Remark 3.11. The short exact sequence (3.10.1) is not exact on the right, because "integrating" a power series in $\mathcal{O}\langle z\rangle$ no longer belong to $\mathcal{O}\langle z\rangle$ (inverting p will not fix this problem either due to existence of arbitrarily large p-powers in the denominators). To get the surjectivity, one needs to work with functions on open disks (as opposed to closed disks), or to take certain limit. Since we will not use this later, we leave this as an exercise for the readers.

The following is an analogue of the classical Atkin–Lehner theory.

Proposition 3.12 (Atkin–Lehner involution)**.** *Let $\psi = \psi_1 \times \psi_2$ be a character of Δ^2 (where we allow $\psi_1 = \psi_2$). Put $\psi'' = \psi_2 \times \psi_1$ and $\varepsilon'' = \varepsilon \cdot \psi'' \cdot \psi^{-1} = \omega^{-s_{\varepsilon''}} \times \omega^{a+s_{\varepsilon''}}$ (with $s_{\varepsilon''} = \{k-2-a-s_\varepsilon\}$).*

(1) *Then we have a well-defined natural morphism*

$$\begin{aligned} \mathrm{AL}_{(k,\psi)} : \mathrm{S}_k^{\mathrm{Iw}}(\psi) &\longrightarrow \mathrm{S}_k^{\mathrm{Iw}}(\psi'') \\ \varphi &\longmapsto \Big(\mathrm{AL}_{(k,\psi)}(\varphi) : x \mapsto \varphi\big(x\big(\begin{smallmatrix} 0 & p^{-1} \\ 1 & 0 \end{smallmatrix}\big)\big)\big|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} \Big). \end{aligned} \tag{3.12.1}$$

Here the last $|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)}$ is the usual action on $\mathcal{O}[z]^{\leq k-2}$ and the trivial *action on the factor ψ''.*

(2) *We have an explicit formula, for $i = 1, 2$ and any j,*

$$\mathrm{AL}_{(k,\psi)}(e_i^* z^j) = p^{k-2-j} \cdot e_{3-i}^* z^{k-2-j}.$$

Or equivalently we have

$$\mathrm{AL}_{(k,\psi)}(\mathbf{e}_\ell^{(\varepsilon)}) = p^{k-2-\deg \mathbf{e}_\ell^{(\varepsilon)}} \mathbf{e}^{(\varepsilon'')}_{d_k^{\mathrm{Iw}}(\psi'')+1-\ell}, \tag{3.12.2}$$

where we added superscripts to the power basis element to indicate the corresponding character. In particular, we have

$$\mathrm{AL}_{(k,\psi'')} \circ \mathrm{AL}_{(k,\psi)} = p^{k-2}. \tag{3.12.3}$$

So $\mathrm{AL}_{(k,\psi)}$ induces an isomorphism $\mathrm{AL}_{(k,\psi)} : \mathrm{S}_k^{\mathrm{Iw}}(\psi) \otimes E \xrightarrow{\cong} \mathrm{S}_k^{\mathrm{Iw}}(\psi'') \otimes E$.

(3) *When $\psi_1 \neq \psi_2$ (or equivalently $k \not\equiv k_\varepsilon \bmod (p-1)$), we have an equality*

$$U_p \circ \mathrm{AL}_{(k,\psi)} \circ U_p = p^{k-1} \cdot \mathrm{AL}_{(k,\psi)} \tag{3.12.4}$$

as maps from $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ to $\mathrm{S}_k^{\mathrm{Iw}}(\psi'')$. Consequently, when $\psi_1 \neq \psi_2$, we can pair the slopes for the U_p-action on $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ and the slopes for the U_p-action on $\mathrm{S}_k^{\mathrm{Iw}}(\psi'')$ so that each pair adds up to $k-1$. In particular all slopes belong to $[0, k-1]$.

Proof. This is well known, but not documented in the literature for our abstract setup. We include a proof for completeness. First of all, $\psi = \varepsilon \cdot (1 \times \omega^{2-k}) = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon+2-k}$ and

$$\varepsilon'' = \varepsilon \cdot \psi'' \cdot \psi^{-1} = \omega^{-s_\varepsilon + (a+s_\varepsilon+2-k)+s_\varepsilon} \times \omega^{a+s_\varepsilon+s_\varepsilon-(a+s_\varepsilon+2-k)} = \omega^{-s_{\varepsilon''}} \times \omega^{a+s_{\varepsilon''}},$$

where $s_{\varepsilon''} = \{k-2-a-s_\varepsilon\}$.

(1) We need to show that if $\varphi \in \mathrm{S}_k^{\mathrm{Iw}}(\psi)$, $\mathrm{AL}_{(k,\psi)}(\varphi)$ as a map from $\widetilde{\mathrm{H}}$ to $\mathcal{O}[z]^{\leq k-2} \otimes \psi''$ is equivariant for the Iw_p-action. Indeed, for $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathrm{Iw}_p$ and $x \in \widetilde{\mathrm{H}}$,

$$\begin{aligned}\mathrm{AL}_{(k,\psi)}(\varphi)\big(x\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)\big) &= \varphi\big(x\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & p^{-1} \\ 1 & 0 \end{smallmatrix}\right)\big)\big|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} = \varphi\big(x\left(\begin{smallmatrix} 0 & p^{-1} \\ 1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} \delta & p^{-1}\gamma \\ p\beta & \alpha \end{smallmatrix}\right)\big)\big|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} \\ &= \psi(\bar{\delta}, \bar{\alpha}) \cdot \varphi\big(x\left(\begin{smallmatrix} 0 & p^{-1} \\ 1 & 0 \end{smallmatrix}\right)\big)\big|_{\left(\begin{smallmatrix} \delta & p^{-1}\gamma \\ p\beta & \alpha \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} = \psi''(\bar{\alpha}, \bar{\delta}) \cdot \varphi\big(x\left(\begin{smallmatrix} 0 & p^{-1} \\ 1 & 0 \end{smallmatrix}\right)\big)\big|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)}.\end{aligned}$$

Here the third equality uses the fact that φ is equivariant for the right action of $\left(\begin{smallmatrix} \delta & p^{-1}\gamma \\ p\beta & \alpha \end{smallmatrix}\right) \in \mathrm{Iw}_p$ on $\widetilde{\mathrm{H}}$ and the product right action of $\left(\begin{smallmatrix} \delta & p^{-1}\gamma \\ p\beta & \alpha \end{smallmatrix}\right) \in \mathrm{Iw}_p$ on $\mathcal{O}[z]^{\leq k-2} \otimes \psi$.

(2) Recall from Notation 3.4 that the generators of $\widetilde{\mathrm{H}}$ are chosen so that $e_1\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right) = e_2$ and $e_2\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right) = e_1$ (as $\left(\begin{smallmatrix} p & 0 \\ 0 & p \end{smallmatrix}\right)$ acts trivially on $\widetilde{\mathrm{H}}$ by Hypothesis 3.1). So if $\varphi = e_i^* z^j \in \mathrm{S}_k^{\mathrm{Iw}}(\psi)$, then

$$\mathrm{AL}_{(k,\psi)}(\varphi)(e_{i'}) = \varphi\big(e_{i'}\left(\begin{smallmatrix} 0 & p^{-1} \\ 1 & 0 \end{smallmatrix}\right)\big)\big|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} = \varphi(e_{3-i'})\big|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} = \begin{cases} z^j\big|_{\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} = (pz)^{k-2-j} & \text{if } i = 3-i' \\ 0 & \text{if } i = i'. \end{cases}$$

(3) Let us check that $\psi_1 = \psi_2$ if and only if $k \equiv k_\varepsilon \bmod (p-1)$. Indeed, $\psi = \varepsilon \cdot (1 \times \omega^{2-k}) = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon+2-k}$. So $\psi_1 = \psi_2$ if and only if $-s_\varepsilon \equiv a + s_\varepsilon + 2 - k \bmod (p-1)$, which is further equivalent to $k \equiv a + 2s_\varepsilon + 2 \equiv k_\varepsilon \bmod (p-1)$.

Now we assume $\psi_1 \neq \psi_2$ and check the equality (3.12.4) and then the last sentence is clearly a corollary of this. Thanks to (3.12.3), we prove $U_p \circ \mathrm{AL}_{(k,\psi)} \circ U_p \circ \mathrm{AL}_{(k,\psi'')} = p^{2k-3}$ instead. We compute using the formula (2.5.1), for $\varphi \in \mathrm{S}_k^{\mathrm{Iw}}(\psi)$,

$$\begin{aligned} &(3.12.5) \qquad U_p \circ \mathrm{AL}_{k,\psi} \circ U_p \circ \mathrm{AL}_{k,\psi''}(\varphi) \\ &= \sum_{i,j=0}^{p-1} \varphi\Big(\bullet \left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)^{-1}\left(\begin{smallmatrix} p & 0 \\ ip & 1 \end{smallmatrix}\right)^{-1}\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)^{-1}\left(\begin{smallmatrix} p & 0 \\ jp & 1 \end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix} p & 0 \\ jp & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} p & 0 \\ ip & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ p & 0 \end{smallmatrix}\right)} \\ &= \sum_{i,j=0}^{p-1} \varphi\Big(\bullet \left(\begin{smallmatrix} p & 0 \\ 0 & p \end{smallmatrix}\right)^{-2}\left(\begin{smallmatrix} 1 & i \\ j & 1+ij \end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix} 1 & i \\ j & 1+ij \end{smallmatrix}\right)\left(\begin{smallmatrix} p & 0 \\ 0 & p \end{smallmatrix}\right)^2} \\ &= p^{2k-4} \sum_{i,j=0}^{p-1} \varphi\Big(\bullet \left(\begin{smallmatrix} 1 & i \\ j & 1+ij \end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix} 1 & i \\ j & 1+ij \end{smallmatrix}\right)} \end{aligned}$$

We separate the discussion depending on whether $1 + ij$ is divisible by p. When $ij \equiv -1 \bmod p$,

$$\begin{aligned}\sum_{j=1}^{p-1} \varphi\Big(\bullet \left(\begin{smallmatrix} 1 & i_j \\ j & 0 \end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix} 1 & i_j \\ j & 0 \end{smallmatrix}\right)} &= \sum_{j=1}^{p-1} \varphi\Big(\bullet \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)^{-1}\left(\begin{smallmatrix} -i_j & 1 \\ 0 & j \end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix} -i_j & 1 \\ 0 & j \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)} \\ &= \sum_{j=1}^{p-1} \psi_1(\bar{j})\psi_2(\bar{j})^{-1}\varphi\Big(\bullet \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)}\end{aligned}$$

where i_j denote the unique integer in $\{1, \ldots, p-1\}$ such that $ji_j \equiv -1 \bmod p$. Since $\psi_1 \neq \psi_2$, the sum of $\psi_1(\bar{j})\psi_2(\bar{j})^{-1}$ over all $\bar{j} \in \Delta$ is zero. So the sum above is zero.

When $ij \not\equiv -1 \bmod p$, we may write

$$\begin{pmatrix}1 & i\\ j & 1+ij\end{pmatrix} = \begin{pmatrix}1/(1+ij) & i\\ 0 & 1+ij\end{pmatrix}\begin{pmatrix}1 & 0\\ j/(1+ij) & 1\end{pmatrix}.$$

So we deduce

$$\begin{aligned}
&\sum_{ij\not\equiv -1 \bmod p} \varphi\Big(\bullet \left(\begin{smallmatrix}1 & i\\ j & 1+ij\end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix}1 & i\\ j & 1+ij\end{smallmatrix}\right)} \\
= &\sum_{ij\not\equiv -1 \bmod p} \varphi\Big(\bullet \left(\begin{smallmatrix}1 & 0\\ j/(1+ij) & 1\end{smallmatrix}\right)^{-1}\left(\begin{smallmatrix}1/(1+ij) & i\\ 0 & 1+ij\end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix}1/(1+ij) & i\\ 0 & 1+ij\end{smallmatrix}\right)\left(\begin{smallmatrix}1 & 0\\ j/(1+ij) & 1\end{smallmatrix}\right)} \\
= &\sum_{ij\not\equiv -1 \bmod p} \psi_1(\overline{1+ij})\psi_2(\overline{1+ij})^{-1}\varphi\Big(\bullet \left(\begin{smallmatrix}1 & 0\\ j/(1+ij) & 1\end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix}1 & 0\\ j/(1+ij) & 1\end{smallmatrix}\right)}
\end{aligned} \tag{3.12.6}$$

Since $\psi_1 \neq \psi_2$, the sum of $\psi_1(\bar\alpha)\psi_2(\bar\alpha)^{-1}$ over all $\bar\alpha \in \Delta$ is zero. So when $j \neq 0$, if we consider the above sum with $j/(1+ij)$ fixed yet letting $1+ij$ to run over all nonzero residual classes mod p, we would get zero. Indeed, there is a bijection

$$\begin{aligned}
\{\bar i, \bar j \in \mathbb{F}_p \mid \bar j \neq 0 \text{ and } 1+\bar i\bar j \neq 0\} &\longleftrightarrow \{(\bar\alpha, \bar\beta) \in (\mathbb{F}_p^\times)^2\}\\
(\bar i,\, \bar j) &\longmapsto (1+\bar i\bar j,\, \bar j/(1+\bar i\bar j))\\
(\bar\alpha\bar\beta,\, (\bar\alpha-1)/\bar\alpha\bar\beta) &\longleftarrow\!\shortmid (\bar\alpha,\, \bar\beta).
\end{aligned}$$

So the sum in (3.12.6) is equal to the sum over the pairs $(i, 0)$, i.e.

$$\sum_{ij\not\equiv -1 \bmod p} \varphi\Big(\bullet \left(\begin{smallmatrix}1 & i\\ j & 1+ij\end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix}1 & i\\ j & 1+ij\end{smallmatrix}\right)} = \sum_{i=0}^{p-1}\varphi\Big(\bullet \left(\begin{smallmatrix}1 & i\\ 0 & 1\end{smallmatrix}\right)^{-1}\Big)\Big|_{\left(\begin{smallmatrix}1 & i\\ 0 & 1\end{smallmatrix}\right)} = p\varphi.$$

Combining this with the discussion when $ij \equiv -1 \bmod p$ and (3.12.5), we deduce that

$$U_p \circ \mathrm{AL}_{k,\psi} \circ U_p \circ \mathrm{AL}_{k,\psi''}(\varphi) = p^{2k-4}\cdot p\varphi = p^{2k-3}\varphi.$$

This concludes the proof of the Proposition. □

Remark 3.13. (1) We point out that $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ and $\mathrm{S}_k^{\mathrm{Iw}}(\psi'')$ often correspond to points on *different* weight disks. More precisely, if $\varepsilon = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon}$ is a character of Δ^2 relevant to $\bar\rho$. Then applying above to $\psi = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon-k+2}$ we get an Atkin–Lehner involution, for $\varepsilon'' = \omega^{-s''_\varepsilon} \times \omega^{a+s''_\varepsilon}$ with $s''_\varepsilon := \{k-2-a-s_\varepsilon\}$,

$$\mathrm{AL}_{(k,\psi)} : \mathrm{S}_k^{\mathrm{Iw}}\big(\varepsilon\cdot(1\times\omega^{2-k})\big) \to \mathrm{S}_k^{\mathrm{Iw}}\big(\varepsilon''\cdot(1\times\omega^{2-k})\big).$$

In particular, for different k's, the Atkin–Lehner involutions are between different pairs of disks. (This phenomenon was known before, see e.g. [LWX14, Step III of §3.23].)

(2) One can also deduce an Atkin–Lehner involution for abstract classical forms of higher level at p (with primitive nebentypus characters). We do not need this generalization in this paper, so we leave it as an exercise for interested readers.

(3) For the formulas in Propositions 3.10 and 3.12, we essentially used the fact that the action of $\left(\begin{smallmatrix}p & 0\\ 0 & p\end{smallmatrix}\right)$ on $\widetilde{\mathrm{H}}$ is given by multiplication with $\xi = 1$. If $\xi \neq 1$ in general, the formulas above should be modified to

$$\mathrm{AL}_{(k,\psi'')} \circ \mathrm{AL}_{(k,\psi)} = p^{k-2}\xi^{-1}, \quad \text{and} \quad U_p \circ \mathrm{AL}_{(k,\psi)} \circ U_p = p^{k-1}\xi^{-1}\cdot \mathrm{AL}_{(k,\psi)}.$$

(4) The case when $\psi_1 \neq \psi_2$ is the usual Atkin–Lehner involution. However, what is particularly useful to us is the case when $\psi_1 = \psi_2$. While the equality (3.12.4) no longer holds, the operator $\mathrm{AL}_{(k,\psi)}$ is closely related to the p-stabilization and T_p- and U_p-operators. We refer to the sequel to this paper for a detailed discussion.

4. Basic properties of ghost series

In this section, we discuss some basic properties of the ghost series:

- We give the dimension formulas for the space of abstract classical forms (Propositions 4.1 and 4.7).
- From the dimension formulas, we deduce a formula in Proposition 4.11 describing the increment in degrees of the coefficients of the ghost series, comparing this to the halo estimate ([LWX14, Theorem 3.16]) for the modified Mahler basis. In particular, we see that the halo estimate for the modified Mahler basis is "surprisingly sharp".
- Next, we discuss the compatibility of the ghost series with theta maps, the Atkin–Lehner involutions, and the p-stabilizations (Proposition 4.18 (1)–(3)).
- At the same time, we prove the most important property of ghost series, namely the *ghost duality*, near a weight point where the space of abstract p-new forms is nonzero. We refer to Proposition 4.18 (4) for the statement.
- Finally, we show that the "old form slopes" of the ghost series at a classical weight k is $\leq \lfloor \frac{k-1}{p+1} \rfloor$ (Proposition 4.28). This is the essential condition for proving folklore conjecture of Gouvêa on maximal old form slopes from local ghost conjecture.

Some of the proofs are rather combinatorially involved and somewhat tedious; we recommend passing the technical proofs for the first reading.

Recall that we have fixed $\bar\rho = \begin{pmatrix} \omega_1^{a+1} & *\neq 0 \\ & 1 \end{pmatrix}$ with $1 \leq a \leq p-4$, and $\widetilde{\mathrm{H}}$ a *primitive* $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar\rho$ on which $\begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix}$ acts trivially. Fix a character $\varepsilon = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon}$ of Δ^2 relevant to $\bar\rho$ with $s_\varepsilon \in \{0, \dots, p-2\}$ (as in Notation 2.24 (2)); in particular, $\varepsilon_1 = e^{-s_\varepsilon}$.

We first deduce the dimension formula for $\mathrm{S}_k^{\mathrm{Iw}}(\psi)$ as a corollary of Proposition 3.7.

Proposition 4.1. *We have the following dimension formulas*

$$d_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big) = \Big\lfloor \frac{k-2-s_\varepsilon}{p-1} \Big\rfloor + \Big\lfloor \frac{k-2-\{a+s_\varepsilon\}}{p-1} \Big\rfloor + 2. \tag{4.1.1}$$

We may write this differently as

- *When $a + s_\varepsilon < p-1$,*

$$d_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big) = 2\Big\lfloor \frac{k-2-s_\varepsilon}{p-1} \Big\rfloor + 1 + \begin{cases} 1 & \text{if } \{k-2-s_\varepsilon\} \geq a \\ 0 & \text{otherwise.} \end{cases}$$

- *When $a + s_\varepsilon \geq p-1$,*

$$d_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big) = 2\Big\lfloor \frac{k-2-\{a+s_\varepsilon\}}{p-1} \Big\rfloor + 1 + \begin{cases} 1 & \text{if } \{k-2-\{a+s_\varepsilon\}\} \geq p-1-a \\ 0 & \text{otherwise.} \end{cases}$$

As a special case of either case, we have

$$d_{k+p-1}^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big) - d_k^{\mathrm{Iw}}\big(\varepsilon \cdot (1 \times \omega^{2-k})\big) = 2.$$

Asymptotically, we see that

$$d_k^{\mathrm{Iw}}\big(\varepsilon\cdot(1\times\omega^{2-k})\big)=\frac{2k}{p-1}+O(1).$$

Proof. The dimension formula (4.1.1) follows from Proposition 3.7 by a simple counting. The alternative writing also follows because

- when $a+s_\varepsilon<p-1$, $\{a+s_\varepsilon\}=a+s_\varepsilon>s_\varepsilon$; and
- when $a+s_\varepsilon\geq p-1$, $\{a+s_\varepsilon\}=a+s_\varepsilon-(p-1)<s_\varepsilon$.

The last two statements are clear from the dimension formulas. □

Notation 4.2. For ε relevant, we set

$$\delta_\varepsilon=\Big\lfloor\frac{s_\varepsilon+\{a+s_\varepsilon\}}{p-1}\Big\rfloor=\begin{cases}0 & \text{if } s_\varepsilon+\{a+s_\varepsilon\}<p-1,\\ 1 & \text{if } s_\varepsilon+\{a+s_\varepsilon\}\geq p-1.\end{cases}$$

We shall often consider the case when $k\equiv k_\varepsilon=2+\{a+2s_\varepsilon\}\bmod (p-1)$ (in which case $\varepsilon\cdot(1\times\omega^{2-k})=\tilde\varepsilon_1$). In this case, we set

$$k=k_\varepsilon+k_\bullet(p-1)\quad\text{for } k_\bullet\in\mathbb{Z}.$$

Remark 4.3. An equality frequently used later is

$$(p-1)\delta_\varepsilon+\{a+2s_\varepsilon\}=s_\varepsilon+\{a+s_\varepsilon\}.$$

Corollary 4.4. *When $k=k_\varepsilon+k_\bullet(p-1)$, we have*

$$d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)=2k_\bullet+2-2\delta_\varepsilon.$$

In particular, the dimensions $d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)$ and $d_k^{\mathrm{new}}(\varepsilon_1)$ are even integers, and asymptotically,

$$d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)=\frac{2k}{p-1}+O(1).$$

Proof. When $k\equiv k_\varepsilon\bmod (p-1)$, we have

$$\{k-2\}=\{k_\varepsilon-2\}=\{a+2s_\varepsilon\}\equiv s_\varepsilon+\{a+s_\varepsilon\}\bmod (p-1).$$

So by (4.1.1),

- when $s_\varepsilon+\{a+s_\varepsilon\}<p-1$, we have $\{k-2\}=s_\varepsilon+\{a+s_\varepsilon\}\geq\max\big(s_\varepsilon,\{a+s_\varepsilon\}\big)$, in which case $d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)=2\cdot\big\lfloor\frac{k-2}{p-1}\big\rfloor+2=2k_\bullet+2$,
- when $s_\varepsilon+\{a+s_\varepsilon\}\geq p-1$, we have $\{k-2\}=s_\varepsilon+\{a+s_\varepsilon\}-p-1<\min\big(s_\varepsilon,\{a+s_\varepsilon\}\big)$, in which case $d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)=2\cdot\big\lfloor\frac{k-2}{p-1}\big\rfloor=2k_\bullet$.

In either case, $d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)$ is an even integer; so $d_k^{\mathrm{new}}(\varepsilon_1)=d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)-2d_k^{\mathrm{ur}}(\tilde\varepsilon_1)$ is also an even integer. The asymptotic of $d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)$ is clear. □

Notation 4.5. For a (right) $\mathbb{F}$-representation σ of $\mathrm{GL}_2(\mathbb{F}_p)$, we write $\mathrm{JH}(\sigma)$ for the multiset of its Jordan–Hölder factors. For a Serre weight $\sigma_{a',b'}$ (see Notation 2.21), we write $\mathrm{Mult}_{\sigma_{a',b'}}(\sigma)$ for the multiplicity of $\sigma_{a',b'}$ in $\mathrm{JH}(\sigma)$.

4.6. **Explicit computation of $\mathrm{S}_k^{\mathrm{ur}}$.** Let $\widetilde{\mathrm{H}}$ and ε be as above. Since $\overline{\mathrm{H}}$ is a (right) projective $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$-module, we have the following sequence of equalities, for $k \equiv k_\varepsilon \bmod (p-1)$

$$\begin{aligned}(4.6.1)\qquad d_k^{\mathrm{ur}}(\varepsilon_1) &= \mathrm{rank}_{\mathcal{O}}\, \mathrm{S}_k^{\mathrm{ur}}(\varepsilon_1) = \mathrm{rank}\, \mathrm{Hom}_{\mathcal{O}[\mathrm{K}_p]}\big(\widetilde{\mathrm{H}},\, \mathrm{Sym}^{k-2}\,\mathcal{O}^{\oplus 2} \otimes \varepsilon_1 \circ \det\big)\\ &= \dim_{\mathbb{F}} \mathrm{Hom}_{\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]}\big(\overline{\mathrm{H}}, \sigma_{k-2,-s_\varepsilon}\big) = \dim_{\mathbb{F}} \mathrm{Hom}_{\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]}\big(\mathrm{Proj}_{a,s_\varepsilon}, \sigma_{k-2,0}\big)\\ &= \mathrm{Mult}_{\sigma_{a,s_\varepsilon}}(\sigma_{k-2,0}).\end{aligned}$$

The last equality used the exactness of $\mathrm{Hom}_{\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]}(\mathrm{Proj}_{a,s_\varepsilon}, -)$. Write $k = k_\varepsilon + k_\bullet(p-1)$ as in Notation 4.2. Recall the definition of δ_ε from Corollary 4.4. Using the notation above, we introduce two integers $t_1 = t_1^{(\varepsilon)}$ and $t_2 = t_2^{(\varepsilon)}$ as follows:

- when $a + s_\varepsilon < p-1$, $t_1 = s_\varepsilon + \delta_\varepsilon$ and $t_2 = a + s_\varepsilon + \delta_\varepsilon + 2$;
- when $a + s_\varepsilon \geq p-1$, $t_1 = \{a + s_\varepsilon\} + \delta_\varepsilon + 1$ and $t_2 = s_\varepsilon + \delta_\varepsilon + 1$.

Proposition 4.7. *We have*

$$\begin{aligned}(4.7.1)\qquad d_k^{\mathrm{ur}}(\varepsilon_1) &= \Big\lfloor \frac{k_\bullet - t_1}{p+1} \Big\rfloor + \Big\lfloor \frac{k_\bullet - t_2}{p+1} \Big\rfloor + 2\\ &= 2\Big\lfloor \frac{k_\bullet - t_1}{p+1} \Big\rfloor + 1 + \begin{cases} 1 & \text{when } k_\bullet - (p+1)\lfloor \frac{k_\bullet - t_1}{p+1} \rfloor \geq t_2\\ 0 & \text{otherwise.}\end{cases}\end{aligned}$$

In particular, $d_k^{\mathrm{ur}}(\varepsilon_1)$ is non-decreasing in the class $k \equiv k_\varepsilon \bmod (p-1)$, and we have

$$d_{k+p^2-1}^{\mathrm{ur}}(\varepsilon_1) - d_k^{\mathrm{ur}}(\varepsilon_1) = 2.$$

Asymptotically, $d_k^{\mathrm{ur}}(\varepsilon_1) = \frac{2k}{p^2-1} + O(1)$ and $d_k^{\mathrm{new}}(\varepsilon_1) = \frac{2k}{p+1} + O(1)$.

Proof. It is enough to compute the Jordan–Hölder factors of $\sigma_{k-2,0}$. By Lemmas A.6 and A.4 (1) (and the congruence $k_\varepsilon - 2 \equiv a + 2s_\varepsilon \bmod (p-1)$), we have the following equality in the Grothendieck group $\mathrm{Groth}(\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)])$ of (right) $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$-modules:

$$[\sigma_{k-2,0}] - [\sigma_{k-2-(p+1),1}] = [\sigma_{\{a+2s_\varepsilon\},0}] + [\sigma_{p-1-\{a+2s_\varepsilon\},\{a+2s_\varepsilon\}}].$$

If we replace $k-2$ by $k-2-(p+1)$ in the above equality and then twist by the character det, we deduce

$$[\sigma_{k-2-(p+1),1}] - [\sigma_{k-2-2(p+1),2}] = [\sigma_{\{a+2s_\varepsilon-2\},1}] + [\sigma_{p-1-\{a+2s_\varepsilon-2\},\{a+2s_\varepsilon\}-1}].$$

Now assume that $k-2 \geq (p+1)(p-1)$, then we can repeat the above computation $p-1$ times to get $p-1$ equalities:

$$\begin{aligned}(4.7.2)\qquad [\sigma_{k-2,0}] - [\sigma_{k-2-(p+1),1}] &= [\sigma_{\{a+2s_\varepsilon\},0}] + [\sigma_{p-1-\{a+2s_\varepsilon\},\{a+2s_\varepsilon\}}]\\ [\sigma_{k-2-(p+1),1}] - [\sigma_{k-2-2(p+1),2}] &= [\sigma_{\{a+2s_\varepsilon-2\},1}] + [\sigma_{p-1-\{a+2s_\varepsilon-2\},\{a+2s_\varepsilon\}-1}].\\ &\cdots\cdots\cdots\\ [\sigma_{k-2-(p-2)(p+1),p-2}] - [\sigma_{k-2-(p-1)(p+1),0}] &= [\sigma_{\{a+2s_\varepsilon+2\},p-2}] + [\sigma_{p-1-\{a+2s_\varepsilon+2\},\{a+2s_\varepsilon\}-(p-2)}].\end{aligned}$$

The sum of the left hand sides of (4.7.2) is equal to $[\sigma_{k-2,0}] - [\sigma_{k-2-(p-1)(p+1),0}]$. On the right sides of (4.7.2), the Serre weight σ_{a,s_ε} appears exactly twice: one time, it appears in the first summand of the $(s_\varepsilon+1)$th equality, and at the other time, it appears in the second summand of the $(\{a+s_\varepsilon\}+1)$th equality. So combining this with (4.6.1), we deduce that

$$d_k^{\mathrm{ur}}(\varepsilon_1) = d_{k-(p-1)(p+1)}^{\mathrm{ur}}(\varepsilon_1) + 2.$$

It then remains to prove (4.7.1) when $k-2<p^2-1$ (or equivalently $k_\bullet<p+1$).

In this case, we write $k-2=s(p+1)+r$ with $s\in\{0,\dots,p-2\}$ and $r\in\{0,\dots,p\}$. We can then repeat the argument above but only using the first s rows of (4.7.2) to deduce that $[\sigma_{k-2,0}]-[\sigma_{r,s}]$ is equal to the sum of the right hand sides of first s rows of (4.7.2).

(1) The "remainder" term $[\sigma_{r,s}]$ contains σ_{a,s_ε} as a Jordan–Hölder factor if and only if $s=s_\varepsilon$ and $a\equiv r \bmod (p-1)$ (note that the latter includes the case that $a=1$ and $r=p$). We immediately remark that when $s=s_\varepsilon$, since $\{a+2s_\varepsilon\}+k_\bullet(p-1)=s_\varepsilon(p+1)+r$, the condition $a\equiv r\bmod (p-1)$ is automatic.

(2) If $s\geq s_\varepsilon+1$, the Serre weight σ_{a,s_ε} will appear in the first summand of the right hand side of (the $(s_\varepsilon+1)$th row) of (4.7.2). This will contribute a σ_{a,s_ε} in the Jordan–Hölder factors of $[\sigma_{k-2,0}]$.

(3) If $s\geq\{a+s_\varepsilon\}+1$, the Serre weight σ_{a,s_ε} will appear in the second summand of the right hand side of (the $(\{a+s_\varepsilon\}+1)$th row) of (4.7.2). This will contribute a σ_{a,s_ε} in the Jordan–Hölder factors of $[\sigma_{k-2,0}]$.

Combining (1) and (2), precisely when $s\geq s_\varepsilon$, (1) and (2) will contribute one σ_{a,s_ε} to the Jordan–Hölder factor of $[\sigma_{k-2,0}]$. Note that $\{a+2s_\varepsilon\}+k_\bullet(p-1)\equiv s_\varepsilon(p+1)+a\equiv a+2s_\varepsilon \bmod (p-1)$. Since $\lfloor\frac{\{a+2s_\varepsilon\}+k_\bullet(p-1)}{p+1}\rfloor=\lfloor\frac{k-2}{p+1}\rfloor=s$ and $\lfloor\frac{s_\varepsilon(p+1)+a}{p+1}\rfloor=s_\varepsilon>\lfloor\frac{s_\varepsilon(p+1)+a-(p-1)}{p+1}\rfloor$, it is straightforward to see that $s\geq s_\varepsilon$ if and only if the following equivalent conditions hold:

$$\{a+2s_\varepsilon\}+k_\bullet(p-1)\geq s_\varepsilon(p+1)+a \quad\Longleftrightarrow\quad k_\bullet\geq s_\varepsilon+\frac{a+2s_\varepsilon-\{a+2s_\varepsilon\}}{p-1}. \tag{4.7.3}$$

On the other hand, the condition $s\geq\{a+s_\varepsilon\}+1$ of (3) is equivalent to

$$\{a+2s_\varepsilon\}+k_\bullet(p-1)\geq(p+1)\big(\{a+s_\varepsilon\}+1\big),$$

which is further equivalent to

$$k_\bullet\geq\Big\lceil\frac{(p+1)\{a+s_\varepsilon\}-\{a+2s_\varepsilon\}+p+1}{p-1}\Big\rceil=\{a+s_\varepsilon\}+2+\frac{2\{a+s_\varepsilon\}-\{a+2s_\varepsilon\}-a}{p-1}. \tag{4.7.4}$$

Now, when $a+s_\varepsilon<p-1$, (4.7.3) is equivalent to $k_\bullet\geq s_\varepsilon+\delta_\varepsilon=t_1$ and (4.7.4) is equivalent to $k_\bullet\geq a+s_\varepsilon+2+\delta_\varepsilon=t_2$. So (4.7.1) follows. When $a+s_\varepsilon\geq p-1$, (4.7.3) is equivalent to $k_\bullet\geq s_\varepsilon+\delta_\varepsilon+1=t_2$ and (4.7.4) is equivalent to $k_\bullet\geq\{a+s_\varepsilon\}+1+\delta_\varepsilon=t_1$. So (4.7.1) follows. ☐

Corollary 4.8. *For ε a relevant character for $\bar\rho$ and $\widetilde{\mathrm{H}}$ a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module, the dimensions $d_k^{\mathrm{Iw}}(\tilde\varepsilon_1)$ and $d_k^{\mathrm{ur}}(\varepsilon_1)$ for every $k\in\mathbb{N}_{\geq 2}$ depend only on $\bar\rho$, k, and ε, but not on (the $\mathrm{GL}_2(\mathbb{Q}_p)$-action on) $\widetilde{\mathrm{H}}$.*

Moreover, for a fixed n, the ghost multiplicity $m_n^{(\varepsilon)}(k)$ in Definition 2.25 is zero when $d_k^{\mathrm{ur}}(\varepsilon_1)\geq n$ (which is certainly the case when $k/(p^2-1)>n/2+2$). So the coefficients of the ghost series $g_n^{(\varepsilon)}(w)$ are polynomials in w.

Proof. This is clear from Propositions 4.1 and 4.7. ☐

Remark 4.9. (1) If $\widetilde{\mathrm{H}}'$ is an $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar\rho$ with multiplicity $m(\widetilde{\mathrm{H}}')$, we note that as $\mathcal{O}[\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$-modules, $\widetilde{\mathrm{H}}'\simeq\widetilde{\mathrm{H}}^{\oplus m(\widetilde{\mathrm{H}}')}$. Yet formulas (4.1.1) and (4.6.1) only makes use of the $\mathcal{O}[\![\mathrm{GL}_2(\mathbb{Z}_p)]\!]$-module structure of $\widetilde{\mathrm{H}}$. It then follows that, for $k \in \mathbb{N}_{\geq 2}$, a character ψ of Δ^2, and a character ε_1 of Δ,

$$\operatorname{rank}_{\mathcal{O}} \mathrm{S}^{\mathrm{Iw}}_{\widetilde{\mathrm{H}}',k}(\psi) = m(\widetilde{\mathrm{H}}')d_k^{\mathrm{Iw}}(\psi) \quad \text{and} \quad \operatorname{rank}_{\mathcal{O}} \mathrm{S}^{\mathrm{ur}}_{\widetilde{\mathrm{H}}',k}(\varepsilon_1) = m(\widetilde{\mathrm{H}}')d_k^{\mathrm{ur}}(\varepsilon_1).$$

So these ranks also only depend on $\bar{\rho}$, ψ, ε_1, k, and (linearly on) $m(\widetilde{\mathrm{H}}')$.

(2) The fact that $d_k^{\mathrm{ur}}(\varepsilon_1)$ is increasing in k relies on the assumption that the parameter a is generic. The analogous statement for non-generic a is false (as in the original examples considered by Bergdall–Pollack [BP19a]).

Now, we move to discuss the slopes when $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$ has valuation $v_p(w_\star) \in (0,1)$, the so-called *spectral halo range*. Note that for every $k \in \mathbb{Z}$, $v_p(w_k) = v_p(\exp(p(k-2)) - 1) \geq 1$; so $v_p(w_\star - w_k) = v_p(w_\star)$ in this case, and thus, for the nth term of the ghost series

$$v_p(g_n^{(\varepsilon)}(w_\star)) = v_p(w_\star) \cdot \deg g_n^{(\varepsilon)}(w).$$

In particular, this valuation is linear in $v_p(w_\star)$. This is a direct analogue of Coleman–Mazur–Buzzard–Kilford's spectral halo conjecture, as partially proved in [LWX14]. The key to that proof is to provide a lower bound on the Newton polygon of the characteristic power series, called the *Hodge polygon* in that paper. Formally translating the construction from that paper, we expect that when $v_p(w_\star) \in (0,1)$, the Newton polygon of the ghost series should lie above the polygon, which we call *ghost Hodge polygon*, whose nth slope is given by $v_p(w_\star) \cdot \lambda_n^{(\varepsilon)}$ with

$$\lambda_n^{(\varepsilon)}(w) = \deg \mathbf{e}_n^{(\varepsilon)} - \left\lfloor \tfrac{\deg \mathbf{e}_n^{(\varepsilon)}}{p} \right\rfloor.$$

It turns out that $\mathrm{NP}(C(w_\star, -))$ is "very close" to this ghost Hodge polygon, as proved in Proposition 4.11 below.

Notation 4.10. For $n \in \mathbb{Z}$ and ε a relevant character, we set $\beta_{[n]}^{(\varepsilon)} = \begin{cases} t_1^{(\varepsilon)} & \text{if } n \text{ is even} \\ t_2^{(\varepsilon)} - \frac{p+1}{2} & \text{if } n \text{ is odd} \end{cases}$.

Since this depends only on the parity of n, we often write $\beta_{\mathrm{even}}^{(\varepsilon)}$ and $\beta_{\mathrm{odd}}^{(\varepsilon)}$ for the values.

Proposition 4.11. *Fix ε a relevant character. If $a + s_\varepsilon < p-1$, then we have*

$$\deg g_{n+1}^{(\varepsilon)}(w) - \deg g_n^{(\varepsilon)}(w) - \lambda_{n+1}^{(\varepsilon)} = \begin{cases} 1 & \text{if } n - 2s_\varepsilon \equiv 1, 3, \dots, 2a+1 \bmod 2p, \\ -1 & \text{if } n - 2s_\varepsilon \equiv 2, 4, \dots, 2a+2 \bmod 2p, \\ 0 & \text{otherwise.} \end{cases} \tag{4.11.1}$$

If $a + s_\varepsilon \geq p-1$, then we have

$$\deg g_{n+1}^{(\varepsilon)}(w) - \deg g_n^{(\varepsilon)}(w) - \lambda_{n+1}^{(\varepsilon)} = \begin{cases} 1 & \text{if } n - 2s_\varepsilon \equiv 2, 4, \dots, 2a+2 \bmod 2p, \\ -1 & \text{if } n - 2s_\varepsilon \equiv 3, 5, \dots, 2a+3 \bmod 2p, \\ 0 & \text{otherwise.} \end{cases} \tag{4.11.2}$$

In either case, we have

$$\deg g_n^{(\varepsilon)}(w) - (\lambda_1^{(\varepsilon)} + \cdots + \lambda_n^{(\varepsilon)}) = \begin{cases} 0 & \text{if } \deg \mathbf{e}_{n+1} - \deg \mathbf{e}_n = a, \\ 0 \text{ or } 1 & \text{if } \deg \mathbf{e}_{n+1} - \deg \mathbf{e}_n = p-1-a. \end{cases} \tag{4.11.3}$$

Moreover, the differences $\deg g_{n+1}^{(\varepsilon)}(w) - \deg g_n^{(\varepsilon)}(w)$ are strictly increasing in n.

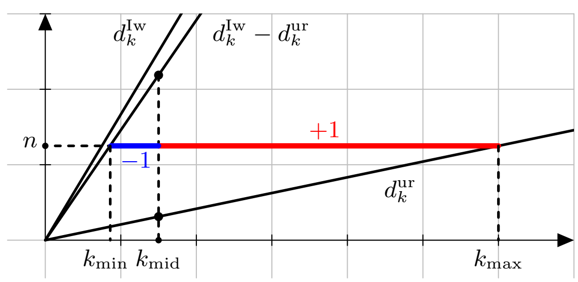
Proof. In this proof, we shall suppress the ε from the notation when no confusion arises. Since $\deg g_n(w)$ is the sum of ghost multiplicities $m_n(k)$ for all $k \equiv k_\varepsilon \bmod (p-1)$, the formula (2.25.3) for the increments of the ghost multiplicities implies that

$$\deg g_{n+1}(w) - \deg g_n(w) = \sum_{\substack{k \equiv k_\varepsilon \bmod (p-1)\\ d_k^{\mathrm{ur}} \leq n < \frac{1}{2} d_k^{\mathrm{Iw}}}} 1 \quad - \sum_{\substack{k \equiv k_\varepsilon \bmod (p-1)\\ \frac{1}{2} d_k^{\mathrm{Iw}} \leq n < d_k^{\mathrm{Iw}} - d_k^{\mathrm{ur}}}} 1. \tag{4.11.4}$$

Note that d_k^{ur}, d_k^{Iw}, and $d_k^{\mathrm{Iw}} - d_k^{\mathrm{ur}}$ are increasing functions in k within the congruence class (for the last one, as k is increased by $p-1$, d_k^{Iw} is increased by 2 and d_k^{ur} is increased by at most 1).

The following picture illustrates the contribution to the degree increment $\deg g_{n+1}(w) - \deg g_n(w)$. The contribution of the blue interval from k_{min} to k_{mid} is negative and the contribution of the red interval from k_{mid} to k_{max} is positive. More precisely, we define these three integers as

$$\begin{aligned} k_{\mathrm{min}} = k_{\mathrm{min}}(n) &:= \min\{k \mid k \equiv k_\varepsilon \bmod (p-1) \text{ and } d_k^{\mathrm{Iw}} - d_k^{\mathrm{ur}} > n\} \\ k_{\mathrm{mid}} = k_{\mathrm{mid}}(n) &:= \min\{k \mid k \equiv k_\varepsilon \bmod (p-1) \text{ and } \tfrac{1}{2} d_k^{\mathrm{Iw}} \leq n\} \\ k_{\mathrm{max}} = k_{\mathrm{max}}(n) &:= \min\{k \mid k \equiv k_\varepsilon \bmod (p-1) \text{ and } d_k^{\mathrm{ur}} \leq n\} \end{aligned} \tag{4.11.5}$$



Graph 2: contribution to increments of degrees.

It remains to express $k_{\mathrm{mid}\bullet}$, $k_{\mathrm{max}\bullet}$, and $k_{\mathrm{min}\bullet}$ in terms of n. Since this result will be used later, we state (a more refined version of) it separately here.

Lemma-Notation 4.12. *We only consider those integers $k \equiv k_\varepsilon \bmod (p-1)$ and write $k = k_\varepsilon + (p-1)k_\bullet$. Fix $n \in \mathbb{Z}_{\geq 0}$. Recall the definition of δ_ε in Notation 4.2, $t_1^{(\varepsilon)}$ and $t_2^{(\varepsilon)}$ from Proposition 4.7, and $\beta_{[n]}^{(\varepsilon)}$ from Notation 4.10*

(1) *There is a unique integer $k_\bullet$ such that $n = \frac{1}{2} d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1)$; it is*

$$k_\bullet = k_{\mathrm{mid}\bullet}^{(\varepsilon)}(n) := n + \delta_\varepsilon - 1.$$

(2) *If $k_\bullet \in \mathbb{Z}_{\geq 0}$ satisfies $n = d_k^{\mathrm{ur}}(\varepsilon_1)$, then*

$$\begin{aligned} &\textit{when } n \textit{ is even,} \quad (p+1)\tfrac{n-2}{2} + t_2^{(\varepsilon)} \leq k_\bullet < (p+1)\tfrac{n}{2} + t_1^{(\varepsilon)}, \\ &\textit{when } n \textit{ is odd,} \quad (p+1)\tfrac{n-1}{2} + t_1^{(\varepsilon)} \leq k_\bullet < (p+1)\tfrac{n-1}{2} + t_2^{(\varepsilon)}. \end{aligned}$$

We denote the largest such $k_\bullet$ by

$$k_{\mathrm{max}\bullet}^{(\varepsilon)}(n) = \frac{p+1}{2} n + \beta_{[n]}^{(\varepsilon)} - 1.$$

(3) *There is at most one $k_\bullet \in \mathbb{Z}_{\geq 0}$ such that $n = d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1)$. Such $k_\bullet$ exists if and only if the interval*

$$\big((p+1)(\tfrac{n-2}{2}+\delta_\varepsilon)-t_1^{(\varepsilon)},\ (p+1)(\tfrac{n}{2}+\delta_\varepsilon)-t_2^{(\varepsilon)}\big] \text{ if } n \text{ is even, and} \tag{4.12.1}$$

$$\big((p+1)(\tfrac{n-1}{2}+\delta_\varepsilon)-t_2^{(\varepsilon)},\ (p+1)(\tfrac{n-1}{2}+\delta_\varepsilon)-t_1^{(\varepsilon)}\big] \text{ if } n \text{ is odd}$$

contains a multiple of p, which would be $pk_\bullet$. We write

$$\tilde{k}_{\min\bullet}^{(\varepsilon)}(n) = \frac{p+1}{2}(n-1+2\delta_\varepsilon) - \beta_{[n-1]}^{(\varepsilon)} + 1$$

for the right end *of the above interval plus 1; and set $k_{\min\bullet}^{(\varepsilon)}(n) = \lceil \tilde{k}_{\min\bullet}^{(\varepsilon)}(n)/p \rceil$.*

Warning: despite the name, $k_{\min\bullet}^{(\varepsilon)}(n)$ is never a $k_\bullet$ such that $n = d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1)$.

Finally, for $? =$ mid, max, and min, we put $k_?^{(\varepsilon)}(n) = k_\varepsilon + (p-1)k_{?\bullet}^{(\varepsilon)}(n)$. We set

$$\tilde{k}_{\min}^{(\varepsilon)}(n) = pk_\varepsilon + (p-1)\tilde{k}_{\min\bullet}^{(\varepsilon)}(n).$$

Remark 4.13. From the definition of $k_{\mathrm{mid}\bullet}^{(\varepsilon)}(n)$, $k_{\min\bullet}^{(\varepsilon)}(n)$, and $k_{\max\bullet}^{(\varepsilon)}(n)$, we see that

- $\frac{1}{2}d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) = n \Leftrightarrow k_\bullet = k_{\mathrm{mid}\bullet}^{(\varepsilon)}(n)$.
- $d_k^{\mathrm{ur}}(\varepsilon_1) \leq n \Leftrightarrow k_\bullet \leq k_{\max\bullet}^{(\varepsilon)}(n)$ and $d_k^{\mathrm{ur}}(\varepsilon_1) \geq n \Leftrightarrow k_\bullet > k_{\max\bullet}^{(\varepsilon)}(n-1)$.
- $d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1) \leq n \Leftrightarrow k_\bullet < k_{\min\bullet}^{(\varepsilon)}(n)$ and $d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1) \geq n \Leftrightarrow k_\bullet \geq k_{\min\bullet}^{(\varepsilon)}(n-1)$.

In particular, we see that the numbers defined in Lemma-Notation 4.12 coincide with those defined in (4.11.5).

Proof. (1) This follows from the dimension formula in Corollary 4.4: $d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) = 2k_\bullet + 2 - 2\delta_\varepsilon$.

(2) This follows from inverting the dimension formula in Proposition 4.7. (There is no problem with the statement when $n = 0$, as we asked for $k_\bullet \geq 0$.)

(3) By Corollary 4.4, $d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) = 2k_\bullet + 2 - 2\delta_\varepsilon$. So the equality $n = d_k^{\mathrm{Iw}}(\tilde{\varepsilon}_1) - d_k^{\mathrm{ur}}(\varepsilon_1)$ is equivalent to

$$d_k^{\mathrm{ur}}(\varepsilon_1) \ = (2k_\bullet + 2 - 2\delta_\varepsilon) - n.$$

When n is even, this means that

$$(p+1)\tfrac{(2k_\bullet+2-2\delta_\varepsilon)-n-2}{2} + t_2 \leq k_\bullet < (p+1)\tfrac{(2k_\bullet+2-2\delta_\varepsilon)-n}{2} + t_1,$$

or equivalently

$$(p+1)(\tfrac{n-2}{2}+\delta_\varepsilon) - t_1 < pk_\bullet \leq (p+1)(\tfrac{n}{2}+\delta_\varepsilon) - t_2.$$

The case when n is odd follows from the same argument. □

Now, we continue the proof of Proposition 4.11. By (4.11.4), we write

$$\deg g_{n+1}(w) - \deg g_n(w) = \big(k_{\max\bullet}(n) - k_{\mathrm{mid}\bullet}(n)\big) - \big(k_{\mathrm{mid}\bullet}(n) - k_{\min\bullet}(n) + 1\big).$$

We shall compare $k_{\max\bullet}(n) - k_{\mathrm{mid}\bullet}(n)$ with $\deg \mathbf{e}_{n+1}$ and $k_{\mathrm{mid}\bullet}(n) - k_{\min\bullet}(n)$ with $\lfloor \frac{\deg \mathbf{e}_{n+1}}{p} \rfloor$.

By Lemma-Notation 4.12 (1)(2), $k_{\max\bullet}(n) - k_{\mathrm{mid}\bullet}(n) = \frac{p-1}{2}\cdot n + \beta_{[n]} - \delta_\varepsilon$, we list the values of $k_{\max\bullet}(n) - k_{\mathrm{mid}\bullet}(n)$ and $\deg \mathbf{e}_{n+1}$ in the following table:

		$k_{\max\bullet}(n)-k_{\mathrm{mid}\bullet}(n)$	$\deg \mathbf{e}_{n+1}$
$a+s_\varepsilon<p-1$	$n=2n_0$	$(p-1)n_0+s_\varepsilon$	$s_\varepsilon+(p-1)n_0$
	$n=2n_0+1$	$(p-1)n_0+a+s_\varepsilon+1$	$a+s_\varepsilon+(p-1)n_0$
$a+s_\varepsilon\geq p-1$	$n=2n_0$	$(p-1)n_0+\{a+s_\varepsilon\}+1$	$\{a+s_\varepsilon\}+(p-1)n_0$
	$n=2n_0+1$	$(p-1)n_0+s_\varepsilon$	$s_\varepsilon+(p-1)n_0$

In other words, by the explicit description of power basis (3.7.1),

$$k_{\max\bullet}(n)-k_{\mathrm{mid}\bullet}(n)-\deg \mathbf{e}_{n+1}=\begin{cases}0 & \text{if } \mathbf{e}_{n+1}=e_1^*z^{s_\varepsilon+(p-1)\lfloor n/2\rfloor},\\ 1 & \text{if } \mathbf{e}_{n+1}=e_2^*z^{\{a+s_\varepsilon\}+(p-1)\lfloor n/2\rfloor}.\end{cases} \tag{4.13.1}$$

By Lemma-Notation 4.12 (1)(3), we have

$$k_{\mathrm{mid}\bullet}(n)-k_{\min\bullet}(n)+1=n-\Big\lceil\frac{\frac{p+1}{2}(n-1)+\delta_\varepsilon-\beta_{[n-1]}+1}{p}\Big\rceil.$$

We list the values of $k_{\mathrm{mid}\bullet}(n)-k_{\min\bullet}(n)+1$ and $\lfloor\frac{\deg \mathbf{e}_{n+1}}{p}\rfloor$ in the following table:

		$k_{\mathrm{mid}\bullet}(n)-k_{\min\bullet}(n)+1$	$\lfloor\frac{\deg \mathbf{e}_{n+1}}{p}\rfloor$
$a+s_\varepsilon<p-1$	$n=2n_0$	$n_0-\lceil\frac{n_0-a-s_\varepsilon-1}{p}\rceil$	$n_0-\lceil\frac{n_0-s_\varepsilon}{p}\rceil$
	$n=2n_0+1$	$n_0+1-\lceil\frac{n_0-s_\varepsilon+1}{p}\rceil$	$n_0-\lceil\frac{n_0-a-s_\varepsilon}{p}\rceil$
$a+s_\varepsilon\geq p-1$	$n=2n_0$	$n_0-\lceil\frac{n_0-s_\varepsilon}{p}\rceil$	$n_0-\lceil\frac{n_0-\{a+s_\varepsilon\}}{p}\rceil$
	$n=2n_0+1$	$n_0+1-\lceil\frac{n_0-\{a+s_\varepsilon\}}{p}\rceil$	$n_0-\lceil\frac{n_0-s_\varepsilon}{p}\rceil$

Here we used the obvious formula $\lfloor\frac{(p-1)a+b}{p}\rfloor=a-\lceil\frac{a-b}{p}\rceil$. The equalities (4.11.1) and (4.11.2) follow from this by a case-by-case argument. We only spell out the case when $a+s_\varepsilon<p-1$ (that is (4.11.1)) and the other case is similar.

When $a+s_\varepsilon<p-1$,

$$k_{\mathrm{mid}\bullet}(n)-k_{\min\bullet}(n)+1-\Big\lfloor\frac{\deg \mathbf{e}_{n+1}}{p}\Big\rfloor=\begin{cases}0 & \text{if } n=2n_0 \text{ and } n_0-s_\varepsilon\equiv a+2,\dots,p \bmod p\\ & \text{or } n=2n_0+1 \text{ and } n_0-s_\varepsilon\equiv 0,\dots,a \bmod p\\ 1 & \text{otherwise}\end{cases}$$

Combining this with (4.13.1), we deduce that

$$\deg g_{n+1}(w)-\deg g_n(w)-\lambda_{n+1}=\begin{cases}1 & \text{if } n-2s_\varepsilon\equiv 1,3,\dots,2a+1 \bmod 2p,\\ -1 & \text{if } n-2s_\varepsilon\equiv 2,4,\dots,2a+2 \bmod 2p\\ 0 & \text{otherwise.}\end{cases}$$

The statement (4.11.3) follows from (4.11.1) and (4.11.2) immediately, by observing that the difference being 1 only appears in the case when $\deg \mathbf{e}_{n+1}-\deg \mathbf{e}_n=p-1-a$.

To see the last statement, we note that, as shown in the table above

$$\big(k_{\max\bullet}(n+1)-k_{\mathrm{mid}\bullet}(n+1)\big)-\big(k_{\max\bullet}(n)-k_{\mathrm{mid}\bullet}(n)\big)$$

is always equal to either $a+1$ or $p-2-a$, which is ≥ 2 under the genericity condition $1\leq a\leq p-4$. On the other hand, the next table shows that

$$\Big|\big(k_{\mathrm{mid}\bullet}(n+1)-k_{\min\bullet}(n+1)\big)-\big(k_{\mathrm{mid}\bullet}(n)-k_{\min\bullet}(n)\big)\Big|\leq 1.$$

Combining these two, we deduce that the increments of degrees $\deg g_{n+1}(w) - \deg g_n(w)$ is always strictly increasing. $\square$

Corollary 4.14 (Ghost conjecture versus spectral halo conjecture)**.** *Let ε be a relevant character, then for $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$ with $v_p(w_\star) \in (0,1)$, the nth slope of $\mathrm{NP}(G(w_\star, -))$ is*

$$v_p(w_\star) \cdot (\deg g_{n+1}^{(\varepsilon)} - \deg g_n^{(\varepsilon)})$$

Moreover, these slopes are strictly increasing with respect to n.

Example 4.15. In the setup of Example 2.26, the following table indicates the difference between $\deg g_{n+1}^{(\varepsilon)} - \deg g_n^{(\varepsilon)}$ and the halo Hodge slope $\lambda_{n+1} := \deg \mathbf{e}_{n+1}^{(\varepsilon)} - \lfloor \deg \mathbf{e}_{n+1}^{(\varepsilon)}/p \rfloor$, on two of the disks.

	$\varepsilon = 1 \times \omega^2$ disk																
$\deg g_{n+1}^{(\varepsilon)} - \deg g_n^{(\varepsilon)}$	0	3	5	8	10	13	15	18	21	23	26	28	31	33	36	39	...
$\deg \mathbf{e}_{n+1}^{(\varepsilon)}$	0	2	6	8	12	14	18	20	24	26	30	32	36	38	42	44	...
$\lambda_{n+1}^{(\varepsilon)}$	0	2	6	7	11	12	16	18	21	23	26	28	31	33	36	38	...
	$\varepsilon = \omega^2 \times 1$ disk																
$\deg g_{n+1}^{(\varepsilon)} - \deg g_n^{(\varepsilon)}$	1	3	6	9	11	14	16	19	21	24	27	29	32	34	37	39	...
$\deg \mathbf{e}_{n+1}^{(\varepsilon)}$	0	4	6	10	12	16	18	22	24	28	30	34	36	40	42	46	...
$\lambda_{n+1}^{(\varepsilon)}$	0	4	6	9	11	14	16	19	21	24	26	30	31	35	36	40	...

Remark 4.16. As indicated by this Corollary, the halo bound is surprising sharp "after we distributed it over to each weight disk". It seems to be possible to give a much more robust proof of the halo conjecture similar to [LWX14]. We will return to this topic in a future work.

The next part of this section is devoted to checking that the slopes of the ghost series are compatible with the theta maps, the Atkin–Lehner involutions, and p-stabilizations (which will be elaborated in the forthcoming paper; but we do not need them logically here). In other words,

- the existence of the theta maps predicts that the slopes of abstract overconvergent but non-classical forms of weight k_0 is $k_0 - 1$ plus the slopes of abstract overconvergent forms of weight $2 - k_0$,
- the Atkin–Lehner involution predicts that the slopes of abstract classical forms of weight k_0 with "opposite characters" can be paired so that the sum is equal to $k_0 - 1$, and
- the p-stabilization process predicts that the slopes of abstract p-old forms of weight k_0 can be paired so that the sum is equal to $k_0 - 1$.

Below, we show that the ghost series satisfies the analogous properties. In addition, we will prove a very important yet mysterious property of the ghost series, which we call the *ghost duality*. Roughly speaking, for a fixed weight $k_0 \equiv k_\varepsilon \bmod (p-1)$, not only the slopes of the abstract p-new forms are equal to $\frac{k_0-2}{2}$, but certain slopes computed using appropriate derivatives in w of the ghost series have slopes $\frac{k_0-2}{2}$. This was first observed by Bergdall and Pollack in their computational data, and was communicated to us. As in many other places, we thank them for sharing their ideas on this project.

Notation 4.17. For $k \equiv k_\varepsilon \bmod (p-1)$, we write

$$g_{n,\hat{k}}^{(\varepsilon)}(w) := g_n^{(\varepsilon)}(w)/(w-w_k)^{m_n^{(\varepsilon)}(k)}.$$

In particular, $g_{n,\hat{k}}^{(\varepsilon)}(w_k)$ is the leading coefficient of the Taylor expansion of $g_n^{(\varepsilon)}(w)$ at $w = w_k$.

More generally, if $\mathbf{k} = \{k_1, \dots, k_r\}$ is a set of ghost zeros with each $k_i \equiv k_\varepsilon \bmod (p-1)$, we put

$$g_{n,\hat{\mathbf{k}}}^{(\varepsilon)}(w) := g_n^{(\varepsilon)}(w)\Big/\prod_{i=1}^{r}(w-w_{k_i})^{m_n^{(\varepsilon)}(k_i)}.$$

We point out a formula that we will use frequently in this section and later:

$$v_p(w_{k_1}-w_{k_2}) = v_p\big(\exp(p(k_2-2))\cdot(\exp(p(k_1-k_2))-1)\big) = 1+v_p(k_1-k_2). \tag{4.17.1}$$

Proposition 4.18. *Fix $k_0 \geq 2$ and a character $\varepsilon = \omega^{-s_\varepsilon} \times \omega^{a+s_\varepsilon}$ of Δ^2 relevant to $\bar{\rho}$ as before. Write $d := d_{k_0}^{\mathrm{Iw}}(\varepsilon\cdot(1\times\omega^{2-k_0}))$ in this theorem.*

1. *(Compatibility with theta maps) Put $\varepsilon' := \varepsilon\cdot(\omega^{k_0-1}\times\omega^{1-k_0})$ with $s_{\varepsilon'} = \{s_\varepsilon+1-k_0\}$. For every $\ell \geq 1$, the $(d+\ell)$th slope of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ is k_0-1 plus the ℓth slope of $\mathrm{NP}(G^{(\varepsilon')}(w_{k_0},-))$. In particular, the $(d+\ell)$th slope of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ is at least k_0-1.*
2. *(Compatibility with Atkin–Lehner involutions) Assume that $k_0 \not\equiv k_\varepsilon \bmod (p-1)$. Put $\varepsilon'' = \omega^{-s_{\varepsilon''}}\times\omega^{a+s_{\varepsilon''}}$ with $s_{\varepsilon''} := \{k_0-2-a-s_\varepsilon\}$. Then for every $\ell \in \{1,\dots,d\}$, the sum of the ℓth slope of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ and the $(d-\ell+1)$th slope of $\mathrm{NP}(G^{(\varepsilon'')}(w_{k_0},-))$ is exactly k_0-1. In particular, the ℓth slope of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ is at most k_0-1.*
3. *(Compatibility with p-stabilizations) Assume that $k_0 \equiv k_\varepsilon \bmod (p-1)$. Then for every $\ell \in \{1,\dots,d_{k_0}^{\mathrm{ur}}(\varepsilon_1)\}$, the sum of the ℓth slope of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ and the $(d-\ell+1)$th slope of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ is exactly k_0-1. In particular, the ℓth slope of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ is at most k_0-1.*
4. *(Ghost duality) Assume that $k_0 \equiv k_\varepsilon \bmod (p-1)$. Write $d_{k_0}^{\mathrm{ur}}$, $d_{k_0}^{\mathrm{Iw}}$, and $d_{k_0}^{\mathrm{new}}$ for $d_{k_0}^{\mathrm{ur}}(\varepsilon_1)$, $d_{k_0}^{\mathrm{Iw}}(\tilde{\varepsilon}_1)$, and $d_{k_0}^{\mathrm{new}}(\varepsilon_1)$, respectively. Then for each $\ell = 0,\dots,\frac{1}{2}d_{k_0}^{\mathrm{new}}-1$,*

$$v_p\big(g^{(\varepsilon)}_{d_{k_0}^{\mathrm{Iw}}-d_{k_0}^{\mathrm{ur}}-\ell,\hat{k}_0}(w_{k_0})\big) - v_p\big(g^{(\varepsilon)}_{d_{k_0}^{\mathrm{ur}}+\ell,\hat{k}_0}(w_{k_0})\big) = (k_0-2)\cdot(\tfrac{1}{2}d_{k_0}^{\mathrm{new}}-\ell). \tag{4.18.1}$$

In particular, the $(d_{k_0}^{\mathrm{ur}}+1)$th to the $(d_{k_0}^{\mathrm{Iw}}-d_{k_0}^{\mathrm{ur}})$th slopes of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0},-))$ are all equal to $\frac{k_0-2}{2}$.

For an alternative explanation of name "ghost duality", we refer to Notation 5.1 later.

Proof of Proposition 4.18. We first list the concrete statements on $g_n^{(\varepsilon)}(w)$ that we shall prove in order to deduce the Theorem:

(a) In the setup of (1), we will prove in § 4.24 that

$$v_p(g^{(\varepsilon)}_{d+\ell+1}(w_{k_0})) - v_p(g^{(\varepsilon)}_{d+\ell}(w_{k_0})) = v_p(g^{(\varepsilon')}_{\ell+1}(w_{2-k_0})) - v_p(g^{(\varepsilon')}_{\ell}(w_{2-k_0})) + k_0 - 1. \tag{4.18.2}$$

(b) Set $\varepsilon'' = \omega^{-s_{\varepsilon''}}\times\omega^{a+s_{\varepsilon''}}$ with $s_{\varepsilon''} := \{k_0-2-a-s_\varepsilon\}$ (allowing $k_0 \equiv k_\varepsilon \bmod (p-1)$ in which case $\varepsilon'' = \varepsilon$). Then for every integer $\ell = 1,\dots,d$, we will prove in § 4.22

that

$$(4.18.3)\qquad \big(v_p(g^{(\varepsilon)}_{d+1-\ell,\hat{k}_0}(w_{k_0})) - v_p(g^{(\varepsilon)}_{d-\ell,\hat{k}_0}(w_{k_0}))\big) + \big(v_p(g^{(\varepsilon'')}_{\ell,\hat{k}_0}(w_{k_0})) - v_p(g^{(\varepsilon'')}_{\ell-1,\hat{k}_0}(w_{k_0}))\big)$$
$$= \begin{cases} k_0-2 & \text{if } k_0 \equiv k_\varepsilon \bmod (p-1) \text{ and } d^{\mathrm{ur}}_{k_0}+1 \leq \ell \leq d^{\mathrm{Iw}}_{k_0} - d^{\mathrm{ur}}_{k_0} \\ k_0-1 & \text{otherwise.} \end{cases}$$

Replacing ℓ by ℓ' in (4.18.3), the equality (4.18.1) follows from summing the resulting equality for ℓ''s being $d^{\mathrm{ur}}_{k_0}+\ell+1, d^{\mathrm{ur}}_{k_0}+\ell+2, \dots, \frac{1}{2}d$. To deduce (1) and (2) from (4.18.3) and (4.18.2) rigorously, we still need to show that the point $(d, v_p(g^{(\varepsilon)}_d(w_{k_0})))$ actually lies on the Newton polygon $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0}, -))$, which we prove now.

Replacing ℓ by ℓ' in (4.18.2) and summing the resulting equality for $\ell' = 0, 1, \dots, \ell-1$ implies that for all $\ell \geq 1$,

$$v_p(g^{(\varepsilon)}_{d+\ell}(w_{k_0})) - v_p(g^{(\varepsilon)}_d(w_{k_0})) = v_p(g^{(\varepsilon')}_\ell(w_{2-k_0})) + \ell(k_0-1) \geq (k_0-1)\ell.$$

Similarly, taking appropriate sums of (4.18.3) implies that, for $\ell = 1, \dots, d$ if $k_0 \not\equiv k_\varepsilon \bmod (p-1)$ and for $\ell = 1, \dots, d^{\mathrm{ur}}_{k_0}$ if $k_0 \equiv k_\varepsilon \bmod (p-1)$,

$$\big(v_p(g^{(\varepsilon)}_d(w_{k_0})) - v_p(g^{(\varepsilon)}_{d-\ell}(w_{k_0}))\big) + v_p(g^{(\varepsilon'')}_\ell(w_{k_0})) = (k_0-1)\ell.$$

In particular, the first summand is less than or equal to $(k_0-1)\ell$. These two estimates together show that Newton polygon $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0}, -))$ passes through the point $(d, v_p(g^{(\varepsilon)}_d(w_{k_0})))$, and the slopes are less than or equal to k_0-1 before this point, and are greater than or equal to k_0-1 after this point.

Finally, to address (3) and the last statement of (4), we need to show that the first d^{ur}_kth slopes of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0}, -))$ are strictly less than $\frac{k_0-2}{2}$. In fact, we will prove these slopes are less than or equal to $\frac{k_0-3}{p+1}$ in Proposition 4.28 below. Moreover, in the proof of Proposition 4.28, we obtain the inequality

$$v_p(g^{(\varepsilon)}_{d^{\mathrm{ur}}_{k_0}}(w_{k_0})) - v_p(g^{(\varepsilon)}_{d^{\mathrm{ur}}_{k_0}-i}(w_{k_0})) \leq i \cdot \frac{k_0-3}{p+1}.$$

for $i = 1, \dots, d^{\mathrm{ur}}_{k_0}$. It follows that $(d^{\mathrm{ur}}_{k_0}, v_p(g^{(\varepsilon)}_{d^{\mathrm{ur}}_{k_0}}(w_{k_0})))$ is a vertex of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0}), -)$. Combining with (4.18.3), we see that $(d - d^{\mathrm{ur}}_{k_0}, v_p(g^{(\varepsilon)}_{d-d^{\mathrm{ur}}_{k_0}}(w_{k_0})))$ is also a vertex of $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0}), -)$.

This then concludes the proof of the Theorem assuming (4.18.2), (4.18.3) and Proposition 4.28, which will be proved below after a couple of lemmas first. $\square$

Lemma 4.19. *Fix $k_0 \in \mathbb{Z}$ and a relevant character ε. We keep the convention that $k = 2 + \{k_\varepsilon - 2\} + (p-1)k_\bullet$. Recall $k^{(\varepsilon)}_{\mathrm{mid}}(n)$, $k^{(\varepsilon)}_{\max}(n)$, and $k^{(\varepsilon)}_{\min}(n)$ from Lemma-Notation 4.12. For $n \geq 0$, we have*

$$(4.19.1)\quad v_p(g^{(\varepsilon)}_{n+1,\hat{k}_0}(w_{k_0})) - v_p(g^{(\varepsilon)}_{n,\hat{k}_0}(w_{k_0})) = \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon)}_{\mathrm{mid}\bullet}(n) < k_\bullet \leq k^{(\varepsilon)}_{\max\bullet}(n)}} \big(v_p(k-k_0)+1\big) - \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon)}_{\min\bullet}(n) \leq k_\bullet \leq k^{(\varepsilon)}_{\mathrm{mid}\bullet}(n)}} \big(v_p(k-k_0)+1\big).$$

For $n \geq 1$, we have

$$v_p(g^{(\varepsilon)}_{n+1,\hat{k}_0}(w_{k_0})) - 2v_p(g^{(\varepsilon)}_{n,\hat{k}_0}(w_{k_0})) + v_p(g^{(\varepsilon)}_{n-1,\hat{k}_0}(w_{k_0})) \tag{4.19.2}$$
$$= \sum_{\substack{k\neq k_0\\ k^{(\varepsilon)}_{\max\bullet}(n-1)<k_\bullet\leq k^{(\varepsilon)}_{\max\bullet}(n)}} \big(v_p(k-k_0)+1\big) + \sum_{\substack{k\neq k_0\\ k^{(\varepsilon)}_{\min\bullet}(n-1)\leq k_\bullet< k^{(\varepsilon)}_{\min\bullet}(n)}} \big(v_p(k-k_0)+1\big) - 2\big(v_p(k^{(\varepsilon)}_{\mathrm{mid}}(n)-k_0)+1\big),$$

where the last term is undefined when $k^{(\varepsilon)}_{\mathrm{mid}}(n) = k_0$, in which case we interpret it as zero.

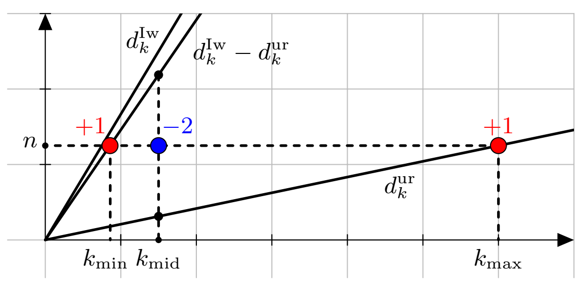
Proof. We first prove (4.19.1). By definition, we have

$$\begin{aligned} v_p(g^{(\varepsilon)}_{n+1,\hat{k}_0}(w_{k_0})) - v_p(g^{(\varepsilon)}_{n,\hat{k}_0}(w_{k_0})) &= \sum_{\substack{k\neq k_0\\ k\equiv k_\varepsilon \bmod (p-1)}} v_p(w_k - w_{k_0})\cdot(m^{(\varepsilon)}_{n+1}(k) - m^{(\varepsilon)}_n(k)) \\ &= \sum_{\substack{k\neq k_0\\ k\equiv k_\varepsilon \bmod (p-1)\\ d^{\mathrm{ur}}_k(\varepsilon_1)\leq n<\frac12 d^{\mathrm{Iw}}_k(\tilde\varepsilon_1)}} \big(v_p(k-k_0)+1\big) - \sum_{\substack{k\neq k_0\\ k\equiv k_\varepsilon \bmod (p-1)\\ \frac12 d^{\mathrm{Iw}}_k(\tilde\varepsilon_1)\leq n< d^{\mathrm{Iw}}_k(\tilde\varepsilon_1)-d^{\mathrm{ur}}_k(\varepsilon_1)}} \big(v_p(k-k_0)+1\big). \end{aligned}$$

The second equality uses the expression for (2.25.3) to compute $m^{(\varepsilon)}_{n+1}(k) - m^{(\varepsilon)}_n(k)$ and the equality $v_p(w_k - w_{k_0}) = v_p(k-k_0)+1$ from (4.17.1). Combining this with Lemma-Notation 4.12, we deduce (4.19.1).

Using a similar argument and citing (2.25.4) in place of (2.25.3), we deduce (4.19.2). □

Remark 4.20. For (4.19.1), the contribution of various k is similar to the case of degrees, as shown in Graph 2 in the proof of Proposition 4.11, namely the blue interval from $k_{\min}$ to k_{mid} contributes negatively and the red interval from k_{mid} to $k_{\max}$ contributes positively. For (4.19.2), the contribution is shown in the picture below, namely, the contributions of the red points at $k_{\min}$ and $k_{\max}$ are positive and the contribution of the blue point at k_{mid} is negative with a multiplier of 2.



Graph 3: contribution to the second difference of p-adic valuations of ghost series

Lemma 4.21. *We have the following equalities*

$$(p-1)(d+\delta_\varepsilon+\delta_{\varepsilon''}-2) = 2(k_0-2) - \{k_\varepsilon - 2\} - \{k_{\varepsilon''}-2\}. \tag{4.21.1}$$
$$(p-1)(d+\delta_\varepsilon-\delta_{\varepsilon'}) = 2(k_0-1) - \{k_\varepsilon - 2\} + \{k_{\varepsilon'}-2\}. \tag{4.21.2}$$

Here we recall

$$\delta_? = \Big\lfloor \frac{s_? + \{a+s_?\}}{p-1}\Big\rfloor = \begin{cases} 0 & \text{if } s_? + \{a+s_?\} < p-1, \\ 1 & \text{if } s_? + \{a+s_?\} \geq p-1. \end{cases}$$

for $? = \varepsilon$, ε' or ε''.

Proof. By the dimension formula of $d = d^{\mathrm{Iw}}_{k_0}(\varepsilon \cdot (1 \times \omega^{2-k_0}))$, we have

$$\begin{aligned} d &= 2 + \frac{k_0 - 2 - s_\varepsilon - \{k_0 - 2 - s_\varepsilon\}}{p-1} + \frac{k_0 - 2 - \{a + s_\varepsilon\} - \{k_0 - 2 - a - s_\varepsilon\}}{p-1} \\ &= 2 + \frac{2(k_0-2) - s_\varepsilon - \{a + s_\varepsilon\} - s_{\varepsilon''} - \{a + s_{\varepsilon''}\}}{p-1} \\ &= 2 - \delta_\varepsilon - \delta_{\varepsilon''} + \frac{2(k_0-2) - \{k_\varepsilon - 2\} - \{k_{\varepsilon''} - 2\}}{p-1}, \end{aligned}$$

where the last equality makes use of the equality (see Remark 4.3)

$$\{k_\varepsilon - 2\} = s_\varepsilon + \{a + s_\varepsilon\} - (p-1)\delta_\varepsilon \tag{4.21.3}$$

and the similar equality with ε'' in place of ε. This proves (4.21.1). We may alternatively use the equality $\{c\} + \{-1-c\} = p-2$ for any integer c to rewrite the first row above as

$$\begin{aligned} d &= \frac{k_0 - 1 - s_\varepsilon + \{s_\varepsilon + 1 - k_0\}}{p-1} + \frac{k_0 - 1 - \{a + s_\varepsilon\} + \{a + s_\varepsilon + 1 - k_0\}}{p-1} \\ &= -\delta_\varepsilon + \delta_{\varepsilon'} + \frac{2(k_0-1) - \{k_\varepsilon - 2\} + \{k_{\varepsilon'} - 2\}}{p-1}. \end{aligned}$$

This proves (4.21.2). $\square$

Now we are ready to prove the two key equalities (4.18.2) and (4.18.3).

4.22. **Proof of** (4.18.3)**.** For $\ell = 1, \dots, d$, applying formula (4.19.1) to the case of $n = d - \ell$ with ε, and the case of $n = \ell - 1$ with ε'', we deduce

$$\begin{aligned} &\big(v_p(g^{(\varepsilon)}_{d-\ell+1,\hat{k}_0}(w_{k_0})) - v_p(g^{(\varepsilon)}_{d-\ell,\hat{k}_0}(w_{k_0}))\big) + \big(v_p(g^{(\varepsilon'')}_{\ell,\hat{k}_0}(w_{k_0})) - v_p(g^{(\varepsilon'')}_{\ell-1,\hat{k}_0}(w_{k_0}))\big) \\ &= \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d-\ell) < k_\bullet \leq k^{(\varepsilon)}_{\mathrm{max}\bullet}(d-\ell)}} \big(v_p(k-k_0)+1\big) - \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon)}_{\mathrm{min}\bullet}(d-\ell) \leq k_\bullet \leq k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d-\ell)}} \big(v_p(k-k_0)+1\big) \\ &\quad + \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon'')}_{\mathrm{mid}\bullet}(\ell-1) < k_\bullet \leq k^{(\varepsilon'')}_{\mathrm{max}\bullet}(\ell-1)}} \big(v_p(k-k_0)+1\big) - \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon'')}_{\mathrm{min}\bullet}(\ell-1) \leq k_\bullet \leq k^{(\varepsilon'')}_{\mathrm{mid}\bullet}(\ell-1)}} \big(v_p(k-k_0)+1\big). \end{aligned} \tag{4.22.1}$$

For the first and the third terms, we separate out the 1's in the sum; and for the second and the fourth terms, we include the 1's into the valuation as a multiple of p. This way, (after rearranging the terms and possibly replacing $v_p(c)$ by $v_p(-c)$,) the sum above becomes the following sum.

$$\big(k^{(\varepsilon)}_{\mathrm{max}\bullet}(d-\ell) - k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d-\ell)\big) + \big(k^{(\varepsilon'')}_{\mathrm{max}\bullet}(\ell-1) - k^{(\varepsilon'')}_{\mathrm{mid}\bullet}(\ell-1)\big) - v \tag{4.22.2}$$

$$+ \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d-\ell) < k_\bullet \leq k^{(\varepsilon)}_{\mathrm{max}\bullet}(d-\ell)}} v_p(k-k_0) + \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon'')}_{\mathrm{mid}\bullet}(\ell-1) < k_\bullet \leq k^{(\varepsilon'')}_{\mathrm{max}\bullet}(\ell-1)}} v_p(k_0-k) \tag{4.22.3}$$

$$- \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon)}_{\mathrm{min}\bullet}(d-\ell) \leq k_\bullet \leq k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d-\ell)}} v_p(pk-pk_0) - \sum_{\substack{k \neq k_0 \\ k^{(\varepsilon'')}_{\mathrm{min}\bullet}(\ell-1) \leq k_\bullet \leq k^{(\varepsilon'')}_{\mathrm{mid}\bullet}(\ell-1)}} v_p(pk_0-pk), \tag{4.22.4}$$

where term υ is zero unless $k_0 \equiv k_\varepsilon \bmod (p-1)$ and $d_{k_0}^{\mathrm{ur}} + 1 \leq \ell \leq d_{k_0}^{\mathrm{Iw}} - d_{k_0}^{\mathrm{ur}}$, in which case $\upsilon = 1$ because the extra 1 in the sum of (4.22.1) is one less due to the removal of the factor when $k = k_0$.

Then to prove (4.18.3), it suffices to show that (4.22.2) is equal to $k_0 - 1 - \upsilon$, and the sum (4.22.3) cancels with the sum (4.22.4).

We prove the former statement. Indeed, by Lemma-Notation 4.12, we need to show that

$$\big(\tfrac{p+1}{2}(d-\ell)+\beta^{(\varepsilon)}_{[d-\ell]}-1\big)-(d-\ell+\delta_\varepsilon-1)+\big(\tfrac{p+1}{2}(\ell-1)+\beta^{(\varepsilon'')}_{[\ell-1]}-1\big)-(\ell-1+\delta_{\varepsilon''}-1)=k_0-1.$$

This is equivalent to proving

$$k_0-1-\tfrac{p-1}{2}(d-1)+\delta_\varepsilon+\delta_{\varepsilon''}=\beta^{(\varepsilon)}_{[d-\ell]}+\beta^{(\varepsilon'')}_{[\ell-1]}.$$

By Lemma 4.21, this is further equivalent to proving

$$\beta^{(\varepsilon)}_{[d-\ell]}+\beta^{(\varepsilon'')}_{[\ell-1]} \;=\tfrac{1}{2}\big(\{k_\varepsilon-2\}+\{k_{\varepsilon''}-2\}\big)+\tfrac{p+1}{2}(\delta_\varepsilon+\delta_{\varepsilon''})-\tfrac{p-3}{2}.$$

This boils down to a case-by-case check, which we do in Lemma 4.23 later.

Now we show that the sum (4.22.3) cancels with the sum (4.22.4). For this, we need to establish three identities.

(1) $k^{(\varepsilon)}_{\mathrm{mid}}(d-\ell)+p-1-k_0=k_0-k^{(\varepsilon'')}_{\mathrm{mid}}(\ell-1)$,
(2) $k^{(\varepsilon)}_{\max}(d-\ell)-k_0=pk_0-\tilde{k}^{(\varepsilon'')}_{\min}(\ell-1)$, and
(3) $\tilde{k}^{(\varepsilon)}_{\min}(d-\ell)-pk_0=k_0-k^{(\varepsilon'')}_{\max}(\ell-1)$.

Assume these identities for a moment. Then the first sum in (4.22.3) is exactly the sum of the valuations of the numbers in the sequence (with step $p-1$) from $k^{(\varepsilon)}_{\max}(d-\ell)-k_0$ (downwards) to $k^{(\varepsilon)}_{\mathrm{mid}}(d-\ell)+(p-1)-k_0 \overset{(1)}{=} k_0-k^{(\varepsilon'')}_{\mathrm{mid}}(\ell-1)$, and then is continued with the second sum of (4.22.3) which is the sum of valuations of the numbers (with step size $p-1$) from $k_0-(k^{(\varepsilon'')}_{\mathrm{mid}}(\ell-1)+p-1)$ to $k_0-k^{(\varepsilon'')}_{\max}(\ell-1)$. In this sum we remove the term if the number happens to be zero. In comparison, in the expression (4.22.4), the second sum is the sum of valuations of only p-multiples in the sequence (with step size $p-1$) from $pk_0-\tilde{k}^{(\varepsilon'')}_{\min}(\ell-1)$ to $pk_0-pk^{(\varepsilon'')}_{\mathrm{mid}\bullet}(\ell-1)\overset{(1)}{=}pk^{(\varepsilon)}_{\mathrm{mid}\bullet}(d-\ell)-pk_0+p(p-1)$; whereas the first sum is the sum of the p-multiples in the continuing sequence (with step size $p-1$) from $pk^{(\varepsilon)}_{\mathrm{mid}\bullet}(d-\ell)-pk_0$ to $\tilde{k}^{(\varepsilon'')}_{\min}(d-\ell)-pk_0$. Again we skip the valuation of zero in the sequence (if it appears). So both the sums (4.22.3) and (4.22.4) are exactly the sum of valuations of (the p-multiples in) the same sequence from $k^{(\varepsilon)}_{\max}(d-\ell)-k_0 \overset{(2)}{=} pk_0-\tilde{k}^{(\varepsilon'')}_{\min}(\ell-1)$ to $k_0-k^{(\varepsilon'')}_{\max}(\ell-1)\overset{(3)}{=}\tilde{k}^{(\varepsilon'')}_{\min}(d-\ell)-pk_0$. Clearly they cancel each other.

It remains to prove the equalities (1), (2), and (3). By Lemma-Notation 4.12, Equality (1) is equivalent to

$$2+\{k_\varepsilon-2\}+(p-1)(d-\ell+\delta_\varepsilon)-k_0=k_0-\big(2+\{k_{\varepsilon''}-2\}+(p-1)(\ell-1+\delta_{\varepsilon''}-1)\big).$$
$$\text{or equivalently, } (p-1)(d+\delta_\varepsilon+\delta_{\varepsilon''}-2)=2k_0-4-\{k_\varepsilon-2\}+\{k_{\varepsilon''}-2\}.$$

But this is exactly Lemma 4.21.

We prove (2). By Lemma-Notation 4.12, this is equivalent to

$$2+\{k_\varepsilon-2\}+(p-1)\big(\tfrac{p+1}{2}(d-\ell)+\beta^{(\varepsilon)}_{[d-\ell]}-1\big)-k_0$$
$$=pk_0-2p-p\{k_{\varepsilon''}-2\}-(p-1)\big(\tfrac{p+1}{2}(\ell-2+2\delta_{\varepsilon''})-\beta^{(\varepsilon'')}_{[\ell]}+1\big).$$

Rearranging, this is equivalent to

$$(p+1)\big(\tfrac{p-1}{2}(d-2+2\delta_{\varepsilon''})-k_0+2\big)+\{k_\varepsilon-2\}+p\{k_{\varepsilon''}-2\}=(p-1)\big(\beta^{(\varepsilon'')}_{[\ell]}-\beta^{(\varepsilon)}_{[d-\ell]}\big).$$

Feeding in the formula (4.21.1), this is further equivalent to

$$\tfrac{p-1}{2}\{k_{\varepsilon''}-2\}-\tfrac{p-1}{2}\{k_\varepsilon-2\}+\tfrac{p^2-1}{2}(\delta_{\varepsilon''}-\delta_\varepsilon)=(p-1)\big(\beta^{(\varepsilon'')}_{[\ell]}-\beta^{(\varepsilon)}_{[d+\ell]}\big).$$

We may cancel the factor $p-1$ on both sides, and then after that, we need to do a case-by-case study which we defer to Lemma 4.23 later.

Finally, we note that (3) is the same as (2) when ε and ε'' are swapped and ℓ is replaced by $d-\ell+1$. This then completes the proof of (4.18.3) assuming the following Lemma. □

Lemma 4.23. *We have the following two identities for every ℓ*

$$\beta^{(\varepsilon)}_{[d+\ell]}-\beta^{(\varepsilon'')}_{[\ell]} = \tfrac12\big(\{k_\varepsilon-2\}-\{k_{\varepsilon''}-2\}\big)+\tfrac{p+1}{2}(\delta_\varepsilon-\delta_{\varepsilon''}),\quad \textit{and}$$
$$\beta^{(\varepsilon)}_{[d-\ell]}+\beta^{(\varepsilon'')}_{[\ell-1]} = \tfrac12\big(\{k_\varepsilon-2\}+\{k_{\varepsilon''}-2\}\big)+\tfrac{p+1}{2}(\delta_\varepsilon+\delta_{\varepsilon''})-\tfrac{p-3}{2}.$$

Proof. In view of the equality $\{k_\varepsilon-2\}+(p-1)\delta_\varepsilon=s_\varepsilon+\{a+s_\varepsilon\}$ and the similar equality for ε'', we need to show that

$$\text{(4.23.1)}\quad \beta^{(\varepsilon)}_{[d+\ell]}-\beta^{(\varepsilon'')}_{[\ell]} = \tfrac12\big(s_\varepsilon+\{a+s_\varepsilon\}\big)-\tfrac12\big(s_{\varepsilon''}+\{a+s_{\varepsilon''}\}\big)+\delta_\varepsilon-\delta_{\varepsilon''},\quad\text{and}$$
$$\text{(4.23.2)}\quad \beta^{(\varepsilon)}_{[d-\ell]}+\beta^{(\varepsilon'')}_{[\ell-1]} = \tfrac12\big(s_\varepsilon+\{a+s_\varepsilon\}\big)+\tfrac12\big(s_{\varepsilon''}+\{a+s_{\varepsilon''}\}\big)+\delta_\varepsilon+\delta_{\varepsilon''}-\tfrac{p-3}{2}.$$

Recall that $s_{\varepsilon''}=\{k_0-2-a-s_\varepsilon\}$. When $a+s_\varepsilon<p-1$, we list our calculation in the following table.

Condition	$s_\varepsilon\le\{k_0-2\}<a+s_\varepsilon$	$\{k_0-2\}<s_\varepsilon$ or $\{k_0-2\}\ge a+s_\varepsilon$
d	odd	even
$a+s_{\varepsilon''}$	$\ge p-1$	$<p-1$
$\beta^{(\varepsilon)}_{\mathrm{even}}$	$s_\varepsilon+\delta_\varepsilon$	$s_\varepsilon+\delta_\varepsilon$
$\beta^{(\varepsilon)}_{\mathrm{odd}}$	$a+s_\varepsilon+\delta_\varepsilon-\frac{p-3}{2}$	$a+s_\varepsilon+\delta_\varepsilon-\frac{p-3}{2}$
$\beta^{(\varepsilon'')}_{\mathrm{even}}$	$\{a+s_{\varepsilon''}\}+\delta_{\varepsilon''}+1$	$s_{\varepsilon''}+\delta_{\varepsilon''}$
$\beta^{(\varepsilon'')}_{\mathrm{odd}}$	$s_{\varepsilon''}+\delta_{\varepsilon''}-\frac{p-1}{2}$	$a+s_{\varepsilon''}+\delta_{\varepsilon''}-\frac{p-3}{2}$
$\beta^{(\varepsilon)}_{[d+\ell]}-\beta^{(\varepsilon'')}_{[\ell]}$	$s_\varepsilon-s_{\varepsilon''}+\delta_\varepsilon-\delta_{\varepsilon''}+\frac{p-1}{2}$	$s_\varepsilon-s_{\varepsilon''}+\delta_\varepsilon-\delta_{\varepsilon''}$
$\beta^{(\varepsilon)}_{[d-\ell]}+\beta^{(\varepsilon'')}_{[\ell-1]}$	$a+s_\varepsilon+s_{\varepsilon''}+\delta_\varepsilon+\delta_{\varepsilon''}+2-p$	$a+s_\varepsilon+s_{\varepsilon''}+\delta_\varepsilon+\delta_{\varepsilon''}-\frac{p-3}{2}$

Note that the expressions in the last two rows are valid regardless of the parity of ℓ. We explain how to get the results in the above table and use them to prove (4.23.1) and (4.23.2). We divide the discussion into two cases. When $s_\varepsilon\le\{k_0-2\}<a+s_\varepsilon$, it follows from Proposition 4.1 that $d:=d^{\mathrm{Iw}}_{k_0}(\varepsilon\cdot(1\times\omega^{2-k_0}))$ is odd. We also have $a+s_{\varepsilon''}\ge p-1$ in this case. We use Notation 4.10 to compute the last six numbers in the table. The penultimate line computes the left hand side of (4.23.1), while the right hand side of (4.23.1) equals to $\frac12(s_\varepsilon+a+s_\varepsilon)-\frac12(s_{\varepsilon''}+a+s_{\varepsilon''}-(p-1))+\delta_\varepsilon-\delta_{\varepsilon''}$, and (4.23.1) is proved in this case.

We use the last line in the table to prove (4.23.2) by a similar computation. The other case when $\{k_0-2\} < s_\varepsilon$ or $\{k_0-2\} \geq a+s_\varepsilon$ can be proved in the same way.

Similarly, when $a+s_\varepsilon \geq p-1$, we have the following table.

Condition	$\{a+s_\varepsilon\} \leq \{k_0-2\} < s_\varepsilon$	$\{k_0-2\} < \{a+s_\varepsilon\}$ or $\{k_0-2\} \geq s_\varepsilon$
d	odd	even
$a+s_{\varepsilon''}$	$< p-1$	$\geq p-1$
$\beta^{(\varepsilon)}_{\mathrm{even}}$	$\{a+s_\varepsilon\}+\delta_\varepsilon+1$	$\{a+s_\varepsilon\}+\delta_\varepsilon+1$
$\beta^{(\varepsilon)}_{\mathrm{odd}}$	$s_\varepsilon+\delta_\varepsilon-\frac{p-1}{2}$	$s_\varepsilon+\delta_\varepsilon-\frac{p-1}{2}$
$\beta^{(\varepsilon'')}_{\mathrm{even}}$	$s_{\varepsilon''}+\delta_{\varepsilon''}$	$\{a+s_{\varepsilon''}\}+\delta_{\varepsilon''}+1$
$\beta^{(\varepsilon'')}_{\mathrm{odd}}$	$a+s_{\varepsilon''}+\delta_{\varepsilon''}-\frac{p-3}{2}$	$s_{\varepsilon''}+\delta_{\varepsilon''}-\frac{p-1}{2}$
$\beta^{(\varepsilon)}_{[d+\ell]}-\beta^{(\varepsilon'')}_{[\ell]}$	$s_\varepsilon-s_{\varepsilon''}+\delta_\varepsilon-\delta_{\varepsilon''}-\frac{p-1}{2}$	$s_\varepsilon-s_{\varepsilon''}+\delta_\varepsilon-\delta_{\varepsilon''}$
$\beta^{(\varepsilon)}_{[d-\ell]}+\beta^{(\varepsilon'')}_{[\ell-1]}$	$a+s_\varepsilon+s_{\varepsilon''}+\delta_\varepsilon+\delta_{\varepsilon''}+2-p$	$a+s_\varepsilon+s_{\varepsilon''}+\delta_\varepsilon+\delta_{\varepsilon''}-\frac{3p-5}{2}$

As above, the expressions in the last two rows are valid regardless of the parity of ℓ. Again the two equalities (4.23.1) and (4.23.2) can be proved via a case-by-case computation. $\square$

4.24. **Proof of** (4.18.2)**.** The proof is similar to § 4.22, so we only sketch the proof and focus on the differences of the two proofs. Applying formula (4.19.1) to the case of $n=d+\ell$ with ε, and the case of $n=\ell$ with ε' and rearranging terms in a way similar to § 4.22, we deduce that

$$\big(v_p(g^{(\varepsilon)}_{d+\ell+1}(w_{k_0}))-v_p(g^{(\varepsilon)}_{d+\ell}(w_{k_0}))\big)-\big(v_p(g^{(\varepsilon')}_{\ell+1}(w_{2-k_0}))-v_p(g^{(\varepsilon')}_{\ell}(w_{2-k_0}))\big)$$

$$=\big(k^{(\varepsilon)}_{\max\bullet}(d+\ell)-k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d+\ell)\big)-\big(k^{(\varepsilon')}_{\max\bullet}(\ell)-k^{(\varepsilon')}_{\mathrm{mid}\bullet}(\ell)\big) \tag{4.24.1}$$

$$+\sum_{k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d+\ell)<k_\bullet\leq k^{(\varepsilon)}_{\max\bullet}(d+\ell)} v_p(k-k_0)\quad+\sum_{k^{(\varepsilon')}_{\min\bullet}(\ell)\leq k_\bullet\leq k^{(\varepsilon')}_{\mathrm{mid}\bullet}(\ell)} v_p(pk-p(2-k_0)) \tag{4.24.2}$$

$$-\sum_{k^{(\varepsilon')}_{\mathrm{mid}\bullet}(\ell)<k_\bullet\leq k^{(\varepsilon')}_{\max\bullet}(\ell)} v_p(k-(2-k_0))\quad-\sum_{k^{(\varepsilon)}_{\min\bullet}(d+\ell)\leq k_\bullet\leq k^{(\varepsilon)}_{\mathrm{mid}\bullet}(d+\ell)} v_p(pk-pk_0). \tag{4.24.3}$$

Then to prove (4.18.2), it suffices to show that (4.24.1) is equal to k_0-1, and the sum (4.24.2) cancels with the sum (4.24.3).

We prove the former statement. Indeed, by Lemma-Notation 4.12, we need to show that

$$\big(\tfrac{p+1}{2}(d+\ell)+\beta^{(\varepsilon)}_{[d+\ell]}-1\big)-(d+\ell+\delta_\varepsilon-1)=(k_0-1)+\big(\tfrac{p+1}{2}\ell+\beta^{(\varepsilon')}_{[\ell]}-1\big)-(\ell+\delta_{\varepsilon'}-1).$$

This is equivalent to proving

$$k_0-1-\tfrac{p-1}{2}d+\delta_\varepsilon-\delta_{\varepsilon'}=\beta^{(\varepsilon)}_{[d+\ell]}-\beta^{(\varepsilon')}_{[\ell]}. \tag{4.24.4}$$

By (4.21.2), we are left to show that

$$\beta^{(\varepsilon)}_{[d+\ell]}-\beta^{(\varepsilon')}_{[\ell]}=\tfrac{p+1}{2}(\delta_\varepsilon-\delta_{\varepsilon'})+\tfrac{1}{2}\big(\{k_\varepsilon-2\}-\{k_{\varepsilon'}-2\}\big).$$

This boils down to a case-by-case check, which we refer to Lemma 4.26 later.

Now we show that the sum (4.24.2) cancels with the sum (4.24.3). For this, we need the following lemma:

Lemma 4.25. *We have the following three identities:*

(1) $k^{(\varepsilon)}_{\mathrm{mid}}(d+\ell) - k_0 = k^{(\varepsilon')}_{\mathrm{mid}}(\ell) - (2-k_0)$,
(2) $k^{(\varepsilon)}_{\max}(d+\ell) - k_0 = \tilde{k}^{(\varepsilon')}_{\min}(\ell) - p(2-k_0) - (p-1)$, *and*
(3) $\tilde{k}^{(\varepsilon)}_{\min}(d+\ell) - pk_0 - (p-1) = k^{(\varepsilon')}_{\max}(\ell) - (2-k_0)$.

Assume these identities for a moment. The first sum of (4.24.2) is the sum of the valuations of the sequence (with step size $p-1$) from $k^{(\varepsilon)}_{\mathrm{mid}}(d+\ell) - k_0 + (p-1)$ to $k^{(\varepsilon)}_{\max}(d+\ell) - k_0 \overset{(2)}{=} \tilde{k}^{(\varepsilon')}_{\min\bullet}(\ell) - p(2-k_0) - (p-1)$, and the second sum of (4.24.2) then continues with the valuations (of only the p-multiples) in the sequence (with step size $p-1$) from $\tilde{k}^{(\varepsilon')}_{\min\bullet}(\ell) - p(2-k_0)$ to $pk^{(\varepsilon')}_{\mathrm{mid}\bullet}(\ell) - p(2-k_0)$. In a similar way, the first sum of (4.24.3) is the sum of the valuations of the sequence (with step size $p-1$) from $k^{(\varepsilon')}_{\mathrm{mid}}(\ell) - (2-k_0) + (p-1)$ to $k^{(\varepsilon')}_{\max}(\ell) - (2-k_0) \overset{(3)}{=} \tilde{k}^{(\varepsilon)}_{\min\bullet}(d+\ell) - pk_0 - (p-1)$, and then the second sum of (4.24.3) continues with the sum of valuations of (only p-multiples in) the sequence (of step size $p-1$) from $\tilde{k}^{(\varepsilon)}_{\min\bullet}(d+\ell) - pk_0$ to $pk^{(\varepsilon)}_{\mathrm{mid}\bullet}(d+\ell) - pk_0$. So these the sums (4.24.2) and (4.24.3) are exactly the sum of valuations of the same sequence from $k^{(\varepsilon)}_{\mathrm{mid}}(d+\ell) - k_0 + p - 1 \overset{(1)}{=} k^{(\varepsilon')}_{\mathrm{mid}}(\ell) - (2-k_0) + p - 1$ to $pk^{(\varepsilon)}_{\mathrm{mid}}(d+\ell) - pk_0 \overset{(1)}{=} pk^{(\varepsilon')}_{\mathrm{mid}}(\ell) - p(2-k_0)$. Clearly they cancel each other.

It remains to prove the equalities (1), (2), and (3) in Lemma 4.25. By Lemma-Notation 4.12, Equality (1) is equivalent to

$$2 + \{k_\varepsilon - 2\} + (p-1)(d+\ell+\delta_\varepsilon - 1) - k_0 = 2 + \{k_{\varepsilon'} - 2\} + (p-1)(\ell + \delta_{\varepsilon'} - 1) - (2-k_0).$$
$$\text{or equivalently, } (p-1)(d+\delta_\varepsilon - \delta_{\varepsilon'}) = 2k_0 - 2 - \{k_\varepsilon - 2\} + \{k_{\varepsilon'} - 2\}.$$

But this is exactly Lemma 4.21.

We prove (2). By Lemma-Notation 4.12, this is equivalent to

$$\begin{aligned} 2 + \{k_\varepsilon - 2\} &+ (p-1)\big(\tfrac{p+1}{2}(d+\ell) + \beta^{(\varepsilon)}_{[d+\ell]} - 1\big) - k_0 \\ &= 2p + p\{k_{\varepsilon'} - 2\} + (p-1)\big(\tfrac{p+1}{2}(\ell - 1 + 2\delta_{\varepsilon'}) - \beta^{(\varepsilon')}_{[\ell-1]}\big) - p(2-k_0). \end{aligned}$$

Rearranging and using Lemma 4.21, this is equivalent to

$$(p-1)(\beta^{(\varepsilon)}_{[d+\ell]} + \beta^{(\varepsilon')}_{[\ell-1]}) = \tfrac{p-1}{2}\{k_\varepsilon - 2\} + \tfrac{p-1}{2}\{k_{\varepsilon'} - 2\} + \tfrac{p^2-1}{2}(\delta_\varepsilon + \delta_{\varepsilon'}) - (p-1)\tfrac{p-3}{2}.$$

Dividing this by $p-1$, this follows from Lemma 4.26 below.

We prove (3). By Lemma-Notation 4.12, this is equivalent to

$$\begin{aligned} 2p + p\{k_\varepsilon - 2\} &+ (p-1)\big(\tfrac{p+1}{2}(d+\ell-1+2\delta_\varepsilon) - \beta^{(\varepsilon)}_{[d+\ell-1]}\big) - pk_0 \\ &= 2 + \{k_{\varepsilon'} - 2\} + (p-1)\big(\tfrac{p+1}{2}\ell + \beta^{(\varepsilon')}_{[\ell]} - 1\big) - (2-k_0). \end{aligned}$$

Rearranging and using Lemma 4.21, this is further equivalent to

$$(p-1)(\beta^{(\varepsilon)}_{[d+\ell-1]} + \beta^{(\varepsilon')}_{[\ell]}) = \tfrac{p-1}{2}\{k_\varepsilon - 2\} + \tfrac{p-1}{2}\{k_{\varepsilon'} - 2\} + \tfrac{p^2-1}{2}(\delta_{\varepsilon'} + \delta_\varepsilon) - (p-1)\tfrac{p-3}{2}.$$

Dividing by $p-1$, this will follow from Lemma 4.26 below.

To sum up, we have proved (4.18.3) provided the following Lemma. $\square$

Lemma 4.26. *We have the following two identities for every* ℓ

$$\begin{aligned} \beta^{(\varepsilon)}_{[d+\ell]} - \beta^{(\varepsilon')}_{[\ell]} &= \tfrac{1}{2}\big(\{k_\varepsilon - 2\} - \{k_{\varepsilon'} - 2\}\big) + \tfrac{p+1}{2}(\delta_\varepsilon - \delta_{\varepsilon'}) \quad \textit{and} \\ \beta^{(\varepsilon)}_{[d+\ell-1]} + \beta^{(\varepsilon')}_{[\ell]} &= \tfrac{1}{2}\big(\{k_\varepsilon - 2\} + \{k_{\varepsilon'} - 2\}\big) + \tfrac{p+1}{2}(\delta_{\varepsilon'} + \delta_\varepsilon) - \tfrac{p-3}{2}. \end{aligned}$$

Proof. This follows from a case-by-case study as in Lemma 4.23. In fact, we get exactly the *same* tables as in Lemma 4.23 with ε' in places of ε'' everywhere. We leave the details to the readers. □

Lastly, we address the last piece needed for Proposition 4.18. This is also an essential ingredient to prove that the local ghost conjecture implies Gouvêa's $\lfloor\frac{k-1}{p+1}\rfloor$-conjecture (see the discussion following Conjecture 1.9). The readers may choose to skip the proof for the first time reading.

Notation 4.27. For a positive integer m, let $\mathrm{Dig}(m)$ denote the sum of all digits in the p-based expression of m. Then the sums of valuations of consecutive integers in $(m_1, m_2]$ with $m_2 > m_1 > 0$ is

$$\sum_{m_1 < i \leq m_2} v_p(i) = \frac{(m_2 - \mathrm{Dig}(m_2)) - (m_1 - \mathrm{Dig}(m_1))}{p-1}. \tag{4.27.1}$$

Proposition 4.28. *Fix a relevant character* ε *of* Δ^2*, and let* $k_0 \equiv k_\varepsilon \bmod (p-1)$ *be a weight. Then the first* $d^{\mathrm{ur}}_{k_0}(\varepsilon_1)$ *slopes of* $\mathrm{NP}(G^{(\varepsilon)}(w_{k_0}, -))$ *are all less than or equal to* $\lfloor\frac{k_0-1-\min\{a+1,p-2-a\}}{p+1}\rfloor$*. More precisely, they are all less than or equal to*

$$\frac{p-1}{2}(d^{\mathrm{ur}}_{k_0}(\varepsilon_1) - 1) - \delta_\varepsilon + \beta_{[d^{\mathrm{ur}}_{k_0}(\varepsilon_1)-1]}. \tag{4.28.1}$$

Proof. In this proof, we shall suppress ε and ε_1 from the notation for simplicity. We first check that (4.28.1) is less than or equal to $\frac{k_0-1-\min\{a+1,p-2-a\}}{p+1}$.

Write $k_0 = 2 + \{a+2s\} + (p-1)k_{0\bullet}$ as before. Using Proposition 4.7 and Remark 4.3, we may separate the argument in the following two cases.

- If $k_{0\bullet} - (p+1)\lfloor\frac{k_{0\bullet}-t_1}{p+1}\rfloor \geq t_2$, then

$$\begin{aligned} (4.28.1) &= (p-1)\Big\lfloor\frac{k_{0\bullet} - t_1}{p+1}\Big\rfloor - \delta + t_2 - 1 \leq (p-1)\frac{(k_{0\bullet} - t_2)}{p+1} - \delta + t_2 - 1 \\ &= \frac{k_0 - 1}{p+1} + \frac{2t_2 - \{a+2s\} - 1}{p+1} - \delta - 1 \\ &= \begin{cases} \frac{k_0-1}{p+1} - \frac{p-2-a}{p+1} & \text{if } a + s_\varepsilon < p-1 \\ \frac{k_0-1}{p+1} - \frac{a+1}{p+1} & \text{if } a + s_\varepsilon \geq p-1. \end{cases} \end{aligned}$$

- If $k_{0\bullet}-(p+1)\lfloor\frac{k_{0\bullet}-t_1}{p+1}\rfloor<t_2$, then

$$\begin{aligned}(4.28.1) &= (p-1)\Big\lfloor\frac{k_{0\bullet}-t_1}{p+1}\Big\rfloor-\delta+t_1\leq(p-1)\frac{(k_{0\bullet}-t_1)}{p+1}-\delta+t_1\\ &=\frac{k_0-1}{p+1}+\frac{2t_1-\{a+2s\}-1}{p+1}-\delta\\ &=\begin{cases}\frac{k_0-1}{p+1}-\frac{a+1}{p+1} & \text{if } a+s_\varepsilon<p-1\\ \frac{k_0-1}{p+1}-\frac{p-2-a}{p+1} & \text{if } a+s_\varepsilon\geq p-1.\end{cases}\end{aligned}$$

The claim follows immediately.

Now, we prove that the first $d_{k_0}^{\mathrm{ur}}$ slopes of $\mathrm{NP}(G(w_{k_0},-))$ are all less than or equal to (4.28.1). It suffices to show that for $i=1,\dots,d_{k_0}^{\mathrm{ur}}$,

$$v_p(g_{d_{k_0}^{\mathrm{ur}}}(w_{k_0}))-v_p(g_{d_{k_0}^{\mathrm{ur}}-i}(w_{k_0}))\leq i\cdot(4.28.1).$$

Recall from (4.19.1) that for $n=d_{k_0}^{\mathrm{ur}}-1,\dots,1,0$ (so $k_{0\bullet}>k_{\max\bullet}(n)$ by Remark 4.13),

$$\begin{aligned}v_p(g_{n+1}(w_{k_0}))-v_p(g_n(w_{k_0})) &= \sum_{k_{\mathrm{mid}\bullet}(n)<k_\bullet\leq k_{\max\bullet}(n)}\big(v_p(k_{0\bullet}-k_\bullet)+1\big)-\sum_{k_{\min\bullet}(n)\leq k_\bullet\leq k_{\mathrm{mid}\bullet}(n)}\big(v_p(k_{0\bullet}-k_\bullet)+1\big)\\ (4.28.2)\quad &\overset{(4.27.1)}{=}\frac{p(k_{\max\bullet}(n)-k_{\mathrm{mid}\bullet}(n))-\mathrm{Dig}(k_{0\bullet}-k_{\mathrm{mid}\bullet}(n)-1)+\mathrm{Dig}(k_{0\bullet}-k_{\max\bullet}(n)-1)}{p-1}\\ &\quad-\frac{p(k_{\mathrm{mid}\bullet}(n)-k_{\min\bullet}(n)+1)-\mathrm{Dig}(k_{0\bullet}-k_{\min\bullet}(n))+\mathrm{Dig}(k_{0\bullet}-k_{\mathrm{mid}\bullet}(n)-1)}{p-1}.\end{aligned}$$

Using basic properties of p-based numbers and Lemma-Notation 4.12, one can deduce that

$$pk_{\min\bullet}(n)+\mathrm{Dig}(k_{0\bullet}-k_{\min\bullet}(n))=\tilde{k}_{\min\bullet}(n)+\mathrm{Dig}(pk_{0\bullet}-\tilde{k}_{\min\bullet}(n)).$$

From this, we see that (4.28.2) is equal to

$$\begin{aligned}&\frac{\tilde{k}_{\min\bullet}(n)+pk_{\max\bullet}(n)-2pk_{\mathrm{mid}\bullet}(n)-p}{p-1}+\frac{\mathrm{Dig}(k_{0\bullet}-k_{\max\bullet}(n)-1)+\mathrm{Dig}(pk_{0\bullet}-\tilde{k}_{\min\bullet}(n))}{p-1}\\ &\quad-\frac{2\,\mathrm{Dig}(k_{0\bullet}-k_{\mathrm{mid}\bullet}(n)-1)}{p-1}.\end{aligned}$$

Plugging in the formulae of Lemma-Notation 4.12 and setting

$$A_n=k_{0\bullet}-k_{\max\bullet}(n)-1,\quad B_n=pk_{0\bullet}-\tilde{k}_{\min\bullet}(n),\quad C_n=k_{0\bullet}-k_{\mathrm{mid}\bullet}(n)-1,$$

we deduce that (4.28.2) is equal to

$$\frac{(p-1)n-1}{2}-\delta_\varepsilon+\frac{p\beta_{[n]}-\beta_{[n-1]}}{p-1}+\frac{\mathrm{Dig}\,A_n+\mathrm{Dig}\,B_n-2\,\mathrm{Dig}\,C_n}{p-1}.$$

Therefore, we have

$$v_p(g_{d_{k_0}^{\mathrm{ur}}}(w_{k_0})) - v_p(g_{d_{k_0}^{\mathrm{ur}}-i}(w_{k_0})) - i\cdot\big(\tfrac{p-1}{2}(d_{k_0}^{\mathrm{ur}}-1) - \delta_\varepsilon + \beta_{[d_{k_0}^{\mathrm{ur}}-1]}\big) \tag{4.28.3}$$
$$= -\tfrac{i(i-1)}{4}(p-1) - \tfrac{i}{2} + \begin{cases} \tfrac{i}{2}(\beta_{[d_{k_0}^{\mathrm{ur}}]} - \beta_{[d_{k_0}^{\mathrm{ur}}-1]}) & \text{if } i \text{ is even} \\ \tfrac{i-1}{2}(\beta_{[d_{k_0}^{\mathrm{ur}}]} - \beta_{[d_{k_0}^{\mathrm{ur}}-1]}) + \frac{\beta_{[d_{k_0}^{\mathrm{ur}}-1]}-\beta_{[d_{k_0}^{\mathrm{ur}}]}}{p-1} & \text{if } i \text{ is odd} \end{cases}$$
$$+ \sum_{1\le j\le i}\left(\frac{\mathrm{Dig}\left(A_{d_{k_0}^{\mathrm{ur}}-j}\right) + \mathrm{Dig}\left(B_{d_{k_0}^{\mathrm{ur}}-j}\right) - 2\,\mathrm{Dig}\left(C_{d_{k_0}^{\mathrm{ur}}-j}\right)}{p-1}\right).$$

Note that $\beta_{[d_{k_0}^{\mathrm{ur}}]} - \beta_{[d_{k_0}^{\mathrm{ur}}-1]} \in \{\pm(t_2 - t_1 - \frac{p+1}{2})\} = \{\pm\frac{p-3-2a}{2}\}$; so it is at most $\frac{p-5}{2}$. Since $p \ge 5$, it is straightforward to see that the second line of (4.28.3) is less than or equal to

$$\begin{cases} -\frac{2}{p-1} & \text{if } i = 1 \\ -i^2 + \frac{1}{2}i & \text{if } i > 1. \end{cases} \tag{4.28.4}$$

Moreover, since (4.28.3) is an integer, it reduces to show

$$\sum_{1\le j\le i}\left(\frac{\mathrm{Dig}\left(A_{d_{k_0}^{\mathrm{ur}}-j}\right) + \mathrm{Dig}\left(B_{d_{k_0}^{\mathrm{ur}}-j}\right) - 2\,\mathrm{Dig}\left(C_{d_{k_0}^{\mathrm{ur}}-j}\right)}{p-1}\right) + (4.28.4) < 1. \tag{4.28.5}$$

From Lemma-Notation 4.12, we have the relation

$$B_n = (p+1)C_n - A_{n+1} - 1, \qquad n \ge 0.$$

For $j = 1, 2, ..., d_{k_0}^{\mathrm{ur}}$, we may compute $A_{d_{k_0}^{\mathrm{ur}}-j}, B_{d_{k_0}^{\mathrm{ur}}-j}, C_{d_{k_0}^{\mathrm{ur}}-j}$ in terms of $A_{d_{k_0}^{\mathrm{ur}}-1}, A_{d_{k_0}^{\mathrm{ur}}-2}$, and $C = C_{d_{k_0}^{\mathrm{ur}}-1}$:

$$A_{d_{k_0}^{\mathrm{ur}}-j} = \begin{cases} A_{d_{k_0}^{\mathrm{ur}}-1} + \frac{p+1}{2}(j-1) & \text{if } j \text{ is odd} \\ A_{d_{k_0}^{\mathrm{ur}}-2} + \frac{p+1}{2}(j-2) & \text{if } j \text{ is even} \end{cases} \text{ and } C_{d_{k_0}^{\mathrm{ur}}-j} = C + (j-1).$$

Hence

$$B_{d_{k_0}^{\mathrm{ur}}-j} = \begin{cases} (p+1)(C + \frac{j-1}{2}) + p - A_{d_{k_0}^{\mathrm{ur}}-2} & \text{if } j \text{ is odd} \\ (p+1)(C + \frac{j-2}{2}) + p - A_{d_{k_0}^{\mathrm{ur}}-1} & \text{if } j \text{ is even.} \end{cases}$$

Note that $k_{\max\bullet}(n)$ is an increasing function in n by Remark 4.13. Thus $A_{d_{k_0}^{\mathrm{ur}}-1} \le A_{d_{k_0}^{\mathrm{ur}}-2} = \frac{1}{2}(p+1) < p$. Then using the inequalities $\mathrm{Dig}(A+B) \le \mathrm{Dig}(A) + \mathrm{Dig}(B)$ and $\mathrm{Dig}((p+1)D) \le 2\,\mathrm{Dig}(D) \le 2D$, we conclude that

$$\mathrm{Dig}\left(A_{d_{k_0}^{\mathrm{ur}}-j}\right) \le \begin{cases} A_{d_{k_0}^{\mathrm{ur}}-1} + (j-1) & \text{if } j \text{ is odd} \\ A_{d_{k_0}^{\mathrm{ur}}-2} + (j-2) & \text{if } j \text{ is even,} \end{cases}$$

and

$$\mathrm{Dig}\left(B_{d_{k_0}^{\mathrm{ur}}-j}\right) \le \begin{cases} \mathrm{Dig}\left((p+1)(C + \frac{j-1}{2})\right) + p - A_{d_{k_0}^{\mathrm{ur}}-2} & \text{if } j \text{ is odd} \\ \mathrm{Dig}\left((p+1)(C + \frac{j-2}{2})\right) + p - A_{d_{k_0}^{\mathrm{ur}}-1} & \text{if } j \text{ is even.} \end{cases}$$

Summing up these inequalities for $j = 1, \ldots, i$, we get

(4.28.6)
$$\sum_{1\le j\le i}\big(\operatorname{Dig} A_{d^{\mathrm{ur}}_{k_0}-j}+\operatorname{Dig} B_{d^{\mathrm{ur}}_{k_0}-j}\big)\le pi+2\lfloor\tfrac{i-1}{2}\rfloor\lfloor\tfrac{i+1}{2}\rfloor+2\sum_{0\le j\le\lfloor\frac{i-1}{2}\rfloor}\operatorname{Dig}\big((p+1)(C+j)\big)$$
$$-\,\delta(i)\big(2\lfloor\tfrac{i-1}{2}\rfloor+\operatorname{Dig}\big((p+1)(C+\lfloor\tfrac{i-1}{2}\rfloor)\big)\big),$$
where $\delta(i)\in\{0,1\}$ is determined by $\delta(i)\equiv i \mod 2$.

We need the following lemma.

Lemma 4.29. *For $p\ge 3$ and $m,N\in\mathbb{N}$, we have*
$$\operatorname{Dig}\big((p+1)N\big)\le\operatorname{Dig}(N)+\operatorname{Dig}(N+m)+(p-2)m$$

Proof. Denote by a and b the number of times we carry over digits to the next one when computing the sums $N+pN$ and $N+m$ respectively. It follows that
$$\operatorname{Dig}\big((p+1)N\big)=\operatorname{Dig}(N)+\operatorname{Dig}(pN)-(p-1)a=2\operatorname{Dig}(N)-(p-1)a$$
and
$$\operatorname{Dig}(N+m)=\operatorname{Dig}(N)+\operatorname{Dig}(m)-(p-1)b.$$
Let d_m be the number of digits of m, namely $\lfloor\frac{\ln m}{\ln p}\rfloor+1$, if in the p-based expression of m the leading term is not $p-1$; otherwise, we define d_m to be one plus the number of digits of m. (Here and later, $\ln(-)$ denotes the natural real logarithmic.) It is not difficult to check that if $b-d_m\ge 1$, then the there must be at least $b-d_m$ consecutive $(p-1)$'s in N, and moreover they will force carrying over digits at least $b-d_m$ times. So $a\ge b-d_m$. Now, combining with the inequalities above, we deduce that
$$\operatorname{Dig}((p+1)N)-\operatorname{Dig}(N)-\operatorname{Dig}(N+m)\le(p-1)(b-a)-\operatorname{Dig}(m)\le(p-1)d_m-\operatorname{Dig}(m).$$
It is not difficult to verify that $(p-1)d_m\le\operatorname{Dig}(m)+(p-2)m$. The lemma is proved. $\square$

Applying Lemma 4.29 to $N=C,C+1,\dots,C+\lfloor\frac{i-1}{2}\rfloor$ and $m=\lfloor\frac{i}{2}\rfloor$ or $\lfloor\frac{i+1}{2}\rfloor$, and adding them up, we deduce that
$$2\sum_{0\le j\le\lfloor\frac{i-1}{2}\rfloor}\operatorname{Dig}\big((p+1)(C+j)\big)-\delta(i)\operatorname{Dig}\big((p+1)(C+\lfloor\tfrac{i-1}{2}\rfloor)\big)$$
$$\le 2\sum_{0\le j\le i-1}\operatorname{Dig}\big((C+j)\big)+(p-2)\lfloor\tfrac{i+1}{2}\rfloor\lfloor\tfrac{i}{2}\rfloor.$$
Note that $\lfloor\frac{i+1}{2}\rfloor-\delta(i)=\lfloor\frac{i}{2}\rfloor$. Combining with (4.28.6), we deduce that
$$\sum_{1\le j\le i}\frac{\operatorname{Dig} A_{d^{\mathrm{ur}}_{k_0}-j}+\operatorname{Dig} B_{d^{\mathrm{ur}}_{k_0}-j}-2\operatorname{Dig} C_{d^{\mathrm{ur}}_{k_0}-j}}{p-1}\le\frac{pi+2\lfloor\frac{i-1}{2}\rfloor\lfloor\frac{i}{2}\rfloor+2(p-2)\lfloor\frac{i+1}{2}\rfloor\lfloor\frac{i}{2}\rfloor}{p-1}$$
$$=2\lfloor\tfrac{i+1}{2}\rfloor\lfloor\tfrac{i}{2}\rfloor+i+\tfrac{\delta(i)}{p-1}.$$

Put
$$S(i):=2\lfloor\tfrac{i+1}{2}\rfloor\lfloor\tfrac{i}{2}\rfloor+i+\tfrac{\delta(i)}{p-1}+(4.28.4).$$
A short computation shows that $S(1)=1-\frac{1}{p-1}$, $S(2)=1$, and for $i\ge 3$,
$$S(i)\le\frac{i^2}{2}+i+(4.28.4)=\frac{-i^2+3i}{2}\le 0.$$

By (4.28.5), we conclude the desired result except for $i=2$. Moreover, if (4.28.5) fails for $i=2$, we must have

$$\mathrm{Dig}\big((p+1)C+(p-A_{d_{k_0}^{\mathrm{ur}}-2})\big)=\mathrm{Dig}\big((p+1)C\big)+p-A_{d_{k_0}^{\mathrm{ur}}-2}$$

and

$$\mathrm{Dig}((p+1)C)=\mathrm{Dig}(C)+\mathrm{Dig}(C+1)+(p-2).$$

From the second equality, we deduce that $\mathrm{Dig}(C+1)<\mathrm{Dig}(C)$, yielding $p|(C+1)$. However, this contradicts the first equality since $p-A_{d_{k_0}^{\mathrm{ur}}-2}=\frac{1}{2}(p-1)>1$. $\square$

5. Vertices of the Newton polygon of ghost series

In this section, we investigate further properties of ghost series. We keep Hypothesis 3.1 on $\widetilde{\mathrm{H}}$, i.e. $\widetilde{\mathrm{H}}$ is a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar{\rho}=\begin{pmatrix}\omega_1^{a+1} & *\neq 0\\ 0 & 1\end{pmatrix}$ with $1\leq a\leq p-4$, on which $\begin{pmatrix}p&0\\0&p\end{pmatrix}$ acts trivially.

We fix $\varepsilon=\omega^{-s_\varepsilon}\otimes\omega^{a+s_\varepsilon}$ a relevant character of Δ^2 in this section. We will suppress ε from all superscripts when no confusion arises. Let $G(w,t)=G^{(\varepsilon)}(w,t)=\sum_{n\geq 0}g_n(w)t^n\in\mathbb{Z}_p[w][\![t]\!]$ denote the corresponding ghost series.

We prove the following two results in this section:

1. For $n\in\mathbb{Z}_{\geq 0}$ and $w_\star\in\mathfrak{m}_{\mathbb{C}_p}$, the point $(n,v_p(g_n(w_\star)))$ is a vertex on the Newton Polygon $\mathrm{NP}(G(w_\star,-))$ if and only if $(n,w_\star)$ is not near-Steinberg (Theorem 5.19). Here, near-Steinberg roughly means that $w_\star$ is "close" to a weight point w_k such that "n lies between d_k^{ur} and $d_k^{\mathrm{Iw}}-d_k^{\mathrm{ur}}$". The precise definition of this concept in Definition 5.11 is an essential tool to parcel the delicate combinatorics of the ghost series.
2. For $k\in\mathbb{Z}$, the slopes of $\mathrm{NP}(G(w_k,-))$ are all integers if a is even and belong to $\frac{1}{2}\mathbb{Z}$ if a is odd (Corollary 5.24). This sheds some lights on the folklore Breuil–Buzzard–Emerton Conjecture (Conjecture 1.12) on slopes of Kisin deformation spaces.

Before diving into the more detailed discussion, let us first explain one of the serious subtleties we face in this section. While most numerical information of ghost series can be easily described asymptotically through linear or quadratic terms, the precise values of these numerical data are somewhat subtle and often involve the p-based expansion of the weight k (or rather $k_\bullet$). This means that for "majority" of the case where the p-based expansion of $k_\bullet$ is not "special", the behavior of the Newton polygon of the ghost series is "reasonable". In contract, near these "special" weights k (e.g. $k_\bullet$ is an integer like $p^{p^p}-p$), we need to be extra careful. See Example 5.25 for a discussion in an example of such pathological case.

Notation 5.1. For integer $k\geq 2$ and $k\equiv k_\varepsilon \bmod (p-1)$, we set

$$\Delta'_{k,\ell}:=v_p\big(g_{\frac{1}{2}d_k^{\mathrm{Iw}}+\ell,\hat{k}}(w_k)\big)-\tfrac{k-2}{2}\ell,\quad\text{for }\ell=-\tfrac{1}{2}d_k^{\mathrm{new}},-\tfrac{1}{2}d_k^{\mathrm{new}}+1,\dots,\tfrac{1}{2}d_k^{\mathrm{new}}.\tag{5.1.1}$$

Then the ghost duality statement in Proposition 4.18 (4) is equivalent to the following equality

$$\Delta'_{k,\ell}=\Delta'_{k,-\ell}\quad\text{for all }\ell=-\tfrac{1}{2}d_k^{\mathrm{new}},\dots,\tfrac{1}{2}d_k^{\mathrm{new}}.\tag{5.1.2}$$

If we connect the points $(\ell,\Delta'_{k,\ell})$ with line segments on the plane to get the graph $\underline{\Delta}'_k$ of a function, the function is in general not convex (although for "majority" k it is; see Lemma 5.6). Consider the convex hull $\underline{\Delta}_k=\underline{\Delta}_k^{(\varepsilon)}$ of such graph and let $(\ell,\Delta_{k,\ell})=(\ell,\Delta_{k,\ell}^{(\varepsilon)})$

denote the corresponding points on this convex hull. In particular, the ghost duality (5.1.2) implies that $\Delta_{k,\ell} = \Delta_{k,-\ell}$.

We start with several technical lemmas giving estimates of $\Delta'_{k,\ell}$ and $\Delta_{k,\ell}$. The proofs of these lemmas have complicated notations though not difficult themselves. The readers may skip the proofs for the first reading (but do read Notation 5.7). The motivation to study this polygon $\underline{\Delta}_k$ is explained by Definition 5.11 and the following remarks and examples.

Lemma 5.2. *For $k = k_\varepsilon + (p-1)k_\bullet$ and $\ell = 1, \dots, \frac{1}{2}d_k^{\mathrm{new}}$, we have*

$$\Delta'_{k,\ell} - \Delta'_{k,\ell-1} \;\geq\; \frac{\min\{a+2, p-1-a\}}{2} + \frac{p-1}{2}(\ell-1) \;\geq\; \frac{3}{2} + \frac{p-1}{2}(\ell-1).$$

Proof. By ghost duality (5.1.2), it is equivalent to consider $\Delta'_{k,-\ell} - \Delta'_{k,-\ell+1}$. We use the formula (4.19.1) for $n = \frac{1}{2}d_k^{\mathrm{Iw}} - \ell = k_\bullet + 1 - \delta_\varepsilon - \ell$, (and write $k_{*\bullet}$ for $k_{*\bullet}(n)$ with $* = \min, \max, \mathrm{mid}$)

$$\Delta'_{k,-\ell} - \Delta'_{k,-\ell+1} = \frac{k-2}{2} - \sum_{\substack{k'_\bullet \neq k_\bullet \\ k_{\mathrm{mid}\bullet} < k'_\bullet \leq k_{\max\bullet}}} \big(v_p(k'_\bullet - k_\bullet) + 1\big) + \sum_{k_{\min\bullet} \leq k'_\bullet \leq k_{\mathrm{mid}\bullet}} \big(v_p(k'_\bullet - k_\bullet) + 1\big).$$

$$\overset{(4.27.1)}{=} \frac{k-2}{2} - \big(k_{\max\bullet} - k_{\mathrm{mid}\bullet} - 1\big) + \big(k_{\mathrm{mid}\bullet} - k_{\min\bullet} + 1\big) - \frac{k_\bullet - k_{\mathrm{mid}\bullet} - 1 - \mathrm{Dig}(k_\bullet - k_{\mathrm{mid}\bullet} - 1)}{p-1}$$
$$- \frac{k_{\max\bullet} - k_\bullet - \mathrm{Dig}(k_{\max\bullet} - k_\bullet)}{p-1} + \frac{k_{\mathrm{mid}\bullet} - k_{\min\bullet} + 1 - \mathrm{Dig}(k_\bullet - k_{\min\bullet}) + \mathrm{Dig}(k_\bullet - k_{\mathrm{mid}\bullet} - 1)}{p-1}$$

(5.2.1)
$$= \frac{k-2}{2} + \frac{p(2k_{\mathrm{mid}\bullet} - k_{\max\bullet} - k_{\min\bullet} + 2)}{p-1} + \frac{\mathrm{Dig}(k_{\max\bullet} - k_\bullet) - \mathrm{Dig}(k_\bullet - k_{\min\bullet}) + 2\,\mathrm{Dig}(\ell-1)}{p-1}.$$

Here $\mathrm{Dig}(m)$ is the sum of all digits in the p-based expression of m, and we made use of formula (4.27.1) for sums of valuations of continuous integers, and also noted that $k_\bullet - k_{\mathrm{mid}\bullet} = \ell$. One can play the digital expansion game to deduce that

$$pk_{\min\bullet} + \mathrm{Dig}(k_\bullet - k_{\min\bullet}) = \tilde{k}_{\min\bullet} + \mathrm{Dig}(pk_\bullet - \tilde{k}_{\min\bullet}). \tag{5.2.2}$$

So (5.2.1) is equal to

(5.2.3)
$$\frac{k-2}{2} + \frac{2pk_{\mathrm{mid}\bullet} - pk_{\max\bullet} - \tilde{k}_{\min\bullet} + 2p}{p-1} + \frac{\mathrm{Dig}(k_{\max\bullet} - k_\bullet) - \mathrm{Dig}(pk_\bullet - \tilde{k}_{\min\bullet}) + 2\,\mathrm{Dig}(\ell-1)}{p-1}.$$

By Lemma-Notation 4.12,

$$k_{\mathrm{mid}\bullet} = k_\bullet - \ell, \quad k_{\max\bullet} = \tfrac{p+1}{2}(k_\bullet + 1 - \delta_\varepsilon - \ell) + \beta_{[n]} - 1,$$
$$\text{and} \quad \tilde{k}_{\min\bullet} = \tfrac{p+1}{2}(k_\bullet - \ell + \delta_\varepsilon) - \beta_{[n-1]} + 1.$$

Then the sum of the first two terms in (5.2.3) is equal to

$$\frac{k-2}{2} + \frac{2p\big(k_\bullet - \ell\big) - p\big(\frac{p+1}{2}(k_\bullet + 1 - \delta_\varepsilon - \ell) + \beta_{[n]} - 1\big) - \big(\frac{p+1}{2}(k_\bullet - \ell + \delta_\varepsilon) - \beta_{[n-1]} + 1\big) + 2p}{p-1}$$
$$= \frac{p-1}{2}(\ell-1) + \frac{k_\varepsilon - 2}{2} + 1 + \frac{p+1}{2}\delta_\varepsilon - \beta_{[n]} + \frac{\beta_{[n-1]} - \beta_{[n]} + \frac{p+1}{2}}{p-1}.$$

We can list the values of $\frac{k_\varepsilon - 2}{2} + 1 + \frac{p+1}{2}\delta_\varepsilon - \beta_{[n]}$ and $\beta_{[n-1]} - \beta_{[n]} + \frac{p+1}{2}$ in the following table.

		$\frac{k_\varepsilon-2}{2}+1+\frac{p+1}{2}\delta_\varepsilon-\beta_{[n]}$	$\beta_{[n-1]}-\beta_{[n]}+\frac{p+1}{2}$
$a+s_\varepsilon<p-1$	n even	$\frac{1}{2}(a+2)$	$a+2$
	n odd	$\frac{1}{2}(p-1-a)$	$p-1-a$
$a+s_\varepsilon\geq p-1$	n even	$\frac{1}{2}(p-1-a)$	$p-1-a$
	n odd	$\frac{1}{2}(a+2)$	$a+2$

We set $\theta := \beta_{[n-1]}-\beta_{[n]}+\frac{p+1}{2}$, which is equal to either $a+2$ or $p-1-a$; in particular, $3\leq\theta\leq p-2$. Then the sum of the first two terms in (5.2.3) is equal to

$$\frac{p-1}{2}(\ell-1)+\frac{1}{2}\cdot\theta+\frac{\theta}{p-1}.$$

On the other hand, for the digital sum in the last term of (5.2.3), if we set

$$\eta := \frac{p-1}{2}k_\bullet-\frac{p+1}{2}\delta_\varepsilon+\beta_{[n]}-1, \tag{5.2.4}$$

then we can write in a uniform fashion that

$$k_{\max\bullet}-k_\bullet=\eta-\frac{p+1}{2}(\ell-1),\quad pk_\bullet-\tilde{k}_{\min\bullet}=\eta+\theta+\frac{p+1}{2}(\ell-1). \tag{5.2.5}$$

To sum up, we deduce from (5.2.3) and the discussion above that $\Delta'_{k,\ell}-\Delta'_{k,\ell-1}$ is equal to

$$\frac{(p-1)(\ell-1)+\theta}{2}+\frac{\theta+\mathrm{Dig}\left(\eta-\frac{p+1}{2}(\ell-1)\right)+2\,\mathrm{Dig}(\ell-1)-\mathrm{Dig}\left(\eta+\theta+\frac{p+1}{2}(\ell-1)\right)}{p-1}. \tag{5.2.6}$$

But it is clear that

$$\begin{aligned}
&\theta+\mathrm{Dig}\left(\eta-\tfrac{p+1}{2}(\ell-1)\right)+2\,\mathrm{Dig}(\ell-1)\\
=\;&\mathrm{Dig}\left(\eta-\tfrac{p+1}{2}(\ell-1)\right)+\mathrm{Dig}(\ell-1)+\mathrm{Dig}(p\ell-p+\theta)\\
\geq\;&\mathrm{Dig}\left(\eta+\theta+\tfrac{p+1}{2}(\ell-1)\right). && (5.2.7)
\end{aligned}$$

It follows that

$$\Delta'_{k,\ell}-\Delta'_{k,\ell-1}\geq\tfrac{1}{2}\big((p-1)(\ell-1)+\theta\big)\geq\tfrac{p-1}{2}(\ell-1)+\tfrac{3}{2}. \qquad\square$$

The following corollary in fact slightly strengthens the previous lemma. Despite the technical nature, it will be used in multiple important proofs later (for Proposition 5.16 and for results in the forthcoming paper).

Corollary 5.3. *Let $k=k_\varepsilon+(p-1)k_\bullet$ and $\ell=1,\dots,\frac{1}{2}d_k^{\mathrm{new}}$ be as above. Let $k'=k_\varepsilon+(p-1)k'_\bullet$ be any weight such that*

$$d_{k'}^{\mathrm{ur}},\ \text{or } d_{k'}^{\mathrm{Iw}}-d_{k'}^{\mathrm{ur}}\ \text{belongs to}\ \left(\tfrac{1}{2}d_k^{\mathrm{Iw}}-\ell,\tfrac{1}{2}d_k^{\mathrm{Iw}}+\ell\right), \tag{5.3.1}$$

or $\frac{1}{2}d_{k'}^{\mathrm{Iw}}\in\left[\frac{1}{2}d_k^{\mathrm{Iw}}-\ell,\frac{1}{2}d_k^{\mathrm{Iw}}+\ell\right]$, then we have

$$\Delta'_{k,\ell}-\Delta'_{k,\ell-1}-v_p(w_k-w_{k'})\geq\frac{1}{2}+\frac{p-1}{2}(\ell-1)-\left\lfloor\frac{\ln((p+1)\ell)}{\ln p}\right\rfloor\geq\frac{1}{2}(2\ell-1), \tag{5.3.2}$$

except when $\ell=1$, the last inequality fails but $\Delta'_{k,1}-\Delta'_{k,0}-v_p(w_k-w_{k'})\geq\frac{1}{2}$ still holds.

When $p\geq 7$ and $\ell\geq 2$, we have $\Delta'_{k,\ell}-\Delta'_{k,\ell-1}-v_p(w_k-w_{k'})\geq\frac{1}{2}(2\ell+1)$.

Proof. The second inequality in (5.3.2) is obvious (when $\ell > 1$); we will prove the first inequality. (Similarly, the last statement also follows from the first inequality.)

We first treat the easy case: $\frac{1}{2}d_{k'}^{\mathrm{Iw}} \in \big[\frac{1}{2}d_k^{\mathrm{Iw}} - \ell,\ \frac{1}{2}d_k^{\mathrm{Iw}} + \ell\big]$. Since $\frac{1}{2}d_k^{\mathrm{Iw}} - \frac{1}{2}d_{k'}^{\mathrm{Iw}} = k_\bullet - k'_\bullet$, we deduce that $|k'_\bullet - k_\bullet| \leq \ell$. In particular, $v_p(w_{k_\bullet} - w_{k'_\bullet}) \leq 1 + \big\lfloor \ln \ell / \ln p \big\rfloor$. So the first inequality of (5.3.2) follows from Lemma 5.2 immediately.

We now assume (5.3.1) holds. First note that, by Remark 4.13, the condition (5.3.1) is equivalent to

$$(5.3.3)\quad k'_\bullet \in \big[k_{\min\bullet}(\tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell), k_{\min\bullet}(\tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell - 1)\big) \cup \big(k_{\max\bullet}(\tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell), k_{\max\bullet}(\tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell - 1)\big],$$

Using the equality (2) in § 4.22:

$$(5.3.4)\qquad k_{\max\bullet}(d_k^{\mathrm{Iw}} - \ell') - k_\bullet = pk_\bullet - \tilde{k}_{\min\bullet}(\ell' - 1), \quad \text{for } \ell' = \tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell + 1 \text{ or } \tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell,$$

we see that for every $k'_\bullet \in \big[k_{\min\bullet}(\frac{1}{2}d_k^{\mathrm{Iw}} - \ell), k_{\min\bullet}(\frac{1}{2}d_k^{\mathrm{Iw}} + \ell - 1)\big)$, there exists $k''_\bullet \in \big(k_{\max\bullet}(\frac{1}{2}d_k^{\mathrm{Iw}} - \ell), k_{\max\bullet}(\frac{1}{2}d_k^{\mathrm{Iw}} + \ell - 1)\big]$ such that $k''_\bullet - k_\bullet = p(k_\bullet - k'_\bullet)$. So to prove the lemma, one may assume that

$$(5.3.5)\qquad k'_\bullet \in \big(k_{\max\bullet}(\tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell), k_{\max\bullet}(\tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell - 1)\big].$$

Now we import the notation from the proof of Lemma 5.2: $n := \frac{1}{2}d_k^{\mathrm{Iw}} - \ell$, $\theta := \beta_{[n-1]} - \beta_{[n]} + \frac{p+1}{2}$, and η as defined in (5.2.4). Then condition (5.3.5) is equivalent to

$$k'_\bullet \in \big(k_{\max\bullet}(n),\ k_{\max\bullet}(n + 2\ell - 1)\big].$$

So by (5.2.5), we deduce that

$$\begin{aligned} k'_\bullet - k_\bullet &> \eta - \tfrac{p+1}{2}(\ell - 1), \qquad \text{and} \\ k'_\bullet - k_\bullet &\leq \eta - \tfrac{p+1}{2}(\ell - 1) + \big(k_{\max\bullet}(n + 2\ell - 1) - k_{\max\bullet}(n)\big) \\ &= \eta - \tfrac{p+1}{2}(\ell - 1) + (p+1)(\ell - 1) + \theta = \eta + \theta + \tfrac{p+1}{2}(\ell - 1), \end{aligned}$$

where the first equality uses the formula for $k_{\max\bullet}(n)$ in Lemma-Notation 4.12(2).

Now we argue as in Lemma 5.2 till (5.2.6) to get

(5.3.6)

$$\Delta'_{k,\ell} - \Delta'_{k,\ell-1} = \frac{(p-1)(\ell-1) + \theta}{2} + \frac{\theta + \mathrm{Dig}\big(\eta - \frac{p+1}{2}(\ell - 1)\big) + 2\,\mathrm{Dig}(\ell - 1) - \mathrm{Dig}\big(\eta + \theta + \frac{p+1}{2}(\ell - 1)\big)}{p-1}.$$

Then we simply note that

$$\begin{aligned} &\tfrac{1}{p-1}\Big(\theta + \mathrm{Dig}\big(\eta - \tfrac{p+1}{2}(\ell - 1)\big) + 2\,\mathrm{Dig}(\ell - 1)\Big) \\ \geq\ &\tfrac{1}{p-1}\Big(\mathrm{Dig}\big(\eta - \tfrac{p+1}{2}(\ell - 1)\big) + \mathrm{Dig}((p+1)(\ell - 1) + \theta)\Big) \\ \geq\ &\tfrac{1}{p-1}\mathrm{Dig}\big(\eta + \theta + \tfrac{p+1}{2}(\ell - 1)\big) + v_p(k'_\bullet - k_\bullet) - \lfloor \ln((p+1)(\ell - 1) + \theta)/\ln p \rfloor. \end{aligned}$$

Here the last inequality uses the following elementary fact (proved in Lemma 5.4 below): for two positive integers A and B, we have

$$\frac{\mathrm{Dig}(A) + \mathrm{Dig}(B) - \mathrm{Dig}(A+B)}{p-1} + \Big\lfloor \frac{\ln B}{\ln p} \Big\rfloor \geq \max_{A < x \leq A+B} \big\{v_p(x)\big\}.$$

From this, we deduce that

$$\Delta'_{k,\ell} - \Delta'_{k,\ell-1} \geq \tfrac{(p-1)(\ell-1)+\theta}{2} + v_p(k'_\bullet - k_\bullet) - \big\lfloor \ln((p+1)(\ell - 1) + \theta)/\ln p \big\rfloor.$$

The corollary follows when $\ell > 1$ because $\theta \in [3, p-2]$.

In the special case of $\ell = 1$, the above inequality becomes

$$\Delta'_{k,1} - \Delta'_{k,0} \geq \frac{\theta}{2} + v_p(k'_\bullet - k_\bullet)$$

as $\theta < p$. So we deduce that

$$\Delta'_{k,1} - \Delta'_{k,0} - v_p(w_k - w_{k'}) \geq \frac{\theta}{2} - 1 \geq \frac{1}{2}. \qquad \square$$

Lemma 5.4. *For two positive integers A and B, we have*

$$\frac{\mathrm{Dig}(A) + \mathrm{Dig}(B) - \mathrm{Dig}(A+B)}{p-1} + \Big\lfloor \frac{\ln B}{\ln p} \Big\rfloor \geq \max_{A < x \leq A+B} \{v_p(x)\}.$$

Proof. Let $n = \lfloor \frac{\ln B}{\ln p} \rfloor$. First note that

$$\frac{\mathrm{Dig}(A) + \mathrm{Dig}(B) - \mathrm{Dig}(A+B)}{p-1} = v_p((A+B)!) - v_p(A!) - v_p(B!) = v_p\Big(\frac{\prod_{j=1}^{B}(A+j)}{B!}\Big).$$

If $\max_{A<x\leq A+B}\{v_p(x)\} \leq n$, then we are done. Otherwise, suppose

$$n < v_p(A+i) = \max_{A<x\leq A+B}\{v_p(x)\}$$

for some $i \in [1, B]$. Note that this implies that $v_p(A+i \pm j) = v_p(j)$ for $j \in [1, B]$. It follows that

$$v_p\Big(\frac{\prod_{j=1}^{B}(A+j)}{B!}\Big) = v_p((i-1)!) + v_p((B-i)!) + v_p(A+i) - v_p(B!).$$

For every $\ell \geq 1$, it is clear that $\lfloor \frac{i-1}{p^\ell} \rfloor + \lfloor \frac{B-i}{p^\ell} \rfloor - \lfloor \frac{B}{p^\ell} \rfloor \geq -1$. We therefore conclude that

$$v_p\Big(\frac{\prod_{j=1}^{B}(A+j)}{B!}\Big) \geq v_p(A+i) - n,$$

yielding the desired result. $\square$

In our forthcoming paper where we prove the local ghost conjecture, we need to establish similar bounds for $\Delta_{k,\ell} - \Delta'_{k,\ell-1}$ as opposed to $\Delta'_{k,\ell} - \Delta'_{k,\ell-1}$. The following three lemmas provide results in this direction.

Lemma 5.5. *Fix $k = k_\varepsilon + (p-1)k_\bullet$ and $\ell \in \{1, 2, \dots, \frac{1}{2}d_k^{\mathrm{new}}\}$, and define η and θ as in the proof of Lemma 5.2. Let β denote the maximal p-adic valuations of integers in $\big[\eta - \frac{p+1}{2}(\ell - 1),\ \eta + \theta + \frac{p+1}{2}(\ell-1)\big]$. Then we have*

$$\Delta'_{k,\ell} - \Delta'_{k,\ell-1} \leq \tfrac{p-1}{2} \cdot \ell + \tfrac{3}{2} + \beta + \lfloor \ln \ell / \ln p \rfloor$$

Proof. We first point out that, when $\ell > 1$, β is automatically greater than or equal to $\lfloor \ln \ell / \ln p \rfloor + 1$. By (5.2.6) and the estimate $\theta \leq p-2$, it remains to show that

$$\frac{\mathrm{Dig}\left(\eta - \frac{p+1}{2}(\ell-1)\right) + \mathrm{Dig}(\ell-1) + \mathrm{Dig}(p(\ell-1)+\theta) - \mathrm{Dig}\left(\eta + \theta + \frac{p+1}{2}(\ell-1)\right)}{p-1} \leq \beta + 2 + \Big\lfloor \frac{\ln \ell}{\ln p} \Big\rfloor.$$

The left hand side exactly counts the number of times we carry over digits to the next one when computing the sum of numbers $\eta - \frac{p+1}{2}(\ell-1)$, $(\ell-1)$, and $p(\ell-1)+\theta$. This is clear. $\square$

Lemma 5.6. *Fix $k = k_\varepsilon + (p-1)k_\bullet$ and $\ell \in \{1, \dots, \frac{1}{2}d_k^{\mathrm{new}} - 1\}$. Let $\theta_\ell \in \{a+2, p-1-a\}$ be the θ as in the proof of Lemma 5.2 for ℓ. Then we have*

$$\Delta'_{k,\ell+1} - 2\Delta'_{k,\ell} + \Delta'_{k,\ell-1} \geq p - 1 - \theta_\ell - 2v_p(\ell) \geq 1 - 2v_p(\ell). \tag{5.6.1}$$

Proof. Write θ_ℓ and η_ℓ (resp. $\theta_{\ell+1}$ and $\eta_{\ell+1}$) as in the proof of Lemma 5.2 for ℓ (resp. $\ell+1$). By (5.2.6), we deduce that

$$\Delta'_{k,\ell+1} - 2\Delta'_{k,\ell} + \Delta'_{k,\ell-1} = \frac{p-1}{2} + \frac{\theta_{\ell+1} - \theta_\ell}{2} + \frac{\mathrm{Dig}\left(\eta_{\ell+1} - \frac{p+1}{2}\ell\right) - \mathrm{Dig}\left(\eta_\ell - \frac{p+1}{2}(\ell-1)\right)}{p-1}$$
$$+ \frac{2\big(\mathrm{Dig}(\ell) - \mathrm{Dig}(\ell-1)\big)}{p-1} + \frac{\theta_{\ell+1} - \theta_\ell + \mathrm{Dig}\left(\eta_\ell + \theta_\ell + \frac{p+1}{2}(\ell-1)\right) - \mathrm{Dig}\left(\eta_{\ell+1} + \theta_{\ell+1} + \frac{p+1}{2}\ell\right)}{p-1}.$$

Write $n = \frac{1}{2}d_k^{\mathrm{Iw}} - \ell$ as in the proof of Lemma 5.2. We note that $\theta_\ell + \theta_{\ell+1} = p+1$ by their definitions and, for the fourth term, we have

$$\frac{2}{p-1}\big(\mathrm{Dig}(\ell) - \mathrm{Dig}(\ell-1)\big) = \frac{2}{p-1} - 2v_p(\ell).$$

For the third term, we note that

$$\left(\eta_\ell - \tfrac{p+1}{2}(\ell-1)\right) - \left(\eta_{\ell+1} - \tfrac{p+1}{2}\ell\right) = \tfrac{p+1}{2} + \eta_\ell - \eta_{\ell+1} = \tfrac{p+1}{2} + \beta_{[n]} - \beta_{[n+1]} = \theta_{\ell+1}.$$

Similarly, for the fifth term, we have

$$\left(\eta_{\ell+1} + \theta_{\ell+1} + \tfrac{p+1}{2}\ell\right) - \left(\eta_\ell + \theta_\ell + \tfrac{p+1}{2}(\ell-1)\right) = \tfrac{p+1}{2} + \beta_{[n+1]} - \beta_{[n]} + \theta_{\ell+1} - \theta_\ell = \theta_{\ell+1}.$$

So the sum of the third term and the fifth term above is greater than or equal to

$$\frac{1}{p-1}\Big(- \theta_{\ell+1} + \big(\theta_{\ell+1} - \theta_\ell\big) - \theta_{\ell+1}\Big) = -\frac{1}{p-1}\big(\theta_{\ell+1} + \theta_\ell\big) = -\frac{1}{p-1} \cdot (p+1).$$

Combining all inequalities above gives

$$\begin{aligned}\Delta'_{k,\ell+1} - 2\Delta'_{k,\ell} + \Delta'_{k,\ell-1} &\geq \frac{p-1}{2} + \frac{p+1-2\theta_\ell}{2} + \Big(\frac{2}{p-1} - 2v_p(\ell)\Big) - \frac{p+1}{p-1} \\ &= p - 1 - \theta_\ell - 2v_p(\ell)\end{aligned}$$

The inequality (5.6.1) follows from combining the above inequalities. □

Notation 5.7. For two points $P = (m, a), Q = (n, b) \in \mathbb{R}^2$, we write $\overline{PQ}$ for the line segment connecting them.

In the rest of this paper, we will frequently use the following concept. Suppose that we have a list of points

$$P_i = (i, A_i) \quad \text{with} \quad A_i \in \mathbb{R}, \text{ for } i = m, m+1, \dots, n$$

(or some subset of this list). We may *shift them down relative to a linear function $y = ax + b$* (for some $a, b \in \mathbb{R}$), by transforming them into

$$Q_i = (i, A_i - ai - b) \quad \text{for} \quad i = m, m+1, \dots, n$$

(or some subset of this list). This way, for every $i \neq j$, the slope of $\overline{Q_iQ_j}$ is precisely the slope of $\overline{P_iP_j}$ minus a. In particular,

(1) if the convex hull of all P_i's is the straight line $\overline{P_mP_n}$, then the convex hull of all Q_i's is the straight line $\overline{Q_mQ_n}$;

(2) for a fixed i_0, the point P_{i_0} is a vertex of the convex hull of the P_i's if and only if the point Q_{i_0} is a vertex of the convex hull of the Q_i's; and
(3) if $i \in [m, n]$, the distance from P_i to the point on $\overline{P_m P_n}$ with x-coordinate i, is the same as the distance from Q_i to the point on $\overline{Q_m Q_n}$ with x-coordinate i.

We shall freely use these elementary facts without make reference for the rest of this paper.

Lemma 5.8. *Assume $p \geq 7$. For $k = k_\varepsilon + (p-1)k_\bullet$ and $\ell = 1, \dots, \frac{1}{2}d_k^{\mathrm{new}}$, we have*

$$\Delta'_{k,\ell} - \Delta_{k,\ell} \leq 3\big(\ln \ell / \ln p\big)^2. \tag{5.8.1}$$

Moreover, if we only assume $p \geq 5$, the following statment holds: when $\ell < 2p$ but $\ell \neq p$, $\Delta'_{k,\ell} = \Delta_{k,\ell}$; and when $\ell = p$, $\Delta'_{k,\ell} - \Delta_{k,\ell} \leq 1$.

Proof. If $(\ell, \Delta'_{k,\ell})$ lies on $\underline{\Delta}_k$, the lemma holds trivially. Now we assume that $(\ell, \Delta'_{k,\ell})$ does not lie on $\underline{\Delta}_k$. Assume that ℓ_- is the largest integer smaller than ℓ such that $P_- := (\ell_-, \Delta'_{k,\ell_-})$ lies on $\underline{\Delta}_k$, and ℓ_+ is the smallest integer larger than ℓ such that $P_+ := (\ell_+, \Delta'_{k,\ell_+})$ lies on $\underline{\Delta}_k$. Let λ denote the slope of the segment $\overline{P_- P_+}$. Then

$$\Delta'_{k,\ell_+} - \Delta'_{k,\ell_+-1} < \lambda < \Delta'_{k,\ell_-+1} - \Delta'_{k,\ell_-}. \tag{5.8.2}$$

By (5.6.1), for every $\ell' \in [\ell_- + 1, \ell_+ - 1]$, $\Delta'_{k,\ell'+1} - \Delta'_{\ell'} \geq \Delta'_{k,\ell'} - \Delta'_{k,\ell'-1} + 1$ whenever $v_p(\ell') = 0$. It is clear from (5.8.2) that there must be some $\ell' \in [\ell_- + 1, \ell_+ - 1]$ whose valuation is strictly positive. Let ℓ_0 denote the $\ell' \in [\ell_- + 1, \ell_+ - 1]$ with maximal valuation. Write $v_p(\ell_0) = m_0$.

We first note that the maximality of $v_p(\ell_0)$ implies that $v_p(\ell_0 + i) = v_p(i)$ for $1 \leq i \leq \ell_+ - \ell_0 - 1$ and $v_p(\ell_0 - i) = v_p(i)$ for $1 \leq i \leq \ell_0 - \ell_- - 1$. Then taking the sum of the first inequality of (5.6.1) over $\ell_- + 1, \ell_- + 2, \dots, \ell_+ - 1$ (and also noting that two consecutive θ_ℓ's have sum equal to $p+1$), we deduce that

$$\begin{aligned}
&(\Delta'_{k,\ell_+} - \Delta'_{k,\ell_+-1}) - (\Delta'_{k,\ell_-+1} - \Delta'_{k,\ell_-})\\
\geq\ & \frac{p-3}{2}(\ell_+ - \ell_- - 1) - \frac{p-5}{2} - 2\Big(\sum_{i=1}^{\ell_0-\ell_--1} v_p(i) + m_0 + \sum_{i=1}^{\ell_+-\ell_0-1} v_p(i)\Big)\\
\geq\ & \frac{p-3}{2}\big(\ell_+ - \ell_- - 1\big) - \frac{p-5}{2} - \frac{2}{p-1}\big((\ell_0 - \ell_- - 1) + (\ell_+ - \ell_0 - 1)\big) - 2m_0\\
\geq\ & \frac{p-4}{2}(\ell_+ - \ell_- - 2) + 1 - 2m_0.
\end{aligned}$$

Here the last inequality uses that $\frac{2}{p-1} \leq \frac{1}{2}$ because $p \geq 5$. Combining with (5.8.2), we obtain

$$m_0 \geq \frac{p-4}{4}(\ell_+ - \ell_- - 2) + 1. \tag{5.8.3}$$

This in particular shows that either $m_0 \geq 2$ (in which case $\ell \geq \ell_- \geq p^{m_0} - 4m_0 \geq 2p$) or $m_0 = 1$ which must be the case when $\ell_+ - 1 = \ell_- + 1 = \ell_0 = \ell$.

We claim that $\Delta'_{k,\ell_0} - \Delta_{k,\ell_0} \leq m_0^2$. The case $\ell = \ell_0$ and the last statement of the lemma follow directly from this inequality. First note that, as explained in Notation 5.7 (3), we may shift all points under consideration by a linear function; this will not affect our discussion (but gives a better visualization). Through such shifting, we may suppose $\Delta'_{k,\ell_0+1} = \Delta'_{k,\ell_0-1}$. Then we can iterate (the second inequality of) (5.6.1) to deduce that, for $1 \le n \le \ell_+ - \ell_0$

$$\Delta'_{k,\ell_0+n} - \Delta'_{k,\ell_0+n-1} \ge -m_0 + n - \tfrac{1}{2} - 2v_p((n-1)!) \ge -m_0 + \tfrac{n}{2}.$$

Symmetrically, for $1 \le n \le \ell_0 - \ell_-$, we have

$$\Delta'_{k,\ell_0+1-n} - \Delta'_{k,\ell_0-n} \le m_0 - \tfrac{n}{2}.$$

Then we can use these to deduce that

$$\max\{\Delta'_{k,\ell_0} - \Delta'_{k,\ell_-}, \Delta'_{k,\ell_0} - \Delta'_{k,\ell_+}\} \le \max_{n_0 \ge 1}\Big(\sum_{n=1}^{n_0}\big(m_0 - \tfrac{n}{2}\big)\Big) \le m_0^2.$$

This yields $\Delta'_{k,\ell_0} - \Delta_{k,\ell_0} \le m_0^2$.

Now we assume $\ell \ne \ell_0$; in particular, $m_0 \ge 2$. Without loss of generality we may assume that $\ell_- < \ell \le \ell_0$. Once again, we shift all points by a linear function with slope λ so that we may assume that $\lambda = 0$ and $\overline{P_-P_+}$ is horizontal. Iterating (5.6.1), for $1 \le n \le \ell_0 - \ell_- - 1$, we obtain

$$\Delta'_{k,\ell_-+n+1} - \Delta'_{k,\ell_-+n} \ge n - 2\sum_{i=\ell_-+1}^{\ell_-+n} v_p(i) \ge n - 2\sum_{i=\ell_-+1}^{\ell_0-1} v_p(i) = n - 2v_p((\ell_0 - \ell_- - 1)!)$$

Using (5.8.3), we get

$$n - 2v_p((\ell_0 - \ell_- - 1)!) \ge n - 2\cdot\frac{4}{(p-1)(p-4)}(m_0 - 1) \ge n - 2(m_0 - 1).$$

We thus deduce that

$$\Delta'_{k,\ell_0} - \Delta'_{k,\ell} \ge \sum_{n=\ell-\ell_-}^{\ell_0-\ell_--1}(n - 2m_0 + 2) > -2m_0^2 + m_0.$$

Since $\lambda = 0$, this implies that

$$\Delta'_{k,\ell} - \Delta_{k,\ell} < 3m_0^2 - m_0. \tag{5.8.4}$$

By (5.8.3) and the assumption $p \ge 7$, we get

$$\ell_0 - \ell \le \ell_+ - \ell_- - 2 < \frac{4}{3}m_0 < 2m_0.$$

As $\ell \ge \ell_0 - 2m_0 \ge p^{m_0} - 2m_0$,

$$\begin{aligned}
3\left(m_0^2 - (\frac{\ln(p^{m_0} - 2m_0)}{\ln p})^2\right) &= 3(m_0 + \frac{\ln(p^{m_0} - 2m_0)}{\ln p})(m_0 - \frac{\ln(p^{m_0} - 2m_0)}{\ln p}) \\
&\le 6m_0(m_0 - \frac{\ln(p^{m_0} - 2m_0)}{\ln p}) = 6m_0^2\cdot(-\frac{\ln(1 - \frac{2m_0}{p^{m_0}})}{m_0 \ln p}) \\
&\le 6m_0^2\cdot\frac{2\cdot\frac{2m_0}{p^{m_0}}}{m_0\ln p} = \frac{24m_0^2}{p^{m_0}\ln p} \le m_0
\end{aligned}$$

holds when $m_0 \ge 2$ and $p \ge 7$ (here we use the inequality $-\ln(1-x) \le 2x$ when $x \in (0, \frac{1}{2})$), the inequality (5.8.1) follows from this and (5.8.4).

□

As a consequence of the above estimate, Lemma 5.2 and Corollary 5.3 can be restated in the form of $\Delta_{k,\ell} - \Delta'_{k,\ell-1}$.

Corollary 5.9. *Let $k = k_\varepsilon + (p-1)k_\bullet$ and $\ell = 1, \dots, \frac{1}{2}d_k^{\mathrm{new}}$ be as above. Let $k' = k_\varepsilon + (p-1)k'_\bullet$ be any weight such that*

$$d_{k'}^{\mathrm{ur}},\ \textit{or}\ d_{k'}^{\mathrm{Iw}} - d_{k'}^{\mathrm{ur}}\ \textit{belongs to}\ \big(\tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell, \tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell\big),$$

or $\frac{1}{2}d_{k'}^{\mathrm{Iw}}$ belongs to $\big[\frac{1}{2}d_k^{\mathrm{Iw}} - \ell, \frac{1}{2}d_k^{\mathrm{Iw}} + \ell\big]$ then we have

$$\Delta_{k,\ell} - \Delta'_{k,\ell-1} - v_p(w_k - w_{k'}) \geq \frac{1}{2}(2\ell - 1). \tag{5.9.1}$$

Moreover, when $p \geq 7$ and $\ell \geq 2$, We have $\Delta_{k,\ell} - \Delta'_{k,\ell-1} - v_p(w_k - w_{k'}) \geq \frac{1}{2}(4\ell - 1)$.

Proof. The general case follows from applying the first statement repeatedly. For the first statement, if $\ell < 2p$ but $\ell \neq p$, $\Delta_{k,\ell} = \Delta'_{k,\ell}$ by Lemma 5.8; so (5.9.1) and the last statement are proved in Corollary 5.3. If $\ell \geq 2p$ or $\ell = p$, (5.9.1) also follows from Corollary 5.3 and Lemma 5.8 by noting that

$$\frac{1}{2} + \frac{p-1}{2}(\ell - 1) - \Big\lfloor \frac{\ln((p+1)\ell)}{\ln p} \Big\rfloor \geq \frac{1}{2}(2\ell - 1) + 3\Big(\frac{\ln \ell}{\ln p}\Big)^2 \tag{5.9.2}$$

holds when $\ell \geq 2p$ or $\ell = p$ if we replace the last term by 1 (citing the special case in Lemma 5.8 instead). When $p \geq 7$, (5.9.2) holds with $2\ell - 1$ replaced by $4\ell - 1$. $\square$

The following slight generalization of Corollary 5.9 is tailored for our need in the sequel of this series.

Corollary 5.10. *Assume $p \geq 7$. Take integers $\ell, \ell', \ell'' \in \{0, 1, \dots, \frac{1}{2}d_k^{\mathrm{new}}\}$ with $\ell \leq \ell' \leq \ell''$ and $\ell'' \geq 1$. Let $k' = k_\varepsilon + (p-1)k'_\bullet$ be a weight such that*

$$d_{k'}^{\mathrm{ur}}\ \textit{or}\ d_{k'}^{\mathrm{Iw}} - d_{k'}^{\mathrm{ur}}\ \textit{belongs to}\ \big[\tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell', \tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell'\big],$$

then we have

$$\Delta_{k,\ell''} - \Delta'_{k,\ell} - (\ell'' - \ell') \cdot v_p(w_k - w_{k'}) \geq (\ell' - \ell) \cdot \Big\lfloor \frac{\ln((p+1)\ell'')}{\ln p} + 1 \Big\rfloor + \frac{1}{2}\big(\ell''^2 - \ell^2\big). \tag{5.10.1}$$

Moreover, if there exists k' such that $v_p(w_{k'} - w_k) \geq \big\lfloor \frac{\ln((p+1)\ell'')}{\ln p} + 2 \big\rfloor$, then there at most two such k' satisfying $v_p(w_{k'} - w_k) \geq \big\lfloor \frac{\ln((p+1)\ell'')}{\ln p} + 2 \big\rfloor$ and

$$d_{k'}^{\mathrm{ur}}\ \textit{or}\ d_{k'}^{\mathrm{Iw}} - d_{k'}^{\mathrm{ur}}\ \textit{belongs to}\ \big(\tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell'', \tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell''\big).$$

Suppose that there are two such k''s, up to swapping k'_1 and k'_2, we have $d_{k'_1}^{\mathrm{ur}}, d_{k'_2}^{\mathrm{Iw}} - d_{k'_2}^{\mathrm{ur}} \in \big[\frac{1}{2}d_k^{\mathrm{Iw}} - \ell', \frac{1}{2}d_k^{\mathrm{Iw}} + \ell'\big]$; and in $\{d_{k'_1}^{\mathrm{ur}}, d_{k'_2}^{\mathrm{Iw}} - d_{k'_2}^{\mathrm{ur}}\}$, one element is $\geq \frac{1}{2}d_k^{\mathrm{Iw}}$ and one element is $\leq \frac{1}{2}d_k^{\mathrm{Iw}}$.

Proof. We first prove the moreover part. By the proof of Corollary 5.3, the condition $d_{k'}^{\mathrm{ur}} \in (\frac{1}{2}d_k^{\mathrm{Iw}} - \ell'', \frac{1}{2}d_k^{\mathrm{Iw}} + \ell'')$ implies that

$$k'_\bullet - k_\bullet \in \Big(\eta - \frac{p+1}{2}(\ell'' - 1),\ \eta + \theta + \frac{p+1}{2}(\ell'' - 1)\Big].$$

So if one such k' satisfies $v_p(w_{k'}-w_k) \geq \lfloor \frac{\ln((p+1)\ell'')}{\ln p} + 2 \rfloor$, there is only such k'. On the other hand, if $d_{k'}^{\mathrm{Iw}} - d_{k'}^{\mathrm{ur}} \in (\frac{1}{2}d_k^{\mathrm{Iw}} - \ell'', \frac{1}{2}d_k^{\mathrm{Iw}} + \ell'')$, then there exists k'' such that

$$k''_\bullet - k_\bullet = p(k_\bullet - k'_\bullet) \quad \text{and} \quad d_{k''}^{\mathrm{ur}} \in \big(\tfrac{1}{2}d_k^{\mathrm{Iw}} - \ell'', \tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell''\big).$$

Moreover, for such k'', $d_{k'}^{\mathrm{Iw}} - d_{k'}^{\mathrm{ur}} \in (\frac{1}{2}d_k^{\mathrm{Iw}} - \ell'', \frac{1}{2}d_k^{\mathrm{Iw}})$ if and only if $d_{k''}^{\mathrm{ur}} \in (\frac{1}{2}d_k^{\mathrm{Iw}}, \frac{1}{2}d_k^{\mathrm{Iw}} + \ell'')$, by § 4.22(2). From this, one easily deduce the moreover part of the corollary.

Now, we move to the general case. If $\ell'' = 1$, (5.10.1) follows from Corollary 5.9. If $\ell'' \geq 2$, we apply the stronger statement in Corollary 5.9 iteratively to deduce that

$$\begin{aligned}\Delta_{k,\ell''} - \Delta'_{k,\ell} - (\ell'' - \ell') \cdot v_p(w_k - w_{k'}) &\geq (\ell''^2 - \ell^2) + (\ell' - \ell) \\ &\geq \frac{1}{2}\big(\ell''^2 - \ell^2\big) + (\ell' - \ell) \cdot \Big\lfloor \frac{\ln((p+1)\ell'')}{\ln p} + 1 \Big\rfloor.\end{aligned}$$

The last inequality follows from $\frac{\ell''+\ell}{2} \geq \lfloor \frac{\ln((p+1)\ell'')}{\ln p} \rfloor$. □

Definition 5.11. Fix ε as above and fix $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$. For each $k = k_\varepsilon + (p-1)k_\bullet$, we let $L_{w_\star,k}$ denote the largest number (if it exists) in $\{1, \dots, \frac{1}{2}d_k^{\mathrm{new}}\}$ such that

$$v_p(w_\star - w_k) \geq \Delta_{k, L_{w_\star,k}} - \Delta_{k, L_{w_\star,k}-1}. \tag{5.11.1}$$

When such $L_{w_\star,k}$ exists (in which case $(L_{w_\star,k}, \Delta_{k,L_{w_\star,k}})$ must be a vertex of $\underline{\Delta}_k$ of Notation 5.1), we call the open interval $\mathrm{nS}_{w_\star,k} := \big(\frac{1}{2}d_k^{\mathrm{Iw}} - L_{w_\star,k}, \frac{1}{2}d_k^{\mathrm{Iw}} + L_{w_\star,k}\big)$ the *near-Steinberg range* for the pair $(w_\star, k)$; we write $\overline{\mathrm{nS}}_{w_\star,k} := \big[\frac{1}{2}d_k^{\mathrm{Iw}} - L_{w_\star,k}, \frac{1}{2}d_k^{\mathrm{Iw}} + L_{w_\star,k}\big]$ for the corresponding closed interval. When no such $L_{w_\star,k}$ exists, $\mathrm{nS}_{w_\star,k} = \overline{\mathrm{nS}}_{w_\star,k} = \emptyset$.

For a positive integer n, we say $(\varepsilon, w_\star, n)$ or simply $(w_\star, n)$ is *near-Steinberg* if n belongs to the near-Steinberg range $\mathrm{nS}_{w_\star,k}$ for some k. This is equivalent to the existence of some $k \equiv k_\varepsilon \bmod (p-1)$ such that $n \in (d_k^{\mathrm{ur}}, d_k^{\mathrm{Iw}} - d_k^{\mathrm{ur}})$ and

$$v_p(w_\star - w_k) \geq \Delta_{k, |n - \frac{1}{2}d_k^{\mathrm{Iw}}|+1} - \Delta_{k, |n - \frac{1}{2}d_k^{\mathrm{Iw}}|}.$$

Remark 5.12. Let us illustrate the meaning of the notion near-Steinberg in a more intuitive way: by Proposition 4.18 (4) the Newton polygon $\mathrm{NP}(G(w_k, -))$ at the weight point w_k has a long straight-line over $(d_k^{\mathrm{ur}}, d_k^{\mathrm{Iw}} - d_k^{\mathrm{ur}})$ with Steinberg form slope, namely $\frac{k-2}{2}$. If a point $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$ is "close" to w_k, the "continuity" of the Newton polygon predicts that $\mathrm{NP}(G(w_\star, -))$ also has a long straight-line over the near-Steinberg range $\mathrm{nS}_{w_\star,k}$ (whose slope is "very often" equal to $\frac{k-2}{2}$ *but need not be so* in general). Condition (5.11.1) exactly quantifies the distance between $w_\star$ and w_k versus the length of the straight line. The ghost duality $\Delta_{k,-\ell} = \Delta_{k,\ell}$ manifests itself as the fact that the x-coordinates of these straight lines are always symmetric about $\frac{1}{2}d_k^{\mathrm{Iw}}$. See Example 5.13 as an illustration of this. Lemma 5.15 below may be viewed as an alternative characterization of the condition (5.11.1).

We also point out that, although not very often, it is possible that for a near-Steinberg pair $(w_\star, n)$, n belongs to the near-Steinberg range for more than one k. See the proof of Theorem 5.19.

Example 5.13. We illustrate the previous remark with a concrete example. Continue with the setup of Example 2.26, with $p = 7$, $a = 2$, and $\epsilon = \omega^2 \times 1$ (so $s_\varepsilon = 4$, corresponding to the "anti-ordinary disc"). The first five terms of the ghost series are

$$\begin{aligned}
g_1(w) &= w - w_6,\\
g_2(w) &= (w-w_{12})(w-w_{18})(w-w_{24})(w-w_{30}),\\
g_3(w) &= (w-w_{18})^2(w-w_{24})^2(w-w_{30})^2(w-w_{36})(w-w_{42})(w-w_{48})(w-w_{54}),\\
g_4(w) &= (w-w_{18})(w-w_{24})^3(w-w_{30})^3(w-w_{36})^2\cdots(w-w_{54})^2(w-w_{60})\cdots(w-w_{78}),\\
g_5(w) &= (w-w_{24})^2(w-w_{30})^4(w-w_{36})^3\cdots(w-w_{54})^3(w-w_{60})^2\cdots(w-w_{78})^2\\
&\quad \cdot(w-w_{84})\cdots(w-w_{102}).
\end{aligned}$$

We are interested in the points $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$ that are close to w_{18}, for which we have $d_{18}^{\mathrm{ur}} = 1$, $\frac{1}{2}d_{18}^{\mathrm{Iw}} = 3$, and $\frac{1}{2}d_{18}^{\mathrm{new}} = 2$. The vertices of the polygon $\underline{\Delta}_{18}$ is given by

	$i=-2$	$i=-1$	$i=0$	$i=1$	$i=2$
$\Delta'_{18,i} = \Delta_{18,i}$	17	11	8	11	17

In particular, we have $\Delta_{18,2} - \Delta_{18,1} = 6$ and $\Delta_{18,1} - \Delta_{18,0} = 3$.

In the following table, we record the minimal valuation $v_p(g_i(w_\star))$ among all $w_\star$ such that $v_p(w_\star - w_{18})$ is the given value r.

when $v_p(w_\star - w_{18}) = r$	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$
$v_p(g_i(w_\star))$ with $r \in (2,\infty)$	1	$3+r$	$8+2r$	$19+r$	33
coordinate at $x=i$ of $\mathrm{NP}(G(w_\star,t))$ when $r\in(6,\infty)$	1	9	17	25	33
coordinate at $x=i$ of $\mathrm{NP}(G(w_\star,t))$ when $r\in(3,6)$	1	$3+r$	$11+r$	$19+r$	33
coordinate at $x=i$ of $\mathrm{NP}(G(w_\star,t))$ when $r\in(2,3)$	1	$3+r$	$8+2r$	$19+r$	33

When $v_p(w_\star - w_{18}) = r > 6$, the Newton polygon $\mathrm{NP}(G(w_\star,t))$ over $x \in [1,5]$ is a straight line of slope $8 = \frac{18-2}{2}$. As r decrease to the range $r \in (3,6)$, $\mathrm{NP}(G(w_\star,t))$ over $x\in[1,5]$ "breaks" into three segments with slopes $2+r$, 8, and $14-r$, and width 1, 2, and 1, respectively. As r further decreases to $r\in(2,3)$, $\mathrm{NP}(G(w_\star,t))$ over $x\in[1,5]$ has four segments, each with a distinct slope and width 1. The "phase-changing points" $r=3$ and $r=6$ correspond exactly to $\Delta_{18,2}-\Delta_{18,1}$ and $\Delta_{18,1}-\Delta_{18,0}$.

Readers might be mislead to think that the middle segment will always have slope $\frac{k-2}{2}$ before it completely breaks into segments of width 1. As pointed out already in Remark 5.12, this may fail when k is large and when the middle segment is "short". See the proof of Proposition 5.16 and Theorem 5.19 for the reasons.

Remark 5.14. We provide additional visualization on near-Steinberg ranges by trying to answer the question in vague terms: for each fixed n (and a fixed ε), what is the set of $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$ such that $(w_\star, n)$ is near-Steinberg?

First of all, there is a unique k, namely $k_{\mathrm{mid}}(n)$ such that $n = \frac{1}{2}d_k^{\mathrm{Iw}}$. So if $v_p(w_\star - w_{k_{\mathrm{mid}}(n)}) \geq \Delta_{k_{\mathrm{mid}}(n),1} - \Delta_{k_{\mathrm{mid}}(n),0}$, then $(w_\star,n)$ is near-Steinberg. By the proof of Lemma 5.2, the difference $\Delta_{k_{\mathrm{mid}}(n),1} - \Delta_{k_{\mathrm{mid}}(n),0} \geq \frac{a+2}{2}$ or $\frac{p-1-a}{2}$ (depending on the parity of n) and this inequality is often an equality (related to the inequality (5.2.7)). So this type of $w_\star$ is a ball around $w_{k_{\mathrm{mid}}(n)}$ of radius roughly $p^{-(a+2)/2}$ or $p^{-(p-1-a)/2}$.

Next, we consider $k = k_{\mathrm{mid}}(n) \pm (p-1)$. Then $(w_\star, n)$ is near-Steinberg for such k if and only if $v_p(w_\star - w_k) \geq \Delta_{k,2} - \Delta_{k,1}$, which by Lemma 5.2, is always greater than or equal to $\frac{p+1}{2}$

(with most of the case being equal). Then such $w_\star$ form a ball around each of $w_{k_{\mathrm{mid}}(n)\pm(p-1)}$ of radius roughly $p^{-(p+1)/2}$.

We may continue this process to see that the near-Steinberg range is roughly a union of these balls with centers at $w_{k_{\mathrm{mid}}(n)\pm(p-1)i}$ of radius roughly $p^{-(p+1)i/4}$. Of course, the precise radii of these balls are combinatorially complicated.

We end this remark with the following observations: for $(w_\star, n)$ to be near-Steinberg, we need $v_p(w_\star) \geq 1$. For a fixed n, if we only consider classical points $w_\star = w_{k'}$, then in the majority of the cases, $(w_{k'}, n)$ is not near-Steinberg (except for about $p^{-(a+2)/2}$ or $p^{-(p-1-a)/2}$ of cases, and this percentage is roughly constant even if n is large), and for the rest of the cases, most near-Steinberg pairs $(w_\star, n)$ correspond to a Steinberg range of length at least 2. Long near-Steinberg ranges are rare when considered with a fixed n.

Lemma 5.15. *The condition* (5.11.1) *implies that the lower convex hull of points*

$$\big(n,\ v_p(g_{n,\hat{k}}(w_k)) + m_n(k)\cdot v_p(w_\star - w_k)\big) \quad \textit{for} \quad n \in \overline{\mathrm{nS}}_{w_\star,k} \tag{5.15.1}$$

is a straight line with slope $\frac{k-2}{2}$.

Proof. Writing $n = \frac{1}{2}d_k^{\mathrm{Iw}} + \ell$, the points (5.15.1) maybe rewritten as

$$\big(\tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell,\ \Delta'_{k,\ell} + \ell\cdot\tfrac{k-2}{2} + (\tfrac{1}{2}d_k^{\mathrm{new}} - |\ell|)\cdot v_p(w_\star - w_k)\big) \quad \text{with} \quad \ell \in \big[-L_{w_\star,k}, L_{w_\star,k}\big].$$

Shifting these points down relative to the linear function (in the sense of Notation 5.7)

$$y = \tfrac{k-2}{2}(x - \tfrac{1}{2}d_k^{\mathrm{Iw}}) + \Delta_{k,L_{w_\star,k}} + (\tfrac{1}{2}d_k^{\mathrm{new}} - L_{w_\star,k})\cdot v_p(w_\star - w_k),$$

these points become

$$\big(\tfrac{1}{2}d_k^{\mathrm{Iw}} + \ell,\ \Delta'_{k,\ell} - \Delta_{k,L_{w_\star,k}} + (L_{w_\star,k} - |\ell|)\cdot v_p(w_\star - w_k)\big) \quad \text{with} \quad \ell \in \big[-L_{w_\star,k}, L_{w_\star,k}\big]. \tag{5.15.2}$$

Now, the condition (5.11.1) $v_p(w_\star - w_k) \geq \Delta_{k,L_{w_\star,k}} - \Delta_{k,L_{w_\star,k}-1}$ implies (and in fact equivalent to!) that

$$(L_{w_\star,k} - |\ell|)\cdot v_p(w_\star - w_k) \geq \Delta_{k,L_{w_\star,k}} - \Delta'_{k,\ell} \quad \text{for} \quad \ell \in [-L_{w_\star,k}, L_{w_\star,k}]$$

and the equality holds when $\ell = \pm L_{w_\star,k}$. So the convex hull of points (5.15.2) is a straight line of slope zero; so the convex hull of points (5.15.1) is $\frac{k-2}{2}$. The lemma is proved. $\square$

Before proceeding, we give a general tool to relate the Newton polygon of ghost series at one point $w_\star$ with that at another "special" point w_k. This will be frequently used in this section.

Proposition 5.16. *Let* $\mathrm{nS}_{w_\star,k} = \big(\frac{1}{2}d_k^{\mathrm{Iw}} - L_{w_\star,k},\ \frac{1}{2}d_k^{\mathrm{Iw}} + L_{w_\star,k}\big)$ *be a near-Steinberg range.*

(1) *For any integer* $k' = k_\varepsilon + (p-1)k'_\bullet \neq k$ *with* $v_p(w_{k'} - w_k) \geq \Delta_{k,L_{w_\star,k}} - \Delta_{k,L_{w_\star,k}-1}$, *we have the following exclusions*

$$\tfrac{1}{2}d_{k'}^{\mathrm{Iw}} \notin \overline{\mathrm{nS}}_{w_\star,k} \quad \textit{and} \quad d_{k'}^{\mathrm{ur}},\ d_{k'}^{\mathrm{Iw}} - d_{k'}^{\mathrm{ur}} \notin \mathrm{nS}_{w_\star,k}.$$

(2) *For any integer* $k' = k_\varepsilon + (p-1)k'_\bullet \neq k$ *such that* $v_p(w_{k'} - w_k) \geq \Delta_{k,L_{w_\star,k}} - \Delta_{k,L_{w_\star,k}-1}$, *the ghost multiplicity* $m_n(k')$ *is linear in* n *when* $n \in \overline{\mathrm{nS}}_{w_\star,k}$.

(3) *The following two lists of points*

$$P_n = \big(n, v_p(g_{n,\hat{k}}(w_\star))\big), \quad Q_n = \big(n, v_p(g_{n,\hat{k}}(w_k))\big) \quad \textit{with } n \in \overline{\mathrm{nS}}_{w_\star,k}$$

are shifts of each other relative to a linear function with some slope in

$$\mathbb{Z} + \mathbb{Z}\cdot\alpha,\ \textit{for } \alpha := \max\{v_p(w_\star - w_{k'}) | w_{k'} \textit{ is a zero of } g_n(w) \textit{ for some } n \in \mathrm{nS}_{w_\star,k}\}. \tag{5.16.1}$$

(4) *More generally, let* $\mathbf{k} := \{k, k_1, \dots, k_r\}$ *with* $k_i \equiv k_\varepsilon \bmod p-1$ *be a set of integers including* k*. Suppose that there is an interval* $[n_-, n_+]$ *such that, for any* $k' = k_\varepsilon + (p-1)k'_\bullet \notin \mathbf{k}$ *with* $v_p(w_{k'} - w_k) \geq v_p(w_\star - w_k)$*, the ghost multiplicity* $m_n(k')$ *is linear in* n *when* $n \in [n_-, n_+]$*. Then for any set of constants* A_n *with* $n \in [n_-, n_+]$*, the two lists of points*

$$P_n = \big(n, A_n + v_p(g_{n,\hat{\mathbf{k}}}(w_\star))\big), \quad Q_n = \big(n, A_n + v_p(g_{n,\hat{\mathbf{k}}}(w_k))\big) \quad \text{with } n \in [n_-, n_+]$$

are shifts of each other by a linear function with some slope in

$$\mathbb{Z}+\mathbb{Z}\cdot\beta, \text{ for } \beta := \max\{v_p(w_\star - w_{k'}) | w_{k'} \text{ is a zero of } g_{n,\hat{\mathbf{k}}}(w) \text{ for some } n \in \mathrm{nS}_{w_\star,k}\}. \tag{5.16.2}$$

(See Notation 4.17 for the definition of $g_{n,\hat{\mathbf{k}}}(w)$*.)*

Proof. (1) Write $L = L_{w_\star,k}$ for simplicity. Suppose that an integer $k' = k_\varepsilon + (p-1)k'_\bullet \neq k$ satisfies either $\frac{1}{2}d_{k'}^{\mathrm{Iw}} \in [\frac{1}{2}d_k^{\mathrm{Iw}} - L, \frac{1}{2}d_k^{\mathrm{Iw}} + L]$, or $d_{k'}^{\mathrm{ur}}$ or $d_{k'}^{\mathrm{Iw}} - d_{k'}^{\mathrm{ur}}$ belongs to $(\frac{1}{2}d_k^{\mathrm{Iw}} - L, \frac{1}{2}d_k^{\mathrm{Iw}} + L)$. Then Corollary 5.9 implies that

$$\Delta_{k,L} - \Delta'_{k,L-1} - v_p(w_{k'} - w_k) \geq \tfrac{1}{2}(2L-1) > 0.$$

In particular, $v_p(w_{k'} - w_k) < \Delta_{k,L} - \Delta_{k,L-1}$. So (1) holds.

(2) follows from (1) and the ghost zero multiplicity pattern in Definition 2.25. (In fact, we just need $\frac{1}{2}d_{k'}^{\mathrm{Iw}} \notin \mathrm{nS}_{w_\star,k}$ instead of $\frac{1}{2}d_{k'}^{\mathrm{Iw}} \notin \overline{\mathrm{nS}}_{w_\star,k}$; the stronger version of (1) is needed for later application.)

We next prove (4). We compare each $v_p(g_{n,\hat{\mathbf{k}}}(w_\star))$ with $v_p(g_{n,\hat{\mathbf{k}}}(w_k))$ by

$$v_p(g_{n,\hat{\mathbf{k}}}(w_\star)) - v_p(g_{n,\hat{\mathbf{k}}}(w_k)) = \sum_{\substack{k' = k_\varepsilon + (p-1)k'_\bullet \\ k' \notin \mathbf{k}}} m_n(k') \cdot \big(v_p(w_\star - w_{k'}) - v_p(w_k - w_{k'})\big). \tag{5.16.3}$$

We consider each term associated to $k' = k_\varepsilon + (p-1)k'_\bullet$, separating into two cases.

- If $v_p(w_{k'} - w_k) < v_p(w_\star - w_k)$ then $v_p(w_\star - w_{k'}) = v_p(w_k - w_{k'})$; thus the corresponding term in (5.16.3) has zero contribution to the sum.
- If $v_p(w_{k'} - w_k) \geq v_p(w_\star - w_k)$, the assumption in (4) implies that $m_n(k')$ is a linear function in n, so the contribution to the sum of this term is linear in n. Moreover, we note that $v_p(w_\star - w_{k'}) - v_p(w_k - w_{k'} \in \mathbb{Z} + \mathbb{Z}\cdot\beta$ by the definition of the number β, and hence the slope of this linear term belongs to (5.16.1).

Summing up, the right hand side of (5.16.3) is a linear function in n; so (4) holds.

Finally, (3) follows from a special case of (4) when $[n_-, n_+] = \overline{\mathrm{nS}}_{w_\star,k}$, $\mathbf{k} = \{k\}$, and (all $A_n = 0$); the condition in (4) is guaranteed by (2) by noting $v_p(w_\star - w_k) \geq \Delta_{k,L_{w_\star,k}} - \Delta_{k,L_{w_\star,k}-1}$. The proposition is proved. □

Remark 5.17. Aside from the standard manipulations of Newton polygons, an essential ingredient of the proof of Proposition 5.16 (1) is to use Corollary 5.3: even though $v_p(w_{k'} - w_k)$ for a ghost zero $w_{k'}$ can be sometimes large, but when that happens, $\Delta'_{k,L} - \Delta'_{k,L-1}$ is also large because it "contains $v_p(w_{k'} - w_k)$ as a summand".

The following immediate corollary of Proposition 5.16 will be the essential ingredient for a proof of that local ghost Conjecture 2.29 implies Breuil–Buzzard–Emerton Conjecture 1.12.

Corollary 5.18. *Let* $\mathrm{nS}_{w_\star,k} = \big(\frac{1}{2}d_k^{\mathrm{Iw}} - L_{w_\star,k}, \frac{1}{2}d_k^{\mathrm{Iw}} + L_{w_\star,k}\big)$ *be a near-Steinberg range.*

(1) *Assume that $w_\star$ is not a zero of $g_n(w)$ for any (or equivalently by Proposition 5.16 (2), some) $n \in \mathrm{nS}_{w_\star,k}$. Then the lower convex hull of points*

$$(n,\ v_p(g_n(w_\star))) \quad \textit{for} \quad n \in \overline{\mathrm{nS}}_{w_\star,k}$$

is a straight line. Moreover, the slope of this straight line belongs to

$$\tfrac{a}{2} + \mathbb{Z} + \mathbb{Z}\cdot\gamma,\ \textit{for } \gamma := \max\{v_p(w_\star - w_{k'}) | w_{k'} \textit{ is a zero of } g_n(w) \textit{ for some } n \in \mathrm{nS}_{w_\star,k}\}. \tag{5.18.1}$$

(2) *Assume that $w_\star = w_{k_1}$ is a zero of $g_n(w)$ for any (or equivalently by Proposition 5.16 (2), some) $n \in \mathrm{nS}_{w_\star,k}$. Then the lower convex hull of points*

$$(n,\ v_p(g_{n,\hat{k}_1}(w_{k_1}))) \quad \textit{for} \quad n \in \overline{\mathrm{nS}}_{w_\star,k}$$

is a straight line of slope in $\frac{a}{2} + \mathbb{Z}$.

Proof. (1) We can write $v_p(g_n(w_\star)) = v_p(g_{n,\hat{k}}(w_\star)) + m_n(k)v_p(w_\star - w_k)$ for each $n \in \overline{\mathrm{nS}}_{w_\star,k}$. Applying Proposition 5.16 (3), we are reduced to prove that the lower convex hull of

$$\big(n,\ v_p(g_{n,\hat{k}}(w_k)) + m_n(k)\cdot v_p(w_\star - w_k)\big) \quad \text{for} \quad n \in \overline{\mathrm{nS}}_{w_\star,k}$$

is a straight line with slopes in (5.18.1). But this is precisely Lemma 5.15 by noting $\frac{k-2}{2} \equiv \frac{k_\varepsilon - 2}{2} \equiv \frac{a+2s_\varepsilon}{2} \equiv \frac{a}{2} \bmod \mathbb{Z}$.

(2) If $k_1 = k$, then $\overline{\mathrm{nS}}_{w_k,k} = [d_k^{\mathrm{ur}}, d_k^{\mathrm{Iw}} - d_k^{\mathrm{ur}}]$ and this is the ghost duality (5.1.2). When $k_1 \neq k$, this follows from an argument similar to (1): for $\mathbf{k} = \{k, k_1\}$, write

$$v_p(g_{n,\hat{k}_1}(w_{k_1})) = v_p(g_{n,\hat{\mathbf{k}}}(w_{k_1})) + m_n(k)v_p(w_k - w_{k_1}).$$

Applying Proposition 5.16 (4) to the case $w_\star = w_{k_1}$, $[n_-, n_+] = \overline{\mathrm{nS}}_{w_{k_1},k}$ and $\mathbf{k}$, we reduce the proof of (2) to proving that the lower convex hull of points

$$\big(n,\ v_p(g_{n,\hat{\mathbf{k}}}(w_k)) + m_n(k)v_p(w_{k_1} - w_k)\big) \quad \text{for} \quad n \in \overline{\mathrm{nS}}_{w_\star,k}$$

is a straight line with slope in $\frac{a}{2} + \mathbb{Z}$. But this list of points is the same as

$$\big(n,\ v_p(g_{n,\hat{k}}(w_k)) + m_n(k)v_p(w_{k_1} - w_k) - m_n(k_1)v_p(w_{k_1} - w_k)\big) \quad \text{for} \quad n \in \overline{\mathrm{nS}}_{w_\star,k}.$$

Proposition 5.16 (2) says that $m_n(k_1)$ is linear in $n \in \overline{\mathrm{nS}}_{w_\star,k}$. So the list above is a linear shift of the list in Lemma 5.15 (with $w_\star = w_{k_1}$), and thus its convex hull is a straight line with slope in $\frac{k-2}{2} + \mathbb{Z} = \frac{a}{2} + \mathbb{Z}$. □

The following theorem, which describes the shapes of $\mathrm{NP}(G(w_\star, -))$, is the main result of this paper.

Theorem 5.19. *Fix a relevant character ε. Fix $w_\star \in \mathfrak{m}_{\mathbb{C}_p}$.*

(1) *The set of near-Steinberg ranges $\mathrm{nS}_{w_\star,k}$ for all k is nested, i.e. for any two such open intervals, either they are disjoint or one is contained in another.*

A near-Steinberg range $\mathrm{nS}_{w_\star,k}$ is called maximal *if it is not contained in other near-Steinberg ranges.*

(2) *The x-coordinates of the vertices of the Newton polygon $\mathrm{NP}(G(w_\star, -))$ are exactly those integers which do not lie in any $\mathrm{nS}_{w_\star,k}$. Equivalently, for an integer $n \geq 1$, the pair $(n, w_\star)$ is near-Steinberg if and only the point $(n, v_p(g_n(w_\star)))$ is not a vertex of $\mathrm{NP}(G(w_\star, -))$.*

(3) *Over the maximal near-Steinberg ranges, the slope of $\mathrm{NP}(G(w_\star, -))$ belongs to (5.18.1).*

Remark 5.20. (1) In Theorem 5.19 (1), it is allowed to have two disjoint open near-Steinberg interval with a common point at their closures.

(2) Combined with the discussion on near-Steinberg range in Remark 5.14, this Theorem implies that, for a fixed n, a "majority" of classical points $w_{k'}$ satisfies that $(n, g_n(w_{k'}))$ is a vertex of $\mathrm{NP}(G(w_{k'}, -))$.

Before proving the theorem, we first prove the following technical lemma on the comparison of sizes of two near-Steinberg ranges whose closures have a nonempty intersection.

Lemma 5.21. *Suppose that two positive integers $k_i = k_\varepsilon + (p-1)k_{i\bullet}$ for $i = 1, 2$ satisfy the condition that the closures of the near-Steinberg ranges $\overline{\mathrm{nS}}_{w_\star,k_i} = [\frac{1}{2}d^{\mathrm{Iw}}_{k_i} - L_{w_\star,k_i}, \frac{1}{2}d^{\mathrm{Iw}}_{k_i} + L_{w_\star,k_i}]$ for $i = 1, 2$ have non-trivial intersection. Without loss of generality, we assume that $L_{w_\star,k_1} \geq L_{w_\star,k_2}$. Then we have $L_{w_\star,k_1} \geq p^{\lceil \frac{1}{2} + \frac{p-1}{2}(L_{w_\star,k_2}-1)\rceil} - L_{w_\star,k_2} > L_{w_\star,k_2}$ and*

$$\Delta_{k_1,L_{w_\star,k_1}} - \Delta_{k_1,L_{w_\star,k_1}-1} > v_p(w_\star - w_{k_2}). \tag{5.21.1}$$

Proof. We can find $n \in \mathbb{N}$ which is contained in both $\overline{\mathrm{nS}}_{w_\star,k_i}$'s for $i = 1, 2$. By Remark 4.13, we have

$$k_{\mathrm{mid}\bullet}(n) \in \big[k_{i\bullet} - L_{w_\star,k_i}, k_{i\bullet} + L_{w_\star,k_i}\big] \quad \text{for} \quad i = 1, 2.$$

It follows that

$$|k_{1\bullet} - k_{2\bullet}| \leq |k_{1\bullet} - k_{\mathrm{mid}\bullet}(n)| + |k_{2\bullet} - k_{\mathrm{mid}\bullet}(n)| \leq L_{w_\star,k_1} + L_{w_\star,k_2}. \tag{5.21.2}$$

On the other hand, condition (5.11.1) and Lemma 5.2 imply that

$$\begin{aligned} v_p(w_\star - w_{k_i}) &\geq \Delta_{k_i,L_{w_\star,k_i}} - \Delta_{k_i,L_{w_\star,k_i}-1} = \Delta'_{k_i,L_{w_\star,k_i}} - \Delta_{k_i,L_{w_\star,k_i}-1} \\ &\geq \Delta'_{k_i,L_{w_\star,k_i}} - \Delta'_{k_i,L_{w_\star,k_i}-1} \geq \tfrac{3}{2} + \tfrac{p-1}{2}(L_{w_\star,k_i} - 1). \end{aligned} \tag{5.21.3}$$

It then follows that

$$v_p(k_{1\bullet} - k_{2\bullet}) \geq \tfrac{1}{2} + \tfrac{p-1}{2}\min\big\{L_{w_\star,k_1} - 1, L_{w_\star,k_2} - 1\big\} = \tfrac{1}{2} + \tfrac{p-1}{2}(L_{w_\star,k_2} - 1). \tag{5.21.4}$$

Combining (5.21.2) and (5.21.4), we see that $L_{w_\star,k_1}$ is much bigger than $L_{w_\star,k_2}$:

$$L_{w_\star,k_1} \geq p^{\lceil \frac{1}{2} + \frac{p-1}{2}(L_{w_\star,k_2}-1)\rceil} - L_{w_\star,k_2} > L_{w_\star,k_2}.$$

This is the first inequality we need to prove. Now we prove (5.21.1). Indeed, otherwise, we may deduce from (5.21.3) that

$$v_p(w_\star - w_{k_i}) \geq \Delta_{k_1,L_{w_\star,k_1}} - \Delta_{k_1,L_{w_\star,k_1}-1} \geq \tfrac{3}{2} + \tfrac{p-1}{2}(L_{w_\star,k_1} - 1).$$

It would then follow that

$$v_p(k_{1\bullet} - k_{2\bullet}) \geq \tfrac{1}{2} + \tfrac{p-1}{2}(L_{w_\star,k_1} - 1) \quad \text{but} \quad |k_{1\bullet} - k_{2\bullet}| < 2L_{w_\star,k_1}.$$

This is impossible. So (5.21.1) holds. □

Now we are ready to prove Theorem 5.19.

Proof. (1) We need to show that, under the setup and notations of Lemma 5.21, $\mathrm{nS}_{w_\star,k_2}$ cannot contain a boundary point $\frac{1}{2}d^{\mathrm{Iw}}_{k_1} - L_{w_\star,k_1}$ or $\frac{1}{2}d^{\mathrm{Iw}}_{k_1} + L_{w_\star,k_1}$ of the bigger near-Steinberg range $\mathrm{nS}_{w_\star,k_1}$. Suppose otherwise. By (5.21.1), we see that $v_p(w_\star - w_{k_1}) > v_p(w_\star - w_{k_2})$. It follows that $v_p(w_{k_1} - w_{k_2}) = v_p(w_\star - w_{k_2})$. So $\mathrm{nS}_{w_\star,k_2}$ is a near-Steinberg range for w_{k_1} too, i.e. $\mathrm{nS}_{w_{k_1},k_2}=\mathrm{nS}_{w_\star,k_2}$. Now apply Corollary 5.18 (2) to $\mathrm{nS}_{w_{k_1},k_2}$, we deduce that the convex hull of points

$$(n,\ v_p(g_{n,\hat{k}_1}(w_{k_1})))\quad\text{for}\quad n\in\overline{\mathrm{nS}}_{w_{k_1},k_2}$$

is a straight line and w_{k_1} is a zero of $g_n(w)$ when $n\in\mathrm{nS}_{w_{k_1},k_2}$. After a linear shift, this means that the convex hull of points $(n,\Delta'_{k_1,n})$ for $n\in\overline{\mathrm{nS}}_{w_{k_1},k_2}$ is a straight line. In particular, $\underline{\Delta}_{k_1}$ (with x-coordinate shifted to the right by $\frac12 d_{k_1}^{\mathrm{Iw}}$) does not have a vertex in $\mathrm{nS}_{w_{k_1},k_2}=\mathrm{nS}_{w_\star,k_2}$. But by the definition of near-Steinberg range (especially the definition of the integer $L_{w_*,k}$ in Definition 5.11), $(\pm L_{w_\star,k_1},\Delta'_{k_1,\pm L_{w_\star,k_1}})$ are vertices of $\underline{\Delta}_{k_1}$. So $\frac12 d_{k_1}^{\mathrm{Iw}}\pm L_{w_\star,k_1}$ cannot be contained in $\mathrm{nS}_{w_\star,k_2}=\mathrm{nS}_{w_{k_1},k_2}$, contradicting our assumption. (1) is proved.

(2) For this, we will prove the following list of statements for each $n\geq 1$:

(a) If n belongs to some near-Steinberg range $\mathrm{nS}_{w_\star,k}$, then $(n,v_p(g_n(w_\star)))$ is not a vertex of $\mathrm{NP}(G(w_\star,-))$.

(b) If n is not contained in any closure of the near-Steinberg range $\overline{\mathrm{nS}}_{w_\star,k}$, then $v_p(g_{n-1}(w_\star))+v_p(g_{n+1}(w_\star))>2v_p(g_n(w_\star))$.

(c) If n is the boundary point of precisely one maximal $\overline{\mathrm{nS}}_{w_\star,k}$, then

$$\text{if } n=\tfrac12 d_k^{\mathrm{Iw}}+L_{w_\star,k},\quad v_p(g_{n+1}(w_\star))-v_p(g_n(w_\star)>\tfrac{1}{2L_{w_\star,k}}\big(v_p(g_n(w_\star))-v_p(g_{n-2L_{w_\star,k}}(w_\star))\big);$$
$$\text{if } n=\tfrac12 d_k^{\mathrm{Iw}}-L_{w_\star,k},\quad \tfrac{1}{2L_{w_\star,k}}\big(v_p(g_{n+2L_{w_\star,k}}(w_\star))-v_p(g_n(w_\star))\big)>v_p(g_n(w_\star))-v_p(g_{n-1}(w_\star).$$

(d) If n is contained in the closure of two maximal near-Steinberg range, i.e. $n=\frac12 d_{k_-}^{\mathrm{Iw}}+L_{w_\star,k_-}=\frac12 d_{k_+}^{\mathrm{Iw}}-L_{w_\star,k_+}$, then

$$\frac{v_p(g_n(w_\star))-v_p(g_{n-2L_{w_\star,k_-}}(w_\star))}{2L_{w_\star,k_-}}<\frac{v_p(g_{n+2L_{w_\star,k_+}}(w_\star))-v_p(g_n(w_\star))}{2L_{w_\star,k_+}}.$$

Statements (b), (c), and (d) imply that for any three consecutive points P_1, P_2 and P_3 in the set $\Omega:=\{(n,v_p(g_n(w_\star)))|\ n$ does not belong to any near-Steinberg range $\mathrm{nS}_{w_\star,k}\}$, then P_2 lies (strictly) below the line segment connecting P_1 and P_3. It follows that the union of the line segments connecting consecutive points in Ω is (lower) convex. Combing with statement (a), we prove (2) immediately.

Statement (a) is proved in Corollary 5.18 (1). We now prove (b). By assumption, n does not belong to the near-Steinberg range $\mathrm{nS}_{w_\star,k}$ for any $k\equiv k_\varepsilon \bmod p-1$. In particular we can take $k=k_{\mathrm{mid}}(n)$ defined in Lemma-Notation 4.12. The non-near-Steinberg condition says that

$$v_p(w_\star-w_k)<\Delta_{k,1}-\Delta_{k,0}\overset{\text{Lemma 5.8}}{=}\Delta'_{k,1}-\Delta'_{k,0}.$$

By ghost duality and (4.19.2), we deduce that

$$2v_p(w_\star-w_k)<\Delta'_{k,1}+\Delta'_{k,-1}-2\Delta'_{k,0}=\sum_{k_{\max\bullet}(n-1)<k'_\bullet\leq k_{\max\bullet}(n)}v_p(w_k-w_{k'})+\sum_{k_{\min\bullet}(n-1)\leq k'_\bullet<k_{\min\bullet}(n)}v_p(w_k-w_{k'}).\tag{5.21.5}$$

We need to show that $v_p(g_{n-1}(w_\star))+v_p(g_{n+1}(w_\star))>2v_p(g_n(w_\star))$. By a formula similar to (4.19.2), it is equivalent to show that

$$\sum_{k_{\max\bullet}(n-1)<k'_\bullet\leq k_{\max\bullet}(n)}v_p(w_\star-w_{k'})+\sum_{k_{\min\bullet}(n-1)\leq k'_\bullet<k_{\min\bullet}(n)}v_p(w_\star-w_{k'})>2v_p(w_\star-w_k).\tag{5.21.6}$$

Note that $k_{\max\bullet}(n) - k_{\max\bullet}(n-1) \in \{a+2, p-1-a\}$. So there are at least 3 terms on the left hand side of (5.21.6). Thus, if $v_p(w_\star - w_k) \leq 1$, the inequality (5.21.6) holds trivially.

Now we assume that $v_p(w_\star - w_k) > 1$. If $v_p(w_\star - w_{k'}) \geq v_p(w_k - w_{k'})$ for every k' appearing in (5.21.6), then (5.21.6) would follow from (5.21.5). Now we assume that this is not the case, and let k' be in the sum (5.21.6) with maximal $v_p(w_k - w_{k'})$ satisfying $v_p(w_k - w_{k'}) > v_p(w_\star - w_{k'}) \geq 1$. Note that by the equality (2) in Lemma 4.25:

$$k_{\max\bullet}(n) - k_\bullet = pk_\bullet - \tilde{k}_{\min\bullet}(n-1) \quad \text{and} \quad k_{\max\bullet}(n-1) - k_\bullet = pk_\bullet - \tilde{k}_{\min\bullet}(n), \tag{5.21.7}$$

the existence of $k''_\bullet \in [k_{\min\bullet}(n-1), k_{\min\bullet}(n))$ implies that there is a $k'_\bullet \in (k_{\max\bullet}(n-1), k_{\max\bullet}(n)]$ such that

$$v_p(k'_\bullet - k_\bullet) = v_p(k_\bullet - k''_\bullet) + 1. \tag{5.21.8}$$

By the maximality of $v_p(w_k - w_{k'})$, we deduce that $k'_\bullet$ must live in $(k_{\max\bullet}(n-1), k_{\max\bullet}(n)]$. Moreover, since $v_p(w_k - w_{k'}) > 1$, using (5.21.7) again, we further deduce that there exists $k''_\bullet \in [k_{\min\bullet}(n-1), k_{\min\bullet}(n))$ satisfying (5.21.8). In particular, we have

$$v_p(w_\star - w_{k''}) \geq \min\{v_p(w_\star - w_k), v_p(w_k - w_{k''})\} > v_p(w_\star - w_k) - 1.$$

In conclusion, using that the first sum in (5.21.6) has at least 3 terms, we deduce that

$$\begin{aligned} \text{l.h.s. of (5.21.6)} &\geq v_p(w_\star - w_{k'}) + 1 + v_p(w_\star - w_{k''}) \\ &> v_p(w_\star - w_k) + 1 + (v_p(w_\star - w_k) - 1) = 2v_p(w_\star - w_k). \end{aligned}$$

This is exactly what we want.

We prove (c) and (d) using a uniform argument, with some abuse of notations:

Notation 5.22. We write $n = \frac{1}{2}d^{\mathrm{Iw}}_{k_-} + L_- = \frac{1}{2}d^{\mathrm{Iw}}_{k_+} - L_+$ for $L_\pm = L_{w_\star, k_\pm}$. If n is not the left (resp. right) boundary of a maximal near-Steinberg range, this is understood as $L_+ = \frac{1}{2}$ (resp. $L_- = \frac{1}{2}$) and $\frac{1}{2}d^{\mathrm{Iw}}_{k_+} = \frac{1}{2}d^{\mathrm{Iw}}_k + \frac{1}{2}$ (resp. $\frac{1}{2}d^{\mathrm{Iw}}_{k_-} = \frac{1}{2}d^{\mathrm{Iw}}_k - \frac{1}{2}$); and we shall not refer to the meaning of k_+ (resp. k_-) in later discussion. Moreover, in this case, we shall understand $\Delta_{k_+, L_+} - \Delta_{k_+, L_+-1} = 1$ (resp. $\Delta_{k_-, L_-} - \Delta_{k_-, L_--1} = 1$).

For the rest of the discussion, we assume that $L_+ \geq L_-$, and the other case can be proved similarly. In this case, $\mathrm{nS}_{w_\star, k_+}$ is a genuine near-Steinberg range, and by (5.21.1) and Lemma 5.2,

$$\Delta_{k_+, L_+} - \Delta_{k_+, L_+-1} > \Delta_{k_-, L_-} - \Delta_{k_-, L_--1}.$$

We need to show that the following three points

$$\left(\tfrac{1}{2}d^{\mathrm{Iw}}_{k_-} - L_-, v_p(g_{n-2L_-}(w_\star))\right), \quad \left(n, v_p(g_n(w_\star))\right), \quad \left(\tfrac{1}{2}d^{\mathrm{Iw}}_{k_+} + L_+, v_p(g_{n+2L_+}(w_\star))\right) \tag{5.22.1}$$

are convex.

Before proceeding, we need a technical result:

Lemma 5.23. *Under the above notations, we have:*

(1) *when $L_+ \geq 2$, there is at most one integer k_1 such that*
- $k_1 \equiv k_\varepsilon \bmod p-1$,
- *either $n = d^{\mathrm{ur}}_{k_1}$ or $n = d^{\mathrm{Iw}}_{k_1} - d^{\mathrm{ur}}_{k_1}$, or equivalently $k_1 \in \big(k_{\max}(n-1), k_{\max}(n)\big] \cup \big[k_{\min}(n-1), k_{\min}(n)\big)$, and*
- *$v_p(w_{k_1} - w_{k_+}) \geq v_p(w_\star - w_{k_+})$ (and hence $v_p(w_{k_1} - w_{k_+}) \geq \Delta_{k_\pm, L_\pm} - \Delta_{k_\pm, L_\pm - 1}$).*

(2) *when* $L_+ = 1$, *either the statement in* (1) *holds, or there exist two integers* k_1 *and* k_1' *such that*
  - $k_1 \equiv k_1' \equiv k_\varepsilon \bmod p-1$,
  - $n = d^{\mathrm{ur}}_{k_1}$ *and* $n = d^{\mathrm{Iw}}_{k_1'} - d^{\mathrm{ur}}_{k_1'}$, *or equivalently* $k_1 \in \big(k_{\max}(n-1), k_{\max}(n)\big]$ *and* $k_1' \in \big[k_{\min}(n-1), k_{\min}(n)\big)$, *and*
  - $v_p(w_{k_1} - w_{k_+}) = 2 \geq v_p(w_\star - w_{k_+})$ *and* $v_p(w_{k_1'} - w_{k_+}) \geq v_p(w_\star - w_{k_+})$

  When the latter case happens, we say that we are in 'special' case.

Proof of Lemma 5.23. To see the uniqueness of k_1, we first note that there is at most one such k_1 in each of the range $\big(k_{\max}(n-1), k_{\max}(n)\big]$ and $\big[k_{\min}(n-1), k_{\min}(n)\big)$ because those are intervals of length $< p^2$. (We warn the reader again of our unusual definition of $k_{\min}$ in Lemma-Notation 4.12.) Suppose that there are two k_1 and k_1' in the two intervals above, respectively. Then

$$v_p(k_{1\bullet} - k_{+\bullet}) \geq v_p(w_\star - w_{k_+}) - 1 \geq \Delta_{k_+, L_+} - \Delta_{k_+, L_+ - 1} - 1 \geq \tfrac{1}{2} + \tfrac{p-1}{2}(L_+ - 1), \tag{5.23.1}$$

where the last inequality is from Lemma 5.2, and the same inequality holds for $v_p(k'_{1\bullet} - k_{+\bullet})$. On the other hand, using the formulae for $k_{\mathrm{mid}\bullet}$ and $k_{\max\bullet}$ from Lemma-Notation 4.12 and that $n = \frac{1}{2} d^{\mathrm{Iw}}_{k_+} - L_+$, we have that

$$k_{+\bullet} = n + L_+ + \delta_\varepsilon - 1, \qquad k_{1\bullet} = (p+1)\tfrac{n}{2} + \beta_{[n]} - 1 - s$$

for some $s \in \{0, \dots, p-\theta\}$, where $\theta = \beta_{[n-1]} - \beta_{[n]} + \frac{p+1}{2}$ as given in the proof of Lemma 5.2. It follows that

$$k_{1\bullet} - k_{+\bullet} - \big(\tfrac{p-1}{2} k_{+\bullet} - \tfrac{p+1}{2}(L_+ + \delta_\varepsilon) - 1\big) = \tfrac{p+1}{2} + \beta_{[n]} - s. \tag{5.23.2}$$

Similarly, using the formulae for $k_{\mathrm{mid}\bullet}$ and $k_{\min\bullet}$ from Lemma-Notation 4.12, we can write $pk'_\bullet = (p+1)(\frac{n}{2} + \delta_\varepsilon) - \beta_{[n]} - s'$, for some $s' \in \{\theta, \dots, p\}$ and get

$$pk_{+\bullet} - pk'_{1\bullet} - \big(\tfrac{p-1}{2} k_{+\bullet} + \tfrac{p+1}{2}(L_+ - \delta_\varepsilon) - 1\big) = \tfrac{p+1}{2} + \beta_{[n]} + s' - p.$$

So we deduce that

$$v_p\big((pk_{+\bullet} - pk'_{1\bullet}) - (k_{1\bullet} - k_{+\bullet})\big) \geq \tfrac{1}{2} + \tfrac{p-1}{2}(L_+ - 1), \quad \text{and}$$
$$\big|(pk_{+\bullet} - pk'_{1\bullet}) - (k_{1\bullet} - k_{+\bullet})\big| = (p+1)L_+ + s + s' - p.$$

This is not possible unless $L_+ = 1$.

Now we assume $L_+ = 1$. Notice that $|s + s' - p| \leq p - \theta$. Thus $v_p\big((pk_{+\bullet} - pk'_{1\bullet}) - (k_{1\bullet} - k_{+\bullet})\big) \leq 1$. This forces $v_p(k_{1\bullet} - k_{+\bullet}) = 1$, and hence $v_p(w_{k_1} - w_{k_+}) = 2$. □

We continue our proof of statements (c) and (d). As suggested by Lemma 5.23, we divide our discussion into two cases: $L_+ \geq 2$ and $L_+ = 1$.

Assume $L_+ \geq 2$ first. Set $\mathbf{k} = \{k_+, k_1\}$ (and keep in mind that it is possible that $k_+ = k_1$). We would like to apply Proposition 5.16 (4) to $k = k_+$, the above $\mathbf{k}$, $w_\star$, and $[n_-, n_+] = [n - 2L_-, n + 2L_+]$. For this, we need to check that for each $k' = k_\varepsilon + (p-1)k'_\bullet \notin \mathbf{k}$ with $v_p(w_{k'} - w_{k_+}) \geq v_p(w_{k_+} - w_\star)$, the numbers $d^{\mathrm{ur}}_{k'}, d^{\mathrm{Iw}}_{k'} - d^{\mathrm{ur}}_{k'}, \frac{1}{2} d^{\mathrm{Iw}}_{k'}$ do not belong to $(n - 2L_-, n + 2L_+)$.

- By Proposition 5.16 (1), they do not belong to $(n, n + 2L_+)$ and $\frac{1}{2} d^{\mathrm{Iw}}_{k'} \neq n$.
- $d^{\mathrm{ur}}_{k'}, d^{\mathrm{Iw}}_{k'} - d^{\mathrm{ur}}_{k'}$ are not equal to n by our choice of $\mathbf{k}$ above.

- When $\mathrm{nS}_{w_\star,L_-}$ is a genuine near-Steinberg range, $\Delta_{k_+,L_+}-\Delta_{k_+,L_+-1}>\Delta_{k_-,L_-}-\Delta_{k_-,L_--1}$ allows us to apply Proposition 5.16 (1) to conclude that $d_{k'}^{\mathrm{ur}}, d_{k'}^{\mathrm{Iw}}-d_{k'}^{\mathrm{ur}}, \frac{1}{2}d_{k'}^{\mathrm{Iw}}$ do not belong to $(n-2L_-,n)$. (When $L_-=\frac{1}{2}$, no more discussion is needed.)

To sum up, we can apply Proposition 5.16 (4) to deduce that the three points in (5.22.1) is a linear shift of the following three points:

$$\Big(j,\ v_p(g_{j,\hat{\mathbf{k}}}(w_{k_+}))+\sum_{k\in\mathbf{k}}m_j(k)v_p(w_\star-w_k)\Big)\quad\text{with}\quad j=n-2L_-,\ n,\ n+2L_+.$$

These points are the same as

$$\Big(j,\ v_p(g_{j,\hat{k_+}}(w_{k_+}))+m_j(k_+)v_p(w_\star-w_{k_+})-m_j(k_1)\big(v_p(w_{k_+}-w_{k_1})-v_p(w_\star-w_{k_1})\big)\Big). \tag{5.23.3}$$

with $j=n-2L_-,n,n+2L_+$, where the last term appears only when k_1 exists and is not equal to k_+ (note that we flip the sign before the sum over $\mathbf{k}$ so that the summands are positive). we write P_-, P, P_+ for these three points, respectively. We separate the discussion into the following three possibilities:

(i) If $k_1=k_+$, that is, $d_{k_+}^{\mathrm{ur}}=n$, then $m_j(k_+)=0$ for $j=n-2L_-,n,n+2L_+$. Proposition 4.18 (applied to the slope of $\overline{PP_+}$) and Proposition 4.28 (applied to the slope of $\overline{P_-P}$) together imply that

$$\text{slope of }\overline{PP_+}-\text{slope of }\overline{P_-P}\geq\frac{k_+-2}{2}-\frac{k_+-1}{p+1}>0.$$

This last inequality needs $k_+\geq 3$; but this is okay because when $k_+=2$ or $2\frac{1}{2}$ (by our convention made in Notation 5.22), $d_{k_+}^{\mathrm{Iw}}$ is at most 1 and we cannot be in the case of (c) and (d). The points P_-, P, and P_+ are clearly convex. Thus (c) and (d) are proved in this case.

(ii) If there is no k_1, then obviously $d_{k_+}^{\mathrm{ur}}\neq n$. Using $\Delta_{k_+,L_+}-\Delta_{k_+,L_+-1}\geq\Delta_{k_-,L_-}-\Delta_{k_-,L_--1}$, we deduce that $d_{k_+}^{\mathrm{ur}}$ does not belong to $(n-2L_-,n)$. Thus $[n-2L_-,n]\subseteq[d_{k_+}^{\mathrm{ur}},\frac{1}{2}d_{k_+}^{\mathrm{Iw}})$; so $m_j(k_+)$ is linear in $j\in[n-2L_-,n]$. By the ghost duality, the slope of $\overline{P_-P}$ is given by

$$\begin{aligned}&\frac{k_+-2}{2}-\frac{\Delta'_{k_+,L_++2L_-}-\Delta'_{k_+,L_+}}{2L_-}+v_p(w_\star-w_{k_+})\\ \leq\ &\frac{k_+-2}{2}-\big(\Delta_{k_+,L_++1}-\Delta_{k_+,L_+}\big)+v_p(w_\star-w_{k_+})<\frac{k_+-2}{2},\end{aligned}$$

where the first inequality uses the fact that (L_+,Δ'_{k_+,L_+}) is a vertex in $\underline{\Delta}_{k_+}$. On the other hand, the slope of $\overline{PP_+}$ is simply $\frac{k_+-2}{2}$. Then (c) and (d) hold in this case too.

(iii) If k_1 exists and $k_1\neq k_+$, then by the uniqueness of k_1, we get $d_{k_+}^{\mathrm{ur}}\neq n$. As in (ii), we first have $[n-2L_-,n]\subseteq[d_{k_+}^{\mathrm{ur}},\frac{1}{2}d_{k_+}^{\mathrm{Iw}})$; so $m_j(k_+)$ is linear in $j\in[n-2L_-,n]$. Regardless of whether $n=d_{k_1}^{\mathrm{ur}}$ or $d_{k_1}^{\mathrm{Iw}}-d_{k_1}^{\mathrm{ur}}$, by (5.23.3),

$$\begin{aligned}&\text{slope of }\overline{PP_+}-\text{slope of }\overline{P_-P}\\ =\ &\frac{k_+-2}{2}-\Big(\frac{k_+-2}{2}-\frac{\Delta'_{k_+,L_++2L_-}-\Delta'_{k_+,L_+}}{2L_-}+v_p(w_\star-w_{k_+})\Big)-\big(v_p(w_{k_+}-w_{k_1})-v_p(w_\star-w_{k_1})\big)\\ \geq\ &\frac{\Delta'_{k_+,L_++2L_-}-\Delta'_{k_+,L_+}}{2L_-}-v_p(w_{k_+}-w_{k_1}),\end{aligned}$$

where the last inequality uses the fact that

$$v_p(w_{k_1}-w_{k_+})\geq v_p(w_\star-w_{k_+}) \quad\Rightarrow\quad v_p(w_\star-w_{k_1})\geq v_p(w_\star-w_{k_+}).$$

Applying Corollary 5.3 to $k=k_+$, $k'=k_1$ $\ell=L_++1,\dots,L_++2L_-$, and noting that $d_{k_1}^{\mathrm{ur}}$ or $d_{k_1}^{\mathrm{Iw}}-d_{k_1}^{\mathrm{ur}}$ belong to $(\frac12 d_{k_+}^{\mathrm{Iw}}-\ell,\frac12 d_{k_+}^{\mathrm{Iw}}+\ell)$ for these ℓ, we then have

$$\Delta'_{k_+,\ell}-\Delta'_{k_+,\ell-1}\geq \tfrac12(2\ell-1)+v_p(w_{k_+}-w_{k_1})>v_p(w_{k_+}-w_{k_1}).$$

Summing this up over $\ell=L_++1,\dots,L_++2L_-$, we obtain

$$\Delta'_{k_+,L_++2L_-}-\Delta'_{k_+,L_+}>2L_-\cdot v_p(w_{k_+}-w_{k_1}).$$

This proves (c) and (d) in the case $L_+\geq 2$.

Now we treat the case for $L_+=1$, and hence $L_-=\frac12$. We can assume that we are in the 'special case'. Replace the set $\mathbf{k}=\{k_+,k_1\}$ by $\mathbf{k}=\{k_+,k_1,k_1'\}$, and a similar argument as above reduces the proof to show that the following three points:

(5.23.4)

$$\Big(j,\ v_p(g_{j,\hat{k_+}}(w_{k_+}))+m_j(k_+)v_p(w_\star-w_{k_+})-\sum_{k=k_1,k_1'}m_j(k)\big(v_p(w_{k_+}-w_k)-v_p(w_\star-w_k)\big)\Big).$$

with $j=n-1,n,n+2$, are convex. We denote these three points by P_-,P,P_+ as before. By a similar argument as in (iii), we have

$$\begin{aligned}&\text{slope of }\overline{PP_+}-\text{slope of }\overline{P_-P}\\ =&\ \frac{k_+-2}{2}-\Big(\frac{k_+-2}{2}-(\Delta'_{k_+,2}-\Delta'_{k_+,1})+v_p(w_\star-w_{k_+})\Big)-\sum_{k=k_1,k_1'}\big(v_p(w_{k_+}-w_k)-v_p(w_\star-w_k)\big)\\ \geq&\ (\Delta'_{k_+,2}-\Delta'_{k_+,1})-v_p(w_{k_+}-w_{k_1'})-\Big(\big(v_p(w_{k_+}-w_{k_1})-v_p(w_\star-w_{k_1})\big)\Big).\end{aligned}$$

Recall $v_p(w_{k_1}-w_{k_+})=2$ and $v_p(w_\star-w_{k_1})\geq 1$, so we have

$$\text{slope of }\overline{PP_+}-\text{slope of }\overline{P_-P}\geq(\Delta'_{k_+,2}-\Delta'_{k_+,1})-v_p(w_{k_+}-w_{k_1'})-1$$

Since $d_{k_1'}^{\mathrm{Iw}}-d_{k_1'}^{\mathrm{ur}}=n\in(\frac12 d_{k_+}^{\mathrm{Iw}}-2,\frac12 d_{k_+}^{\mathrm{Iw}}+2)$, we can apply Corollary 5.3 to $k=k_+$, $k'=k_1'$ and $l=2$, and get

$$\Delta'_{k_+,2}-\Delta'_{k_+,1}-v_p(w_{k_+}-w_{k_1'})\geq\frac32>1.$$

This completes the proof of (c) and (d), and part (2) of the theorem as well.

(3) By (2), over a maximal near-Steinberg range $\overline{\mathrm{nS}}_{w_\star,k}$, the slopes are given by the slopes of the segment connecting

$$\big(\tfrac12 d_k^{\mathrm{Iw}}-L_{w_\star,k},\ v_p(g_{\frac12 d_k^{\mathrm{Iw}}-L_{w_\star,k}}(w_\star))\big)\quad\text{and}\quad\big(\tfrac12 d_k^{\mathrm{Iw}}+L_{w_\star,k},\ v_p(g_{\frac12 d_k^{\mathrm{Iw}}+L_{w_\star,k}}(w_\star))\big).$$

Then (3) follows from Corollary 5.18 (1). ☐

This proposition above implies the following, which is in close relation with Breuil–Buzzard–Emerton's Conjecture 1.12.

Corollary 5.24. *Fix ε a relevant character. Then for any $k\equiv k_\varepsilon \bmod p-1$,*

(1) *the slopes of $\mathrm{NP}(G(w_k,-))$ with multiplicity one are all integers, and*
(2) *other slopes of $\mathrm{NP}(G(w_k,-))$ always have even multiplicity and they belong to $\frac{a}{2}+\mathbb{Z}$.*

Proof. Note that $v_p(g_n(w_k))$ is always an integer by definition. So a slope with multiplicity 1 must be an integer. (2) follows from Theorem 5.19 (3) above. $\square$

Example 5.25. We give a example of a situation where near-Steinberg ranges are nested. A pathological situation could be when there exists a sequence of distinct positive integers $k_{1\bullet}, \dots, k_{r\bullet}$ such that $k_{i+1\bullet} \equiv k_{i\bullet} \bmod p^{pk_{i\bullet}}$. Set $k_i := k_\varepsilon + (p-1)k_{i\bullet}$. If $n = \frac{1}{2}d_{k_r}^{\mathrm{Iw}} - 1$ and $v_p(w_\star - w_{k_r}) \geq k_{r-1\bullet}$, then $(n, w_\star, k_i)$ is near-Steinberg for all i.

Proposition 5.26. *Fix ε a relevant character and $k_0 \equiv k_\varepsilon \bmod p-1$. The following statements are equivalent for $\ell \in \{0, \dots, \frac{1}{2}d_{k_0}^{\mathrm{new}} - 1\}$.*

(1) *The point $(\ell, \Delta'_{k_0,\ell})$ is not a vertex of $\underline{\Delta}_{k_0}$,*
(2) *$(\frac{1}{2}d_{k_0}^{\mathrm{Iw}} + \ell, w_{k_0}, k_1)$ is near-Steinberg for some $k_1 > k_0$, and*
(3) *$(\frac{1}{2}d_{k_0}^{\mathrm{Iw}} - \ell, w_{k_0}, k_2)$ is near-Steinberg for some $k_2 < k_0$.*

Moreover, the slopes of $\underline{\Delta}_{k_0}$ with multiplicity one belong to $\mathbb{Z}$. Other slopes all have even multiplicity and the slopes belong to $\frac{a}{2} + \mathbb{Z}$.

Proof. The implication (2) ⇒ (1) and (3) ⇒ (1) is proved in Corollary 5.18 (2). We will prove (1) ⇒ (2) and the proof of (1) ⇒ (3) is similar. We need to show that, for each integer ℓ, if $\big(\frac{1}{2}d_{k_0}^{\mathrm{Iw}} + \ell, w_{k_0}, k_1\big)$ is not near-Steinberg for any $k_1 > k_0$, then $(\ell, \Delta'_{k_0,\ell})$ is a vertex. Let $2L_+$ (resp. $2L_-$) denote the largest size of near-Steinberg range nS_{w_k,k_+} (resp. nS_{w_k,k_-}) such that $\frac{1}{2}d_{k_0}^{\mathrm{Iw}} + \ell = \frac{1}{2}d_{k_+}^{\mathrm{Iw}} - L_+$ (resp. $\frac{1}{2}d_{k}^{\mathrm{Iw}} + \ell = \frac{1}{2}d_{k_-}^{\mathrm{Iw}} + L_-$). If such near-Steinberg range does not exist, we put $L_+ = \frac{1}{2}$ (resp. $L_- = \frac{1}{2}$). Then we need to show that

$$\frac{v_p(\Delta'_{k_0,\ell-2L_-}) - v_p(\Delta'_{k_0,\ell})}{2L_-} < \frac{v_p(\Delta'_{k_0,\ell+2L_+}) - v_p(\Delta'_{k_0,\ell})}{2L_+}. \tag{5.26.1}$$

We shall indicate how to modify the proof of (b)(c)(d) of Theorem 5.19 (2), to prove the corresponding situation in our case. Indeed, when $L_+ = L_- = \frac{1}{2}$, we may prove (5.26.1) using the exactly same argument as in the proof of (a) of Theorem 5.19 (2) for $w_\star = w_{k_0}$ and $n = \frac{1}{2}d_{k_0}^{\mathrm{Iw}} + \ell$, except that $g_j(w_\star)$ is replaced by $\Delta'_{k_0, j - \frac{1}{2}d_{k_0}^{\mathrm{Iw}}}$ for $\big|j - \big(\frac{1}{2}d_{k_0}^{\mathrm{Iw}} + \ell\big)\big| \leq 1$ and the role of $w_\star$ is played by w_{k_0}.

Now, we assume that at least one of L_+ and L_- is greater or equal to 1. Without loss of generality, we assume that $L_+ \geq L_-$ and the other case can be treated similarly. Then we can argue in a same way as in the proof of (c) and (d) of Theorem 5.19 (2), with $w_\star = w_{k_0}$, $n = \frac{1}{2}d_{k_0}^{\mathrm{Iw}} + \ell$, and $g_j(w_\star)$ replaced by $g_{j,\hat{k}_0}(w_{k_0})$. After we apply Proposition 5.16 (4), the points P_-, P, P_+ in (5.23.3) should become

$$\Big(j,\ v_p(g_{j,\hat{k_+}}(w_{k_+})) + (m_j(k_+) - m_j(k_0))v_p(w_{k_0} - w_{k_+}) - m_j(k_1)\big(v_p(w_{k_+} - w_{k_1}) - v_p(w_{k_0} - w_{k_1})\big)\Big). \tag{5.26.2}$$

But $m_j(k_0)$ is linear in k_0; so we may remove this term, and completely reduce to the proof of (c) and (d) therein. $\square$

Appendix A. Recollection of representation theory of $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$

In this appendix, we recall some basic facts on (right) $\mathbb{F}$-representations of $\mathrm{GL}_2(\mathbb{F}_p)$. Our main references are [Br-notes] and [Pa04] which work with left representations. Our convention will be the corresponding *right representation given by transpose*.

A.1. **Setup.** Throughout this appendix, we fix an *odd* prime number p. We set

$$\bar{\mathrm{G}} = \mathrm{GL}_2(\mathbb{F}_p), \quad \bar{\mathrm{U}} = \begin{pmatrix} 1 & \mathbb{F}_p \\ 0 & 1 \end{pmatrix}, \quad \text{and } \ \bar{\mathrm{B}} = \begin{pmatrix} \mathbb{F}_p^\times & \mathbb{F}_p \\ 0 & \mathbb{F}_p^\times \end{pmatrix}.$$

$$\bar{\mathrm{T}} = \begin{pmatrix} \mathbb{F}_p^\times & 0 \\ 0 & \mathbb{F}_p^\times \end{pmatrix}, \quad \bar{\mathrm{U}}^- = \begin{pmatrix} 1 & 0 \\ \mathbb{F}_p & 1 \end{pmatrix}, \quad \text{and } \ \bar{\mathrm{B}}^- = \begin{pmatrix} \mathbb{F}_p^\times & 0 \\ \mathbb{F}_p & \mathbb{F}_p^\times \end{pmatrix}.$$

The coefficient field will be a finite extension $\mathbb{F}$ of $\mathbb{F}_p$. Write $\omega : \mathbb{F}_p^\times \to \mathbb{F}^\times$ for the character induced by inclusion. For a character $\chi_{a,b} := \omega^a \times \omega^b$ and an $\mathbb{F}$-representation V of $\bar{\mathrm{G}}$, we write $V(\chi)$ for the subspace of V on which $\bar{\mathrm{T}}$ acts by χ.

Let $\bar{s} = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ represent the nontrivial element in the Weyl group of $\bar{\mathrm{G}}$. If χ is a character of $\bar{\mathrm{T}}$, we use χ^s to denote the character defined by

$$\chi^s(\bar{t}) = \chi(\bar{s}^{-1}\bar{t}\bar{s}), \quad \text{for all } \bar{t} \in \bar{\mathrm{T}}.$$

For a pair of non-negative integers (a, b), we use $\sigma_{a,b}$ to denote the *right* $\mathbb{F}$-representation $\mathrm{Sym}^a\, \mathbb{F}^{\oplus 2} \otimes \det^b$ of $\bar{\mathrm{G}}$ (in particular, b is taken modulo $p-1$). The right action is the transpose of the left action in [Br-notes, Pa04]; explicitly, it is, realized on the space of homogeneous polynomials of degree a:

$$h(X,Y)\big|_{\left(\begin{smallmatrix} \bar{\alpha} & \bar{\beta} \\ \bar{\gamma} & \bar{\delta} \end{smallmatrix}\right)} := h(\bar{\alpha}X + \bar{\beta}Y, \bar{\gamma}X + \bar{\delta}Y) \quad \text{for } \left(\begin{smallmatrix} \bar{\alpha} & \bar{\beta} \\ \bar{\gamma} & \bar{\delta} \end{smallmatrix}\right) \in \bar{\mathrm{G}}.$$

When $a \in \{0, \dots, p-1\}$ and $b \in \{0, \dots, p-2\}$, $\sigma_{a,b}$ is irreducible. These representations exhaust all irreducible (right) $\mathbb{F}$-representations of $\mathrm{GL}_2(\mathbb{F}_p)$, and are called *Serre weights.* We use $\mathrm{Proj}_{a,b}$ to denote the projective envelope of $\sigma_{a,b}$ as a (right) $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$-module.

Lemma A.2. *The projective envelope* $\mathrm{Proj}_{a,b}$ *can be described explicitly as follows.*

(1) *For any b,* $\mathrm{Proj}_{0,b}$ *is of dimension p over $\mathbb{F}$, and admits a socle filtration*

$$\mathrm{Proj}_{0,b} := \sigma_{0,b} \ \text{———}\ \sigma_{p-3,b+1} \ \text{———}\ \sigma_{0,b},$$

where the $\sigma_{0,b}$ on the left represents the socle, the $\sigma_{0,b}$ on the right represents the cosocle, each line represents a (unique up to isomorphism) nontrivial extension between the two Jordan–Hölder factors.

(2) *When $1 \leq a \leq p-2$,* $\mathrm{Proj}_{a,b}$ *is of dimensional $2p$ over $\mathbb{F}$, and admits a socle filtration*

$$\begin{tikzcd}[column sep=small, row sep=tiny] & \sigma_{p-1-a,a+b} \arrow[dr, dash] & \\ \mathrm{Proj}_{a,b} := \sigma_{a,b} \arrow[ur, dash] \arrow[dr, dash] & \oplus & \sigma_{a,b}, \\ & \sigma_{p-3-a,a+b+1} \arrow[ur, dash] & \end{tikzcd}$$

where the $\sigma_{a,b}$ on the left represents the socle, the $\sigma_{a,b}$ on the right represents the cosocle, each line represents a (unique up to isomorphism) nontrivial extension between the two factors. (Here when $a = p-2$, the term $\sigma_{p-3-a,a+b+1}$ is interpreted as 0.)

(3) *For any b, the Serre weight $\sigma_{p-1,b}$ is projective, i.e.* $\mathrm{Proj}_{p-1,b} = \sigma_{p-1,b}$. *In particular,* $\dim_{\mathbb{F}} \mathrm{Proj}_{p-1,b} = p$.

Moreover, in either case $\mathrm{Proj}_{a,b}$ *is also the injective envelope of $\sigma_{a,b}$ as an $\mathbb{F}$-representation of* $\mathrm{GL}_2(\mathbb{F}_p)$.

Proof. The dimensions of the projective envelopes follows from [Pa04, Lemma 4.1.7]. The socle filtrations of the projective envelopes in (1) and (2) follows from [BrPa12, Lemma 3.4, Lemma 3.5]. Part (3) follows from [Pa04, Corollary 4.2.22]. □

A.3. **Induced representations.** Consider the following character η of $\bar{\mathrm{B}}$

$$\eta : \bar{\mathrm{B}} \to \mathbb{F}^\times, \qquad \left(\begin{smallmatrix} \bar\alpha & \bar\beta \\ 0 & \bar\delta \end{smallmatrix}\right) \mapsto \bar\alpha.$$

For $k \in \mathbb{Z}$, we have an induced representation

$$\mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}(\eta^k) := \{f : \bar{\mathrm{G}} \to \mathbb{F} \mid f(\bar b \bar g) = \eta^k(\bar b) f(\bar g), \text{ for all } \bar b \in \bar{\mathrm{B}}\},$$

which we equip with the *right action* given by

$$f|_{\bar h}(\bar g) := f(\bar g \bar h^{\mathrm{T}}), \quad \text{for } \bar g, \bar h \in \bar{\mathrm{G}}.$$

This is the transpose of the usual left action.

Lemma A.4. (1) *For any integer $1 \le k \le p-1$, we have an exact sequence of $\bar{\mathrm{G}}$-representations:*

$$0 \to \sigma_{k,0} \to \mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\, \eta^k \to \sigma_{p-1-k,k} \to 0. \tag{A.4.1}$$

The above exact sequence splits if and only if $k = p-1$.

(2) *Taking $\bar U$-invariants of* (A.4.1) *gives an exact sequence of $\bar{\mathrm{U}}$-invariants.*

$$0 \to \sigma_{k,0}^{\bar{\mathrm{U}}} \to \left(\mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\, \eta^k\right)^{\bar{\mathrm{U}}} \to \sigma_{p-1-k,k}^{\bar{\mathrm{U}}} \to 0.$$

(3) *In the description of $\mathrm{Proj}_{a,b}$ of Lemma A.2 (2), the extensions*

$$\sigma_{a,b} \text{ ——— } \sigma_{p-1-a,a+b} \qquad \text{and} \qquad \sigma_{p-1-a,a+b} \text{ ——— } \sigma_{a,b}$$

are twists of induced representations $\mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\, \eta^a \otimes \det^b$ and $\mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\, \eta^{p-1-a} \otimes \det^{a+b}$, respectively.

Proof. (1) follows from [Pa04, Proposition 3.2.2,Lemma 4.1.3]. (2) follows from [Pa04, Lemma 3.1.11]. (3) follows from [Pa04, Lemma 4.1.1]. □

Lemma A.5. *When $1 \le a \le p-2$, in terms of the socle filtration in Lemma A.2 (2), the $\bar{\mathrm{U}}$-invariant subspace of $\mathrm{Proj}_{a,b}$ is a 2-dimensional subspace given by*

$$(\mathrm{Proj}_{a,b})^{\bar{\mathrm{U}}} \cong \sigma_{a,b}^{\bar{\mathrm{U}}} \oplus \sigma_{p-1-a,a+b}^{\bar{\mathrm{U}}},$$

where the first term is the $\sigma_{a,b}$ in the socle. In particular, the actions of $\bar{\mathrm{T}}$ on these two 1-dimensional subspaces are given by the characters

$$\omega^b \times \omega^{a+b} \qquad \text{and} \qquad \omega^{a+b} \times \omega^b, \qquad \text{respectively.}$$

Proof. Since $\mathrm{Proj}_{a,b}$ is a free module over $\mathbb{F}[\bar{\mathrm{U}}]$, we know that $\dim(\mathrm{Proj}_{a,b})^{\bar{\mathrm{U}}} = 2$. But $\mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\, \eta^a \otimes \det^b$ is a subrepresentation of $\mathrm{Proj}_{a,b}$ by Lemma A.4 (3) and has already 2-dimensional $\bar{\mathrm{U}}$-invariants by Lemma A.4 (2). So we have

$$(\mathrm{Proj}_{a,b})^{\bar{\mathrm{U}}} \cong \left(\mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\, \eta^a\right)^{\bar{\mathrm{U}}} \otimes \det^b \cong \sigma_{a,b}^{\bar{\mathrm{U}}} \oplus \sigma_{p-1-a,a+b}^{\bar{\mathrm{U}}}. \qquad \square$$

Lemma A.6. *For any integer $k \ge p+1$, we have an exact sequence of $\bar{\mathrm{G}}$-representations:*

$$0 \to \sigma_{k-(p+1),1} \xrightarrow{i} \sigma_{k,0} \xrightarrow{\pi} \mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\, \eta^k \to 0.$$

Proof. This is [Re10, Proposition 2.4]. For the convenience of the readers, we recall the definition of the two maps. The map i is given by multiplication by the homogeneous polynomial $f_p(X,Y) = X^pY - XY^p$ of degree $p+1$. It is straightforward to check that i is $\bar{\mathrm{G}}$-equivariant and injective. The map π comes from adjunction

$$\mathrm{Hom}_{\bar{\mathrm{G}}}(\sigma_{k,0}, \mathrm{Ind}_{\bar{\mathrm{B}}}^{\bar{\mathrm{G}}}\,\eta^k) \cong \mathrm{Hom}_{\bar{\mathrm{B}}^-}(\sigma_{k,0}, \eta^k).$$

(Note that both $\sigma_{k,0}$ and the induced representations are right representations, coming from the transpose of the natural left action.) The vector $X^k \in \sigma_{k,0}$ is $\bar{\mathrm{U}}^-$-invariants and is an eigenvector of $\bar{\mathrm{T}}$ with eigencharacter η^k. $\square$

We conclude the appendix with the following result that is claimed in Remark 2.23(2).

Proposition A.7. *Let $\bar{\rho} = \begin{pmatrix} \omega_1^{a+b+1} & *\neq 0 \\ 0 & \omega_1^b \end{pmatrix}$ with $1 \leq a \leq p-4$ and $0 \leq b \leq p-2$ be as in Definition 2.22 and let $\widetilde{\mathrm{H}}$ is a primitive $\mathcal{O}[\![K_p]\!]$-projective augmented module of type $\bar{\rho}$. In particular, $\overline{\mathrm{H}} = \widetilde{\mathrm{H}}/(\varpi, \mathrm{I}_{1+p\mathrm{M}_2(\mathbb{Z}_p)})$ is isomorphic to $\mathrm{Proj}_{a,b}$ as a right $\mathbb{F}[\mathrm{GL}_2(\mathbb{F}_p)]$-module. Then $\mathrm{S}_2^{\mathrm{Iw}}(\omega^b \times \omega^{a+b})$ is of rank 1 over $\mathcal{O}$, and the U_p-action on this space is invertible (over $\mathcal{O}$); in particular, the U_p-slope on this space is zero.*

Proof. By Proposition 4.1, $\mathrm{S}_2^{\mathrm{Iw}}(\omega^b \times \omega^{a+b})$ is free of rank 1 over $\mathcal{O}$.

Consider three characters $\varepsilon = 1 \times \omega^a$, $\varepsilon' = \omega^a \times 1$ and $\psi = \omega^a \times \omega^a$. By Corollary 4.4 and Proposition 4.7, we have $d_{p+1-a}^{\mathrm{Iw}}(\psi) = 2$ and $d_{p+1-a}^{\mathrm{ur}}(\psi) = 0$. We choose a set $\{\tilde{u}_j = \begin{pmatrix} 1 & 0 \\ j & 1 \end{pmatrix}, j = 0, \dots, p-1, \tilde{u}_\infty = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\}$ of coset representatives of $\mathrm{Iw}_p \setminus \mathrm{K}_p$. The formula

$$\mathrm{pr}(\varphi)(m) = \sum_{j=0,\dots,p-1,\infty} \varphi(m\tilde{u}_j^{-1})|_{\tilde{u}_j}, \varphi \in \mathrm{S}_{p+1-a}^{\mathrm{Iw}}(\psi), m \in \tilde{\mathrm{H}}$$

defines a map $\mathrm{pr} : \mathrm{S}_{p+1-a}^{\mathrm{Iw}}(\psi) \to \mathrm{S}_{p+1-a}^{\mathrm{ur}}(\psi)$. Here $(\cdot)|_{\tilde{u}_j}$ in the above formula is the right action of K_p defined in (2.20.1). On the other hand, a straightforward computation gives an equality of operators $\mathrm{AL}_{p+1-a,\psi} \circ \mathrm{pr} - \mathrm{AL}_{p+1-a,\psi} = U_p$ on $\mathrm{S}_{p+1-a}^{\mathrm{Iw}}(\psi)$. Since $\mathrm{S}_{p+1-a}^{\mathrm{ur}}(\psi) = 0$, we actually have $\mathrm{U}_p = -\mathrm{AL}_{p+1-a,\psi}$ on $\mathrm{S}_{p+1-a}^{\mathrm{Iw}}(\psi)$. So by Proposition 3.12, the U_p-slopes on $\mathrm{S}_{p+1-a}^{\mathrm{Iw}}(\psi)$ are $\frac{p-1-a}{2}$ with multiplicity 2. By Proposition 3.10, we have $v_p(c_1^{(\varepsilon')}(w_{p+1-a})) \geq \frac{p-1-a}{2} > 1$. Since $c_1^{(\varepsilon')}(w) \in \mathcal{O}[\![w]\!]$ and $v_p(w_{p+1-a}) \geq 1$, $v_p(w_2) \geq 1$, we have $v_p(c_1^{(\varepsilon')}(w_2)) \geq 1$. Since $d_2^{\mathrm{Iw}}(\varepsilon') = 1$, by Proposition 3.10 and Proposition 3.12, we see that the U_p-slope on $\mathrm{S}_2^{\mathrm{Iw}}(\varepsilon')$ is exactly 1. Now by Proposition 3.12, the U_p-slope on $\mathrm{S}_2^{\mathrm{Iw}}(\varepsilon)$ is 0. $\square$

Appendix B. Errata for [LWX14]

We include some errata for [LWX14] here.

B.1. [LWX14, Proposition 3.4]**.** This is just a bad convention, the action $||_{\delta_p}^{[\cdot]}$ is a *right* action; however, [LWX14, Proposition 3.4] uses the column vector convention. Ideally, one should swap the indices of $P_{m,n}(\delta_p)$. This does not create an mathematical error.

B.2. [LWX14, § 5.4]**.** In [LWX14, (5.4.1)], we defined a closed subspace

$$\mathrm{Ind}_{B^{\mathrm{op}}(\mathbb{Z}_p)}^{\mathrm{Iw}_q}([\cdot]')^{\mathrm{mod}} = \widehat{\bigoplus}_{n\geq 0} T^n \Lambda^{>1/p} \cdot \binom{z}{n}$$

of the space of continuous functions $\mathcal{C}^0(\mathbb{Z}_p; \Lambda^{>1/p})$. Unfortunately, as stated, this subspace is *not* stable under the $\mathbf{M}_1$-action. If one insists on having an $\mathbf{M}_1$-action, one has to replace the completed direct sum above with direct product, then the rest of the estimate stays valid, except that the space $S_{\mathrm{int}}^{D,\dagger,1}$ is not a Banach module over $\Lambda^{>1/p}$ in the usual sense; it is the $\Lambda^{>1/p}$-linear dual of a Banach module over $\Lambda^{>1/p}$. The compactness of U_p needs be interpreted otherwise.

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